-**LMP**-	V_XC_S affix	-**RMPŘ**-	illocution category	-**MŢPR**-	perspective category
-**LMPW**-	V_XC_S affix-type	-**RMPÇ**-	validation category	-**MŢPŘ**-	extension category
-**LMPY**-	V_XC_S affix-degree	-**RMPĻ**-	mood category	-**MŢPF**-	plexity category
-**LMPL**-	formative (part of speech)	-**RMPM**-	context category	-**MŢPŢ**-	similarity catetgory
-**LMPR**-	adjunct (part of speech)	-**RMPN**-	aspect category	-**MŢPÇ**-	separability category
-**LMPŘ**-	referential (part of speech)	-**RMPH**-	effect category	-**MŢPS**-	version category
				-**MŢPŠ**-	function category
				-**MŢPH**-	specification category

To name a particular grammatical sub-category (e.g., comitative case), use the appropriate root/stem above in conjunction with one of the following rules:

- To name a noun case or verbal format, use the stem -- above in conjunction with the specialized C_S-root (see Sec. 4.2 of the Design Document) of the Type-2 case-accessor affix for the case in question. For example, the COMITATIVE case would be named ***warmpa ëizyi've*** *'grammatical case/format associated with that which accompanies'*.
- For categories other than noun cases or verbal formats, use the appropriate root/stem from the list above in conjunction with one of the following stems as appropriate, where that stem in turn manifests (i.e., is declined into) the particular morphological category: -**PS**- Stem 1: 'to happen/occur'; -**PS**- Stem 2: 'to be/manifest'. For example, the Ithkuil term for the subjunctive mood would be ***warmpļa hre psi've*** *'grammatical mood associated with what may happen'*.

2.0 COMMON STATES AND ACTS

-**PS**- **'HAPPEN / OCCUR(ENCE) / MANIFEST / EVENT'**		
BSC	**STEM 1:** (to be) an occurrence of something; to happen / occur / take place / transpire / to "pass" [both the content/nature of the event and its transpiring	**STEM 2:** a state/act of manifesting; be or make real / reify
CTE	(to be) a state of something occurring/transpiring of an act/event/state	
CSV	(to be) an act of (causing) something to happen/occur	**STEM 3:** event (spatio-temporal context)
OBJ	(to be) that which happens/occurs/transpires	

-PSM- ‘STEP / ACT / ACTION / PHENOMENON / INCIDENT’		
BSC	**STEM 1:** (to be) a step/act/action (to be) taken; to take a step (to do/accomplish something), to act (in order to do/accomplish something), to take an action (to do something	**STEM 2:** (to be) a phenomenon; for a phenomenon to happen [i.e., an event/occurrence/state so unusual, surprising, unlikely, or consequential as to be significant] **STEM 3:** (to be) an incident [i.e., an event/occurrence with significant and (potentially) adverse/undesirable consequences]
CTE	(to be) a state of a step/act/action occurring/transpiring/being taken	
CSV	(to be) an act of taking a step or action to cause something to happen/occur	
OBJ	(to be) the step/act/action one takes to so something	

-ȚT- ‘FACT / KNOW’		
BSC	**STEM 1:** (to be) an act of knowing an ontological fact; to know an ontological fact [= have awareness/understanding of and ability to recall an ontological fact]	**STEM 2:** (to be) epistemological/conventionalized fact; know [= have awareness/understanding and ability to recall] an epistemological/conventionalized fact **STEM 3:** (to be) something known via experience/proficiency; know via familiarity or experience]
CTE	(to be) the ability/capacity to know an ontological fact	
CSV	(to be) the physical/mental process of (an act of) knowing an ontological fact	
OBJ	(to be) an ontological fact; (to be) the ontological fact one knows	

Derivations: find out, discover, determine, wisdom, insight, enlightenment, scholar

-RȚT- ‘STUDY / LEARN / KNOWLEDGE SOURCE’		
BSC	**STEM 1:** (to be) an act of studying/learning knowledge; to study [CPT version = learn] *	**STEM 2:** (to be) an act of practicing; to practice CPT = be proficient at; have proficiency in **STEM 3:** (to be) a source of knowledge, a resource from which facts can be learned, a knowledge base
CTE	(to be) the ability/capacity to study [CPT version = learn]	
CSV	(to be) a physical act/process of studying [CPT version = learning]	
OBJ	(to be) what one studies or learns about	

* This stem signifies learning from static knowledge sources (e.g., books/documents and/or personal research; for learning from a teacher, see -**DDY**- ‘TEACHING ↔ BEING A STUDENT’

-LŢT- ‘VOCATION / CAREER / AREA OF EXPERTISE OR PROFICIENCY’		
BSC	**STEM 1:** (to be) a state/process of pursuing a vocation; to pursue a vocation [i.e., a skill or subject matter one is dedicated to learning and practicing] CPT Version = (to be) a state/process of having/practicing a vocation; to have/practice a vocation	**STEM 2:** (to be) a state/process of pursing a career; pursue a career [i.e., one’s chosen set of skills/knowledge/expertise for making one’s living] **STEM 3:** (to be) a state/process of learning/studying an area of expertise; to pursue becoming proficient/expert at something CPT = (to be/have) proficiency/expertise in something
CTE	(to be) the ability/capacity to pursue/study [CPT version = practice/engage in] one’s vocation	
CSV	(to be) a state/process of learning/studying/practicing for one’s vocation; to study/learn/practice in furtherance of one’s vocation CPT = to perform or carry out an act as part of one’s vocation	
OBJ	(to be) (the particular set of skills or subject matter which is) one’s vocation	

-ŘŢT- ‘COMPETENCE / QUALIFICATION’		
BSC	**Stem 1:** (to be) a state/instance of being competent; to be competent (i.e., able to meet expected standards of performance in one’s work/duties)	**Stem 2:** (to be) a state/instance of being qualified for something, meeting qualifications for something; to be qualified at/for something, to meet the qualifications for something [i.e., where the qualifying criteria are skill-, knowledge-, or experience-based] **Stem 3:** (to be) a state of/instance of being qualified for something, meeting qualifications for something; to be qualified at/for something, to meet the qualifications for something [i.e., where the qualifying criteria are NOT skill-, knowledge-, or experience-based, but rather are based on an external parameter such as age, height, weight, nationality, one’s religion, one’s location, being in a certain profession, etc.]
CTE	(to be) the feeling/subjective experience of being competent; to experience the feeling of knowing one is competent	
CSV	(to be) a physical manifestation of competency; to do something competently or that demonstrates competence	
OBJ	(to be) what one is competent at	

-ŇŢT- ‘TEST / EVALUATION / VETTING / EXPERIMENT’		
BSC	**STEM 1:** (to be) a state/act of testing a party/entity (i.e., subjecting a party/entity to a process by which it/they must prove/demonstrate that it /they meet some standard or expected level of performance/knowledge/skill/talent, etc.)	**STEM 2:** (to be) a vetting/evaluation process (i.e., a process of investigating/evaluating a party/entity to determine that they meet some standard or expected level of performance/skill/background, etc.); to vet, to evaluate **STEM 3:** (to be) an experiment/test (i.e., a clinical procedure to determine the validity of a specific hypothesis, to determine the presence/absence of (a) particular substance(s), etc.; to conduct an experiment or clinical test
CTE	(to be) a state of being subject to testing	
CSV	(to be) the physical act/process of testing a party/entity; to test, to give a test/exam	
OBJ	(to be) the test/exam itself which the party/entity takes or is subjected to	

-MṬT- ‘DEMONSTRATION / PROOF’		
BSC	**STEM 1:** (to be) a state/act of demonstrating the factuality of something; to demonstrate that something is so CPT Version = verification; to verify	**STEM 2:** (to be) a state/act of proving, presenting evidence that attempts to prove something; to attempt to prove, to present evidence that attempts to prove something CPT Version = proof; to prove **STEM 3:** (to be) a state of establishing/meeting a standard of proof; to establish or meet a standard of proof
CTE	(to be) a state of being subject to demonstration, being demonstrated	
CSV	(to be) the physical act/process of demonstrating something; to do something that demonstrates something is so	
OBJ	(to be) that which is (to be) demonstrated	

-NṬT- ‘SUBJECT / TOPIC / MATTER / THEME / POINT / THESIS’		
BSC	**STEM 1:** (to be) the topic / subject / subject matter / matter at hand; to address/discuss/ deal with a particular topic / subject (i.e., what is at hand, in mind, under consideration, under advisement, at issue, in question, etc.)	**STEM 2:** (to be) the point/argument* one wishes/tries to make (e.g., in a discussion, an essay, an article, a speech, a briefing, etc.); to try to make one’s point [i.e., try to get one’s audience to understand or accept one’s opinion or conclusions]. CPT Version = succeed in making one’s point **STEM 3:** (to be) a thesis; to present a thesis (i.e., a conclusion about something based on a formal analysis and presentation of evidence)
CTE	(to be) a state of there being a topic/subject/subject matter/theme	
CSV	(to be) an act/process of addressing or dealing with a particular topic/subject/theme	
OBJ	(to be) one’s opinion/take/perspective/point of view/contribution on a topic/subject/theme	

-VL- ‘REALIZATION / UNDERSTANDING / COMPREHENSION’		
BSC	**STEM 1:** (to be) an act of realizing something; to realize [= understand via act of insight/intuition]	**STEM 2:** (to be) an act of understanding/comprehending something; to understand or comprehend something [= know via a process of personal and/or shared analysis/contemplation/introspection/discussion] **STEM 3:** (to be) an act of understanding/comprehending something: understand or comprehend something [= due to intelligibility/distinctness/clarity/coherence/legibility/lucidity of input/medium/message/material]
CTE	(to be) the ability/capacity to realize something	
CSV	(to be) the physical/mental process of (an act of) realizing something	
OBJ	(to be) what one realizes	

-ŢK- **'MAKE / CONSTRUCT / INTEGRATE / FORM'** Associated affix: **MAK** (NOTE: This root does not refer to 'make/create' meaning 'emit/secrete/produce'as in 'make noise' or 'make milk', etc.)		
BSC	**STEM 1:** (to be) an instance/act of making/constructing/creating something via integration of material resources according to an implicit or explicit plan/instinct/motive/design	**STEM 2:** (to be) an instance/act of forming or fashioning something via a combination of ingredients, e.g., cooking, mixing together, combining melodies from instruments to make music, tinkering with a variety of parts, etc. **STEM 3:** (to be) an instance/act of bringing something into being via organizing disparate parts into a coherent whole
CTE	(to be) the existential state of creating/making/constructing	
CSV	(to be) the method/steps employed/manifested in making/creating/constructing something	
OBJ	(to be) that which is made/created/constructed; the product of an act of making/creating/constructing	

-ŘŢK- **'COMMERCIAL MANUFACTURING / FABRICATION / BUILDING / PRODUCTION / ESTABLISHING'**		
BSC	**STEM 1:** (to be) an instance/act of commercial manufacturing; to manufacture	**STEM 2:** (to be) an instance/act of commercial fabrication/building; to fabricate or build something **STEM 3:** (to be) an instance/act of formally producing/establishing/ founding something (e.g., publishing a book or piece of music, establishing a church, founding a charity, etc.)
CTE	(to be) the existential state of manufacturing	
CSV	(to be) the method/steps employed/manifested in manufacturing something	
OBJ	(to be) that which is; the product of an act of manufacturing; a manufactured good/product	

-KSF- **'USE / UTILIZATION / CONSUMPTION / EXPENDITURE'**		
BSC	**STEM 1:** (to be) an instance/act of using/utilizing/making use of something/availing oneself of; to use/utilize, to make use of something, to avail oneself of something	**STEM 2:** (to be) an instance/act of serving as, being put into use as, being made to function as; to serve as, to (be) put into use as, to function as **STEM 3:** (to be) a state/act/instance of consumption/expenditure, being consumed/expended/used up; to consume, to spend a resource, to use (up) a resource
CTE	(to be) a state of use/utilization/being used/being made use of	
CSV	(to be) a physical act of using/utilizing/making use of; to physically perform an act of using/utilizing/making use of/availing oneself of	
OBJ	(to be) that which is used/utilize/made use of	

-KŠF- 'ENGAGEMENT / BE IN PLAY / RECOURSE / AGENCY'		
BSC	**STEM 1:** (to be) an instance/act of engagement, putting into action, bringing into play, being brought/put into play; to engage in something, to bring something into action, to put something into play, to have something be part of the action	**STEM 2:** (to be) an instance/act of resorting to something, having recourse to do something, to act on the belief one has to pursue a particular course of action; to resort to something, have recourse to do something, to act on the belief one has to pursue a particular course of action **STEM 3:** (to be) an instance/act of agency/exertion/force/control; to make happen, to cause to happen, to be the agential force behind something, to exert or exercise control over the course of something [i.e., an act of agency which causes/makes/initiates a tangible act/result/change of state affecting another party
CTE	(to be) a state of engagement/being in play/being in action	
CSV	(to be) a physical act of bringing/putting into play/action; to engage	
OBJ	(to be) the action/situation/context in(to) which something is made part of	

-MPR- 'TERMINATION / FINISH / END / FINALITY / COMPLETION'		
BSC	**Stem 1:** (to be) a state/act of ending/finishing/ being over/concluded; to be ending, be at an end, to be about over, to be about finished/concluded (i.e., no longer taking place, no longer extant/present/active, etc.) [CPT version = ended, finished, concluded, over]	**Stem 2:** (to be) a state/act of bringing to a final/ultimate state; to bring to a final/ultimate state, to finalize, to be final, to be reaching a final/ultimate (i.e., quasi-permanent) state [CPT version = be in a final/ultimate state] **Stem 3:** (to be) a state/act of completing; to be reaching a state of completion (i.e., of a goal-oriented process) [CPT version = a completed state; to complete, to reach/achieve completion]
CTE	(to be) the state of being at an end, being finished/concluded	
CSV	(to be) a physical act of finishing, bringing to an end, concluding; to finish, to end, to conclude, to be done (with), to be over	
OBJ	(to be) that which is ended/finished/over	

-NY- 'CHOICE / PREFERENCE'		
BSC	**STEM 1:** (to be) a choice (to be) made; to choose [= both the state of having a choice to make and the choices available to choose from]	**STEM 2:** (to be) an act/process of "weighing"/pondering of a choice; conduct a pro-vs.-con analysis **STEM 3:** (to be) an state/act of settling for something (i.e., reluctant choice to avoid the alternative)
CTE	(to be in) a state of having to choose; to face a choice; to have a choice to make	
CSV	(to be) the act/process of choosing from one's available options/alternatives	
OBJ	(to be) the choice made; to be the entity/alternative/option chosen or selected	

-RNY- ‘SELECTION / ALTERNATIVE / OPTION / VOTING’		
BSC	**STEM 1:** (to be) a selection (to be) made (from limited choices); to select/pick (from limited choices) [= both the state of having a selection to make and the choices available to select from] *	**STEM 2:** (to be) a state of having an alternative or available option; to have an alternative or available option **STEM 3:** (to be) an act of voting; to vote
CTE	(to be in) a state of having to choose; to face a choice; to have a choice to make	
CSV	(to be) the act/process of choosing from one’s available options/alternatives	
OBJ	(to be) the choice made; to be the entity/alternative/option chosen or selected	

* This stem is distinguished from Stem 1 of the the root -**NY**- above in that it implies a limited number of choices available

-SN- ‘CHANGE / VARIATION / VERSION / DEVIATION’		
BSC	**STEM 1:** (to be) a change (from one state to another); to change (from one state to another)	**STEM 2:** (to be) an act of variance or being a differing version of something; to vary, to be a differing version **STEM 3:** (to be) an act of deviation; to deviate
CTE	(to be) the state of transformation which occurs during a change	
CSV	(to be) the physical/tangible process of an act of change (i.e., what an entity physically undergoes while changing); to transform	
OBJ	(to be) that aspect of an entity which changes; what gets changed about an entity	

Derivations: mutate, convert, shift, tamper with, transform, transition

-RSN- ‘ALTERATION / MODIFICATION / DIVERSIFICATION / MODULATION’		
BSC	**STEM 1:** (to be) an act/process of alteration, modification; to alter, to modify (i.e., a small change that does not significantly alter the identity or basic nature/essence/form, etc. of an entity)	**STEM 2:** (to be) a state/act of diversification/permutation; to diversify, permutate **STEM 3:** (to be) a state/act of modulation; to modulate (i.e., control the variance in the state of an entity)
CTE	(to be) the state of transformation which occurs during an alteration/modification	
CSV	(to be) the physical/tangible process of an act of alteration/modification (i.e., what an entity physically undergoes while changing)	
OBJ	(to be) what changes during an alteration/modification, what gets altered/modified, the alteration/modification itself	

-PṬ- ‘SEEK / SEARCH’		
BSC	**STEM 1:** (to be) an act of searching/seeking/looking for something specific whose location is unknown; a search for something specific whose location is; to search/seek/look for something unknown (i.e., its location/existence is uncertain)	**STEM 2:** (to be a) search, seek, look for something lost [i.e., something known to exist which is missing] **STEM 3:** quest / seek to accomplish/achieve something
CTE	(to be in) the psychological state of searching/seeking; to be driven by the need/desire to search/seek	
CSV	(to be) the process or physical act itself of searching/seeking	
OBJ	(to be) what one is seeking or searching/looking for	

-PṬR- ‘EXPLORATION / PURSUIT / HUNTING’		
BSC	**STEM 1:** (to be) a state/act of exploration; to explore [i.e., to sensorily observe and/or examine something unknown or previously unencountered to learn/discover its nature/structure/attributes, etc.]	**STEM 2:** (to be) a state/act of pursuit; to pursue, to track down (i.e., the existence of which is certain but whose location is uncertain) **STEM 3:** (to be) a state/act of hunting; to hunt for (i.e., with the goal of capturing/killing/detaining, etc. that which is hunted)
CTE	(to be) the psychological state of exploration; to be driven by the need/desire to explore	
CSV	(to be) a physical act/process of exploring; to perform the physical act(s) of exploring	
OBJ	(to be) what one explores (i.e., the place/area/realm/surface/volume, etc. one explores through or in)	

-GŠ- ‘LOSE / MISPLACE / SQUANDER / DEPRIVATION / FORFEITURE’		
BSC	**STEM 1:** (to be) a state/act of something being lost/mislaid/missing (i.e., misplaced); to lose something (i.e., to misplace), to misplace	**STEM 2:** (to be) a state/act of losing something/someone; to lose (i.e., to have an entity taken away or for an entity to go away and not be accessible or potentially recoverable) **STEM 3:** a state/act of squandering or wasting something; to squander, to waste (i.e., to allow something to become lost via neglect or irresponsibility)
CTE	(to be) a state of being lost/misplaced/missing/mislaid	
CSV	(to be) an act of misplacing/losing/mislaying something; to do something which causes an entity to be(come) lost/misplaced/missing	
OBJ	(to be) that which is/has been misplaced/lost/mislaid	

-GŠT- ‘DEPRIVATION / BEREAVEMENT / FORFEITURE’		
BSC	**STEM 1:** (to be) a state of deprivation; to be deprived of, to (have to) do without something	**STEM 2:** (to be) a state/act of bereavement; to be bereft/bereaved **STEM 3:** (to be) a state/act of forfeiture; to forfeit
CTE	(to be) a state of there being something that is not available	
CSV	(to be) an act of depriving, to do something which causes one to experience deprivation	
OBJ	(to be) that which one is deprived of, that which one must do without	

-ZD- ‘COMPARISON / CONTRAST / DISTINGUISH / MATCHING’		
BSC	**STEM 1:** (to be) a[n act of] comparison/contrasting; to compare/contrast	**STEM 2:** (to be) an act of distinguishing; to distinguish (= process of careful selection from among various alternatives) **STEM 3:** (to be) an act/process of matching one entity to another; to match something to something else
CTE	(to be) the state of being compared/contrasted	
CSV	(to be) the physical act/process of comparison/contrasting (i.e., step(s) taken to conduct the comparison)	
OBJ	(to be) the object/entity compared or contrasted with something else, that which is (being) compared	

-ZDR- ‘MEASUREMENT / DISCERNMENT / COLLATION’		
BSC	**STEM 1:** (to be) an act/process of measuring; to measure [i.e., to quantify an amount or degree of some attribute of an entity]	**STEM 2:** (to be) an act/process of discerning; to discern [i.e., to carefully observe a pattern or patterns or sub-phenomenon within an entity or phenomenon] **STEM 3:** (to be) an act/process of collation; to collate [i.e., determine the relationships between entities]
CTE	(to be) the state of being measured	
CSV	(to be) the physical act/process of measuring	
OBJ	(to be) that which is measured	

-ŠT- ‘EXPECTATION / SUSPENSE / ANTICIPATION / VIGIL’		
BSC	**STEM 1:** (to be) a state of expectation; to have an expectation of/concerning; to expect something (i.e., to have a degree of certainty that a future occurrence/situation/event will manifest in a particular way)	**STEM 2:** (to be) a state of anticipation/suspense; to experience a sense/feeling of anticipation/suspense; for something to be impending, the wait/expectation of which generates suspense and anticipation **STEM 3:** (to be) a state of vigilance, a vigil, a watch; to be vigilant/watchful, to conduct a vigil
CTE	(to be) the mental/psychological state of a expectation; be in a state of expectation	
CSV	(to be) the physical manifestation of a process of expecting; to endure a period of expectation	
OBJ	(to be) that which one expects, what/whom one is expecting	

-RŠT- ‘PREDICTION / FORESEE / FORECAST’		
BSC	**STEM 1:** (to be) state/act of prediction; to predict	**STEM 2:** (to be) a state/act of foresight; to be something prospective, to be something foreseeable; to foresee **STEM 3:** (to be) a state/act of prognostication/forecasting; to forecast
CTE	(to be) a psychological state of a process of prediction; undergo a state of predicting	
CSV	(to be) the physical act of predicting; to make a prediction	
OBJ	(to be) what one predicts	

-G- ‘AMBULATION’		
BSC	**STEM 1:** (to be) an instance of bodily ambulation; to ambulate (= to use one’s body’s natural method of translative motion along a surface, e.g., walk, hop, crawl, swim, slither, etc.)	**STEM 2:** (to be) an instance of rapid ambulation; to run, gallop, scamper, scurry, etc. **STEM 3:** (to be) an instance of unnatural/affected ambulation, e.g., limp, stagger
CTE	(to be) the translative/directed movement through spacetime from one point to another as accomplished via ambulation	
CSV	(to be) a single iteration/cycle of the particular pattern of bodily movement(s) used in ambulation; to make/undergo a single iteration of the body’s natural ambulatory cycle	
OBJ	(to be) the surface upon which one ambulates (e.g., the ground, the floor, the firmament, the ramp, the stairs, etc.)	

-GTH- ‘MARCHING / PRANCING / RACING / SWAGGERING / SAUNTERING’		
BSC	**STEM 1:** (to be) an instance of marching or prancing; to march or prance (= to ambulate in a deliberately formal and structured way)	**STEM 2:** (to be) an instance of racing [i.e.,competitive speed running] **STEM 3:** (to be) an instance of swaggering or sauntering [i.e., affected ambulation in a deliberately exaggerated manner
CTE	(to be) the translative/directed movement through spacetime from one point to another as accomplished via marching or prancing	
CSV	(to be) a single iteration/cycle of the particular pattern of bodily movement(s) used in marching or prancing; to make/undergo a single iteration of the body’s ambulatory cycle while marching or prancing	
OBJ	(to be) the surface upon which one marches or prances (e.g., the ground, the floor, the firmament, the ramp, the stairs, etc.)	

-MZ- ‘REMEMBER / RECALL / MEMORY / MEMORIZATION’		
BSC	**STEM 1:** (to be) a memory; to remember something (= recall something) [whether unwilled/affective or self-willed/voluntary]	**STEM 2:** (to be) the retention of something past into present; to retain something from the past into the present **STEM 3:** (to be) an act of memorization; to memorize, to remember [i.e., commit to memory]
CTE	(to be) the content of a memory or recollection, what is remembered or recalled	
CSV	(to be) the act/process of recalling something; to recall	
OBJ	(to be) the entity/circumstance/event/state which a memory is about	

MORPHOLOGICAL DERIVATIONS: recollect(ion), remind, memorize, memento, monument, archive, commemorate, commemoration, souvenir/keepsake, relic, to bear in mind, ponder/relish a memory, sear into one’s memory, visual record / recorded image, camera, movie

-MZM- ‘TRACE / VESTIGE / MEMORIAL / ARCHIVAL RECORD’		
BSC	**STEM 1:** (to be) a state of there being a trace/vestige of something occurring [i.e., a physical indication left behind of a previous state or that something has previously occurred]	**STEM 1:** (to be) a memorial (i.e., an (object/action formally recognizing or acknowledging something/someone past) **STEM 3:** (to be) a state/act of having/making a record of something; to have/make a record of something, to archive
CTE	(to be) the trace/vestige itself	
CSV	(to be) the act of leaving a trace/vestige behind	
OBJ	(to be) that which the trace/vestige indicates has occurred	

-ŇL- ‘SOLVE / FIGURE OUT’		
BSC	**STEM 1:** (to be) an act/process of trying to solve a mystery or puzzle, trying to figure out a riddlesome problem; to attempt to solve a mystery/puzzle/dilemma/ riddlesome problem [CPT Version = to solve it, to figure it out]	**STEM 2:** (to be) an act of analytical reasoning/logic as applied to attempting to solve a mystery/puzzle/riddlesome problem/dilemma **STEM 3:** (to be) an act/state of insight/inspiration as part of one’s attempt to solve a puzzle/mystery/riddlesome problem, dilemma
CTE	(to be) a state of mystery, being faced with an unsolved puzzle/mystery/problem/dilemma	
CSV	(to be) the act/process of trying to figure out an explanation/solution; to try to figure something out	
OBJ	(to be) the puzzle/mystery/unsolved issue/dilemma/riddlesome problem itself	

-LLT- ‘WONDERING’		
BSC	**STEM 1:** (to be) a state/act of wondering about something that is potentially knowable; to wonder about something [i.e., desire to know something that can (presumably/theoretically) be known, e.g., *“I wonder what papaya tastes like”, “I wonder if she is religious.”*]	**STEM 2:** (to be) a state act of wondering about something no longer knowable; to wonder about something [i.e., desire to know something that (presumably/theoretically) can no longer be known, e.g., *“I wonder what cooked dinosaur tasted like”, “I wonder what Newton would think of quantum theory.”]* **STEM 3:** (to be) a state/act of wondering about something not yet knowable [i.e., desire to know something unknowable until some future time, e.g., “*I wonder if I’ll ever be able to walk again”, “I wonder if humankind will ever colonize another planet.”*]
CTE	(to be in) a state of wondering, i.e., the desire to know something that can (presumably/theoretically) be known	
CSV	(to be) the act/process of asking oneself (either verbally or in one’s thoughts) something which one desires to know that can (presumably/theoretically) be known; to ask oneself such a question	
OBJ	(to be) what one wonders about	

-CC- ‘MISTAKE / ERROR / MISCONCEPTION / MISCALCULATION’		
BSC	**Stem 1:** (to be) a state/act of being mistaken or in error regarding something; to be wrong about something, to be mistaken or in error about something	**Stem 2:** (to be) a state/act of misconception; to be under a misconception, to act/do something based upon a misconception **Stem 3:** (to be) a state/act of miscalculation or misjudgement; to misjudge or miscalculate something
CTE	(to be) a state of erroneousness, a state of being wrong about something	
CSV	(to be) an act of being mistaken or in error; to make a mistake, to be in error	
OBJ	(to be) that which one is mistaken about; that which one makes/has made an error concerning	

-CT- ‘LOOK OVER / EXAMINE / INSPECT’		
BSC	**Stem 1:** (to be) an act of looking something over; to give something a lookover (i.e., perform a casual or brief visual/tactile observation of an entity in order to form an initial reaction/appraisal)	**Stem 2:** (to be) an act of examination; to examine (i.e., to observe/inspect something in order to determine its physical condition and/or character(istics).) **Stem 3:** (to be) an act of inspection/checking; to inspect, to check (i.e., examine something to determine the degree to which it complies with, adheres to, or deviates from some expectation or standard)
CTE	(to be) the state of being looked over	
CSV	(to be) the physical process of looking over (i.e., the physical act performed which constitutes looking something over)	
OBJ	(to be) that which is looked over	

-RCT- 'OBSERVE / SCRUTINIZE / RECONNAISSANCE'		
BSC	**STEM 1:** (to be) an act of observation; to observe (to watch something closely for the purpose of determining what it does/how it behaves/what happens to it, etc.	**STEM 2:** (to be) an act of scrutiny; to scrutinize **STEM 3:** (to be) an act of reconnaissance; to reconnoiter
CTE	(to be) a state of being under observation	
CSV	(to be) the physical act/process of observing; to make/conduct an act/process of observation	
OBJ	(to be) the object/entity being observed	

-XP- 'SUBSTITUTION / TRANSPOSITION / REPLACEMENT'		
BSC	**STEM 1:** (to be) an act of substitution (= having one entity take the place of another)	**STEM 2:** (to be) an act of transposition/ interchange (= switching places) **STEM 3:** (to be) an act of supplanting/replacement/change-out (= refresh/renew)
CTE	(to be) the entity which is replaced/discarded/removed by an act of substitution	
CSV	(to be) the physical (trans-)action of substitution; to physically act to substitute one entity for another	
OBJ	(to be) the entity which is substituted for or takes the place/role of another	

-ČTL- 'TRANSPORT STOP / STATION / HARBOR'		
BSC	**STEM 1:** (to be) a stop on a transport route; to make a stop at (i.e., where one embarks/disembarks from a means of local transport)	**STEM 2:** (to be) transport station (i.e., centralized one goes to purchase tickets and embark/disembark from a form of local transport **STEM 3:** (to be) harbor, place of anchorage
CTE	(to be) state of embarkation/disembarkation on local transport line	
CSV	(to be) a physical act of embarking/boarding or disembarking/getting out or off of a form of local transport	
OBJ	(to be) a particular stop on a transport route/line	

The Specification pattern of the above root applies to the root below:

-RČTL- 'AIRPORT / SEAPORT / TRANSPORTATION TERMINAL OR HUB'
Stem1. (to be) a location served by a commercial sea line or airline
Stem 2. (to be) a transport terminal
Stem 3. (to be) a port (e.g., airport, seaport, heliport, etc.) or transportation hub

-ÇL- 'PREPARATION / PRELIMINARY MEASURE / READINESS'		
BSC	**STEM 1:** (to be) a state/act of preparation (for something); to prepare (for something) [i.e., to take steps or measures that make one (more) ready to deal with something] (CPT version = 'state of readiness; be ready')	**STEM 2:** (to be) a state/act of establishing preliminary measures necessary for something else to occur; to perform a preliminary step/task/measure, "lay the groundwork for", "lay a foundation for" **STEM 3:** (to be) a state/act of preparing/readying something, putting something together, assembling, making something ready for use or consumption; to prepare/ready something for use (e.g., food, a tent, wood in a fireplace, a painter's palette, a surgeon's operating room, etc.)
CTE	(to be) state of desire to prepare [CPT version = preparedness/readiness]	
CSV	(to be) a physical act of preparation; to do something that prepares an entity for something	
OBJ	(to be) what one is preparing for	

-RÇL- 'PRECAUTIONARY MEASURE / PRECAUTIONARY TRAINING / CONTINGENCY / BACK-UP'
Stem 1: (to be) a state/act of having precaution; take precautionary measure
Stem 2: (to be) a state/act of instruction/training in order to be prepared/ready for something; to instruct/train (a party to be able to perform some function or duty or skill in order to be prepared/ready for something)
Stem 3: (to be) a state/act of having/creating a contingency or back-up measure; to have/create/use a contingency or back-up measure

-ŇÇL- 'PROVISION / SUPPLYING / FURNISHING / EQUIPPING'		
BSC	**STEM 1:** (to be) a state/act of providing; to provide, offer provision [i.e., give something which one can make use of or consume] CPT version = provide	**STEM 2:** (to be) a state/act of supplying/replenishing/furnishing; to supply, to furnish with, to replenish **STEM 3:** (to be) a state/act of equipping, providing resources for the establishment or carrying out of something; to equip, provide resources/equipment for
CTE	(to be) state of lacking something that needs to be provided; to lack, to need	
CSV	(to be) a physical act of offering to provide; to make an offer to provide	
OBJ	(to be) what is being provided, what is being offered as a provision	

-CPH- 'PRESERVATION / CONSERVATION'		
BSC	**Stem 1:** (to be) a state/act of saving/preserving something (from decaying or being damaged through neglect/misuse/mistreatment)	**Stem 2:** (to be) a state/act of saving/preserving something (from being lost or discarded); to save/preserve something (from being lost or discarded) **Stem 3:** (to be) a state/act of preserving/conserving something via a chemical or biological transformation (e.g., freezing, dehydration, pickling, embalming, curing, salting, etc.)
CTE	(to be) a state of being/having been saved/preserved (from decay or damage)	
CSV	(to be) a physical act of preserving/saving something (from decaying or being damaged)	
OBJ	(to be) what is being saved/preserved	

-CPR- 'RESCUE / EXTRICATION / DELIVERANCE / SALVATION / LIBERATION / EMANCIPATION'		
BSC	**Stem 1:** (to be) a state/act of saving/rescuing a party/entity; to save/rescue a party/entity, to come to the rescue of a party/entity, to extricate a party/entity from an undesirable situation	**Stem 2:** (to be) a state/act of deliverance/salvation/redemption (i.e., finding a way to end the conditions which keep a party in a state an undesirable social, psychological, spiritual, or personally harmful state); to save/deliver/redeem **Stem 3:** (to be) a state/act of liberation/emancipation (i.e. putting an end to socio-political/economic conditions which have kept a party/entity in a state of being oppressed); to liberate, to emancipate
CTE	(to be) a state of being/having been saved/rescued (from a harmful or undesirable situation)	
CSV	(to be) a physical act of saving/rescuing; to physically save/rescue a party/entity; to save/rescue someone/something from danger/harm	
OBJ	(to be) the party/entity being saved/rescued	

-CX- 'MANNER / WAY / GUISE / FORM / BEHAVIOR / COMPORT'		
BSC	**STEM 1:** (to be) the manner/way in which something/someone acts/operates/manifests itself	**STEM 2:** (to be) the guise/form taken by a party/entity **STEM 3:** (to be) one's behavior/conduct/comport
CTE	(to be) the inner/intrinsic/inherent motivation/drive/instructions, etc. which control one's manner/way	
CSV	(to be) the outward physical manifestation or discernible praxes of one's manner/way	
OBJ	(to be) that which has or manifests a particular manner/way	

-RCX- 'METHOD / PROCEDURE / POLICY / PLAN / STRATEGY'		
BSC	**STEM 1:** (to be) the method (i.e., proscribed/conventionalized procedure) by which a party/entity performs/accomplishes something	**STEM 2:** (to be) the practice/policy used by a party/entity in operating/manifesting itself **STEM 3:** (to be) a plan (i.e., a series of inter-dependent procedural steps toward accomplishing something)
CTE	(to be) the state of there being a methodical process	
CSV	(to be) a physical step/act in a method(ical procedure); to implement/engage in a step/act that is part of a method(ical procedure)	
OBJ	(to be) that which a method(ical procedure) is intended to accomplish	

Morphological Derivations: technique, process, procedure, regimen, strategy

-ŘK- ‘ALIENABLE POSSESSION / OWNERSHIP’		
BSC	**STEM 1:** (to be) an act of physical possession; to have/possess (temporary/circumstantial and alienable – e.g., a book in one’s hand)	**STEM 2:** (to be) an act of possession; have/possess/own; possession, piece of property (alienable but quasi-permanent by law/right/permission and in one’s immediate possession-, e.g., a library book which I’ve checked out) **STEM 3:** (to be) an act of possession; have/possess/own; possession, piece of property (alienable but quasi-permanent by law/right, not in one’s immediate possession -e.g., a book I bought/own but am not currently holding/reading)
CTE	(to be) the state of alienable possession; possessing something potentially alienable	
CSV	(to be) the physical means by which a state of alienable possession is manifested, e.g., by holding or carrying, by a verbal or written statement, via containment within one’s pocket, etc.)	
OBJ	(to be) the object/entity possessed (temporary/circumstantial and alienable)	

-ŘKW- ‘INALIENABLE POSSESSION / OWNERSHIP’		
BSC	**STEM 1:** (to be) an act of physical possession; to have/possess (circumstantial and affective + (quasi-) inalienable, either permanent or temporary – e.g., a medical condition, an office at work I occupy, a seat at a cinema showing)	**STEM 2:** (to be) an act of possession/ownership; have/possess/own; possession, piece of property (quasi-permanent and legally inalienable except through formal/legal separation process – e.g., a house I own, a pet dog I registered **STEM 3:** (to be) an act of physical possession; to have/possess (genetic, intrinsic and inalienable, but not considered property – e.g., a sibling, a child, a limb, a memory)
CTE	(to be) the state of inalienable possession; possessing something that is potentially inalienable	
CSV	(to be) the physical means by which a state of inalienable possession is manifested, e.g., by biological/genetic association, by physical connection, by law, etc.)	
OBJ	(to be) the object/entity inalienably possessed	

-ŽV- ‘GOOD / BENEFICIAL / ADVANTAGEOUS’		
BSC	**STEM 1:** (to be) something good (= materially beneficial to the context at hand) [both the act/event/situation and its beneficent quality]	**STEM 2:** (to be) something good [= morally right; beneficial in a metaphysical sense] **STEM 3:** (to be) something good [= advantageous or effective]
CTE	(to be) the essential state/quality of material goodness/beneficence manifested in a particular act/state/event/situation/entity, etc.	
CSV	(to be) a particular act/state/event/situation/entity, etc. identified as being good; to do something materially/tangibly good/beneficial	
OBJ	(to be) the particular element(s) (e.g., word(s), action(s), emanation(s), appearance, product, presence of something, lack of something, etc.) which makes something describable/identifiable as being materially/tangibly “good/beneficial”	

-LŽV- ‘FAIRNESS / EQUITABILITY / IMPARTIALITY / REASONABLENESS’		
BSC	**STEM 1:** (to be) something fair/equitable/impartial/reasonable	**STEM 2:** (to be) something just [= legally right; morally proper in a metaphysical sense] **STEM 3:** (to be) something unbiased, without prejudice
CTE	(to be) the essential state/quality of fairness/equitability/impartiality/reasonableness manifested in a particular act/state/event/situation/entity, etc.	
CSV	(to be) a particular act/state/event/situation/entity, etc. identified as being fair/equitable/impartial/ reasonable; to do something fairly/equitably/impartiality/reasonably	
OBJ	(to be) the particular element(s) (e.g., word(s), action(s), emanation(s), appearance, product, presence of something, lack of something, etc.) which makes something describable/identifiable as being fair/equitable	

-TM- ‘POWER / EFFORT / OPERATION / FUNCTIONING’		
BSC	**STEM 1:** (to be) an instance/act of energy expenditure in a particular form; to manifest a display/use of energy in a particular form/manner; exert energy in a particular form/manner	**STEM 2:** (to be) an instance/manifestation of action or effort; to act, to perform, to do, make an effort (i.e., energy expenditure focused on a specific task) **STEM 3:** (to be) an act/state of an entity operating or functioning or “working” (i.e., the proper or expected functioning of a device or machine or process as in *“Does the washing machine work?”)*; to operate, to function
CTE	(to be) the process of energy being expended (i.e., conversion of source fuel into power)	
CSV	(to be) a physical manifestation of energy expenditure (i.e., what action(s) an entity actually does that requires energy to do)	
OBJ	(to be) that to/for which energy is being used/exerted	

-RTM- ‘WORK / LABOR / JOB’		
BSC	**STEM 1:** (to be) an instance of work/labor; to work, to labor (i.e., organized/planned effort over time associated with a commercial or community endeavor)	**STEM 2:** (to be) an act of producing a tangible product via labor/effort **STEM 3:** (to be) a job task; employment-related task; do one’s job
CTE	(to be) the process of energy/effort expended via labor	
CSV	(to be) a physical manifestation of labor/working; to physically do work/labor; to labor	
OBJ	(to be) that which one works on or attempts to accomplish/build/make/change via one’s labor	

-ṬP- ‘FACULTY / ATTRIBUTE / QUALITY / TRAIT / PROPERTY / CHARACTERISTIC / FEATURE / ATTRIBUTE’		
BSC	**Stem 1:** (to be) a state/act of having a particular faculty/capacity/ability; to have a particular faculty/capacity for something or ability to do something	**Stem 2:** (to be) a state/act of having a particular quality, trait, or property (i.e., an inherent and necessary characteristic that is intrinsic to one’s identity), e.g., mass, spherical shape, being alive, being toxic to certain life-forms, being gelatinous, being a vertebrate, etc.) **Stem 3:** (to be) a state/act of having a particular superficial feature, attribute, or characteristic (out of multiple possibilities, e.g., blue-eyed, having freckles, being fat, having stripes, etc.)
CTE	(to be) a faculty as an existential state	
CSV	(to be) a physical act that demonstrates/manifests a particular faculty	
OBJ	(to be) the particular faculty/ability itself	

-ÇD- ‘IMPRESSION / APPEARANCE / “LOOK” / SEMBLANCE / ASPECT / GUISE’		
BSC	**STEM 1:** (to be) an impression; to give off an impression (i.e., to incite a reaction or belief in another based upon one’s appearance/behavior/words/actions, etc., whether intentionally or not)	**STEM 2:** (to be) the appearance or “look” or semblance or aspect of an entity [i.e., the outward subjective impression upon on observer given off by one’s visual dress, physique, manner]; to appear, to look [= have the appearance/look/semblance/aspect of] **STEM 3:** (to be) the guise of an entity; to be in the guise of, have the guise of
CTE	(to be) an impression experienced/received by observing/noticing something; to receive/experience and impression given off by some entity/party	
CSV	(to be) an act/state of giving off/creating an impression to another	
OBJ	(to be) the entity/person/party who gives off an impression or whom one has an impression about	

-GT- OVERSIGHT / RESPONSIBILITY’		
BSC	**Stem 1:** (to be) a state/act/process of oversight of something/someone; to oversee something/someone (i.e., be aware and mindful of how something or someone functions/behaves/transpires)	**Stem 2:** (to be) a state/act of being responsible for something/someone; to look after, to be responsible for (i.e., oversight which carries negative consequences if not performed adequately) **Stem 3:** (to be) an obligation or duty
CTE	(to be) a state of there being oversight	
CSV	(to be) a rule/procedure/custom/praxis associated with the process of overseeing; to carry out a rule/procedure/custom/praxis associated with oversight	
OBJ	(to be) a party who is subject to oversight	

-GTR- ‘SUPERVISION / REGULATION’		
BSC	**STEM 1:** (to be) a state/act/process of being in charge of something; to be in charge, to take charge of something/someone	**STEM 2:** (to be) a state/act/process of supervision; to supervise, have supervisory responsibility for something/someone **STEM 3:** (to be) a state/act/process of regulation; to regulate something (i.e., exercise the authority to make rules for the oversight/functioning of something)
CTE	(to be) a state of there being someone in charge	
CSV	(to be) a rule/procedure/custom/praxis associated with being in charge; to do/perform an act which demonstrates one being in charge	
OBJ	(to be) what one is in charge of	

-TḶP- ‘PERSONAL RESPONSIBILITY / BLAME / LIABILITY’		
BSC	**STEM 1:** (to be) a state/act/process of having personal responsibility for or regarding something or someone	**STEM 2:** (to be) a state/act/process of blame; to be [held] to blame for something, to (be) blame(d) for something **STEM 3:** (to be) a state/act/process of liability; to be [held] liable for something
CTE	(to be) a state of being [held] personally responsible	
CSV	(to be) an act/behavior/practice associated with being personally responsible; to do/perform an act which demonstrates one is (being) personally responsible	
OBJ	(to be) what one is [held] personally responsible for	

-ṬPH- ‘OPENNESS / TRANSPARENCY / STRAIGHTFORWARDNESS / ACCOUNTABIILITY’		
BSC	**STEM 1:** (to be) a state/act/process of being open/transparent to/with others	**STEM 2:** (to be) a state/act of straightforwardness, being straight and to-the-point with others; to be/speak straightforward(ly); to be straight with others, to be direct and to-the-point **STEM 3:** (to be) a state/act/process of accountability; to be (potentially) accountable for something [i.e., responsibility for addressing/fixing/dealing with/explaining/making good on something if something goes wrong]
CTE	(to be) a state of openness/transparency	
CSV	(to be) an act/behavior/practice associated with being open/transparent; to do/perform an act which demonstrates one is (being) open/transparent	
OBJ	(to be) what one is open/transparent about	

-TKH- ‘RULE / PRECEPT / MAXIM / AXIOM / EDICT ’		
BSC	**STEM 1:** (to be) an act/instance/process of using/applying/establishing/enforcing/ `being subject to a rule or precept	**STEM 2:** (to be) a state/act/process of applying/establishing/being subject to a maxim or axiom (i.e., a foundational/underlying presupposition/assumption as to a state of affairs that constitutes an unquestioned rule or fundamental principle for further decisions) **STEM 3:** (to be) a state/act/instance of dictating/establishing/ announcing an edict or ruling [i.e., a declaration by a recognized authority that establishes/enforces a legally-binding rule/law/decision/state of affairs]
CTE	(to be) a state of there being a rule/precept in place that governs a situation or one’s behavior	
CSV	(to be) an act/instance of applying/enforcing/establishing/dictating a rule or precept	
OBJ	(to be) a particular rule/precept	

-MČ- ‘POWER / EFFICACY / POTENCY’		
BSC	**Stem 1:** (to be) a state/act of power (i.e., puissant force derived from energy utilization/expenditure); to (em)power, to energize, galvanize	**Stem 2:** (to be) a state/act of efficacy; be efficacious, be effectual (i.e., ability/capacity to get things done effectively) **Stem 3:** (to be) state/act of potency; be/make potent/powerful
CTE	(to be) a state of having physical power/puissance/might/strength; be powerful/puissant/mighty (in terms of strength/degree of forceful utilization of energy)	
CSV	(to be) an act of physical power; to act in a manner demonstrating/manifesting physical power	
OBJ	(to be) that which is (em)powered	

-RMČ- ‘FORCE / EFFECTIVENESS / PRODUCTIVITY’		
BSC	**STEM 1:** (to be) a state/act of force; use force, act forcefully/with force (i.e., delineated/directed application/utilization of energy)	**STEM 2:** (to be) a state of effectiveness; be/make effective **STEM 3:** (to be) a state/act of productivity; be/make productive
CTE	(to be) a state of being under force, subjected to force	
CSV	(to be) an act of using/applying force; to use/apply force	
OBJ	(to be) the force one uses/applies	

-KSM- GEOPOLITICAL COUNTRY/LAND/ TERRITORY/REGION		
BSC	**Stem 1:** (to be) a geopolitical country/land/territory, i.e., the tangible area of land, bodies of water, and airspace subject to control/rule by a governmental entity, often associated or (quasi-)contiguous with a specific ethno-cultural society of people.	**Stem 2:** (to be) a state or province or subdivision of Stem 1, subject to some degree of regional autonomous governmental administration **Stem 3:** (to be) a geopolitical or cultural region delineated by a distinct culture, ethnicity, language, history, etc. which does not necessarily align with or have its own separate sovereignty or quasi-autonomous governmental administration.
CTE	(to be) the state of being a geopolitical country/land/territory	
CSV	(to be) the particular location and/or geographic/terrestrial features/terrain associated with a geopolitical country/land/territory	
OBJ	(to be) a particular (i.e., named) geopolitical country/land/territory	

-RC- COINCIDENCE / SIMULTANEITY (**NOTE:** The stems of this root, given that they imply a second, associated, or corollary event/act/state, are generally used with an accompanying clause marked by the COMPARATIVE case-frame)		
BSC	**Stem 1:** (to be) a state/instance of coinciding, a coincidence; to coincide [not necessarily simultaneous but auspiciously timed in relation to another event]	**Stem 2:** (to be) a state/instance of simultaneity; to occur simultaneously without knowledge/awareness of the other state/event occurring at the same time **Stem 3:** (to be) a state/instance of synchronicity (i.e., a coincidence so unlikely but nevertheless having significant personal impact on a situation, that one can only marvel and/or be suspicious of its occurrence)
CTE	(to be) a state of coincidence	
CSV	(to be) an act which constitutes a coincidence; to do something that creates/constitutes a coincidence (with something else)	
OBJ	(to be) the other "half" of a coincidence, the other coinciding event/entity or circumstance	

-ZK- 'STRANGENESS / ODDITY / WEIRDNESS'		
BSC	**Stem 1:** (to be) something strange/odd/weird [i.e., having the effect of being unfamiliar, unknown, not previously expected or encountered, and which defies immediate explanation and inspires curiosity]	**Stem 2:** (to be) something strange/odd/weird/disconcerting [i.e., unfamiliar and inexplicable with the effect of inspiring caution, trepidation, apprehension due to being seen as a potential threat] **Stem 3:** (to be) something scary/creepy/frightening [i.e., unfamiliar and inexplicable with the effect of inspiring fear at a perceived threat]
CTE	(to be) strange/odd/weird as a characteristic	
CSV	(to be) a particular behavior or physical aspect which characterizes something or someone as strange/odd/weird	
OBJ	(to be) the sense/feeling that something is strange/odd/weird	

	-SX- ‘COMPLAINT / GRIEVANCE	
BSC	Stem 1: (be) a act/instance of complaining, (be) a complaint; to complain, to make a complaint	Stem 2: (be) a grievance (requiring compensation/accommodation/recompense), i.e., a formal or legalistic complaint Stem 3: (be) a criminal complaint (i.e., a charge/allegation of a criminal offense perpetrated against a party, requiring justice)
CTE	(be) the content of a complaint, i.e., what one is complaining about	
CSV	(be) the physical act (verbal, written, processual, etc.) of making a complaint (irrespective of the content of the complaint	
OBJ	(be) the party to whom one complains; the party from whom relief/resolution of the complaint is requested	

	-ÇPL- ‘SECRETION / RESIN / SAP	
BSC	Stem 1: (be) a act/instance of a bodily secretion	Stem 2: (be) an act/instance of sap secretion from a tree or sap-producing plant Stem 3: (be) an act/instance of secretion of a resin
CTE	(be (in)) the state of having secreted	
CSV	(be) the physical process of secreting	
OBJ	(be) an amount of a secreted substance	

	-PP- ‘PICK UP / TAKE (IN) / AMASSING / ACQUISITION / PROCUREMENT’	
BSC	**Stem 1:** (to be) an act/instance of taking or picking something up for the purpose of (temporarily/circumstantially) keeping/guarding/using it or conveying it elsewhere; to pick something up, take something (in), to obtain something	**Stem 2:** (to be) an act/process of amassing/piling up, adding to, growing in number/amount; to amass, to pile up, to add to, to grow in number/amount **Stem 3:** (to be) an act/instance of procurement or acquiring; to procure/acquire something
CTE	(to be) a state of (temporary/circumstantial) possession as a result of being picked up or taken in	
CSV	(to be) a physical act of picking up or taking in; to pick up, to take in	
OBJ	(to be) that which is picked up or taken (in) or obtained	

-RPP- ‘GATHERING / COLLECTION / ACCUMULATION’		
BSC	**STEM 1:** (to be) an instance of gathering something (up/in) [i.e., bringing separate entities together to form a grouping thereof]	**STEM 2:** (to be) an act/process of collection; to collect **STEM 3:** (to be) an act/process of accumulation/accretion
CTE	(to be) an act/instance of taking (temporary) possession, oversight or guardianship of something/someone	
CSV	(to be) a physical act of gathering (up/in); to gather (up/in)	
OBJ	(to be) that which is gathered (up/in)	

-LPP- ‘ACCRUAL / SAVING UP / HOARDING’		
BSC	**Stem 1:** (to be) a state/act/process of accrual of something; to accrue something [i.e., a periodic or constant obtaining and holding onto (a supply of) something so that its amount/number continues to increase]	**Stem 2:** (to be) a state/act/process of saving money or other financial/monetary resources; to accrue (monetary) savings, to save up **Stem 3:** (to be) a state/act/process of hoarding; to hoard [i.e., obsessive and excessive accrual without a rational basis]
CTE	(to be) a state of there being something that has (been) accrued	
CSV	(to be) a physical act of accruing; to take action to accrue something	
OBJ	(to be) what one accrues	

-LÇM- ‘SIGNATURE / PERSONAL AUTHORIZATION / PERSONAL AUTHENTICATION’		
BSC	**Stem 1:** (to be) a state/act of personally authorization or authentication via signature; to personally authorize something by signing one’s written signature	**Stem 2:** (to be) a state/act personal authorization by means other than signature, e.g., sworn oath, voice command, thumbprint, retinal scan, etc. **Stem 3:** (to be) a state/act of personal authentication (i.e., a process to ensure one’s identity), e.g., signature, fingerprints, voiceprint, DNA test, retinal scan, visual identification by others, etc.
CTE	(to be) a state of personal authorization or authentication via signature	
CSV	(to be) a physical act of signing one’s written signature; to sign one’s written signature	
OBJ	(to be) what one signs (e.g., a contract, a personal check, a document, etc.)	

-ŇF- 'PROBLEM / MYSTERY'		
BSC	Stem 1: (to be) a problem/issue causing inconvenience/impediment; have an issue/problem with	Stem 2: (to be) a mystery or puzzling situation with no readily apparent explanation/solution Stem 3: (to be) something irksome, i.e., an entity or situation that causes annoyance and gives rise to the need/desire to resolve/eliminate the annoyance
CTE	(to be) the issue/problem requiring resolution caused/created by an entity/party or situation	
CSV	(to be) the entity/party or situation itself which has caused/created an issue/problem	
OBJ	(to be) the entity/party affected by or experiencing /perceiving the issue/problem caused/created by another entity/party/situation	

The following root has the same Specification pattern as the root **-ŇF-** above:

-RŇF- 'TROUBLE / MISFORTUNE / DISASTER / CALAMITY'
Stem 1: (to be) a state of being in trouble (i.e., a situation which places/has placed a party in adverse circumstances, e.g., physical danger, legal difficulties, under threat of losing one's livelihood/possessions/reputation/loved ones, etc.); to be in trouble, to be threatened by adverse circumstances, to be facing an adverse situation
Stem 2: (to be) a state of suffering misfortune; to suffer from misfortune, to be in unfortunate circumstances
Stem 3: (to be) a state of suffering disastrous/calamitous circumstances; to be facing a disaster/calamity, to suffer a disaster/calamity

-ŠY- 'RAISE / FOSTER / NURTURE / CULTIVATE'		
BSC	**Stem 1:** (to be) an act of raising/fostering/being guardian to someone; to raise, foster, function as guardian over	**Stem 2:** (to be) an act of nurturing; to nurture (i.e., provide an environment and resources for an entity's growth and well-being) **Stem 3:** (to be) an act of cultivation; to cultivate, to garden
CTE	(to be) the state of being fostered/raised by someone	
CSV	(to be) a physical act of raising/fostering	
OBJ	(to be) the person/entity raised/fostered	

-ÇTR- 'PUZZLE / RIDDLE'		
BSC	**Stem 1:** (to be) a puzzle (i.e., a recreational game/activity requiring one to resolve/figure out/piece together a solution)	**Stem 2:** (to be) a riddle (i.e., a word-based recreational puzzle usually designed to be witty on its initial face) **Stem 3:** (to be) charade (i.e., a recreational puzzle acted out in real-time)
CTE	(to be) the specific content/nature of a puzzle	
CSV	(to be) the form/type/structure of a puzzle	
OBJ	(to be) the party who undertakes to solve a puzzle	

Derivation: 'maze' = Stem 1 above plus the Type-2 concatenated stem for 'pathway/trail'

-ZJ- 'CONCEALMENT / SECRECY'		
BSC	**Stem 1:** (to be) a state/act of concealing/hiding something, act of concealment; to conceal, to hide something [i.e., by being placed somewhere so as not to be found/discovered/encountered/accessible]	**Stem 2:** (to be) a state/act of concealing/hiding/masking something; act of concealment; to conceal/hide/mask [i.e., by masking, covering up, disguising, enclosing so as not to be directly observed, etc.]
CTE	(to be) a state of concealment/being hidden	
CSV	(to be) a physical act of concealing/hiding; to do something to conceal/hide an entity	**Stem 3:** (to be) a state/act of secrecy; to keep something secret [OBJ Specification = 'a secret']
OBJ	(to be) what is (to be) concealed	

-FH- 'FIND / DISCOVER / ENCOUNTER / MEET'		
BSC	**Stem 1:** (to be) a state/act of finding/discovering something (by chance)	**Stem 2:** (to be) a state/act of encountering, coming upon/across something; to encounter or come across/upon something/someone (by chance)
CTE	(to be) a state of discovery (i.e., the state one is in upon having discovered something.	
CSV	(to be) a physical act of finding/discovering something by chance	**Stem 3:** (to be) a state/act of meeting someone (i.e., mutual encounter by chance)
OBJ	(to be) that which is found/discovered by chance	

NOTE: For a find/discovery made through a directed process of inquiry/investigation, see the root -**ND**-.

-MḶ- 'OBSCURITY / AMBIGUITY / STEALTH / SURREPTITIOUSNESS'		
BSC	**Stem 1:** (to be) a state/act of obscuring; render something obscure or unclear [i.e., so that it is not clear or easily discernible as to what it is exactly]	**Stem 2:** (to be) a state/act of ambiguity; to render something ambiguous, to obfuscate
CTE	(to be) a state of obscurity/being obscure	**Stem 3:** (to be) a state/act of stealth/surreptitiousness; to act stealthily/surreptitiously, be stealthful/surreptitious (i.e., so as to render something difficult to observe or notice)
CSV	(to be) a physical act of obscuring; to do something that makes an entity obscure or unclear	
OBJ	(to be) what is (made) obscure/unclear	

-ẒŘ- ‘SLOT / GROOVE / FURROW’		
BSC	**Stem 1:** (to be) a linear indentation/groove/channel; to make a linear indentation/groove/channel	**Stem 2:** (to be) a slot; to make a slot (i.e., a linear opening into which something can be placed/fitted; does NOT mean ‘slot’ in a classificatory/schematic sense as in *“Can we be slotted in between those appointments?”* or *“The morpho-phonology of the Ithkuil formative has a slot structure.”*)
CTE	(to be) the state of being grooved, having a linear indentation/channel/groove	
CSV	(to be) a physical act of making/fashioning a linear indentation/groove/channel	
OBJ	(to be) an implement for use with or placement into a linear indentation/groove/channel	**Stem 3:** (to be) a furrow

-RÇ- ‘AESTHETIC APPEAL / PRETTINESS / LOVELINESS / BEAUTY’		
BSC	**Stem 1:** (to be) a state/act of perceiving/experiencing something aesthetically appealing; to perceive/experience something aesthetically appealing; to like something aesthetically-speaking	**Stem 2:** (to be) a state/act of perceiving/experiencing something pretty/fair/lovely; to perceive/experience something pretty/fair/lovely
CTE	(to be) a state of aesthetic appeal, a state of something being aesthetically appealing	**Stem 3:** (to be) a state/act of perceiving/experiencing something beautiful; to perceive/experience something beautiful
CSV	(to be) an act of perceiving/contemplating/experiencing something aesthetically appealing; to like something aesthetically-speaking	
OBJ	(to be) something aesthetically appealing	

-ĻN- ‘VALUE / EXPEDIENCE / SUITABILITY’		
BSC	**Stem 1:** (to be) a state of something being potentially worthwhile or of worth/value, something worth pursuing / to be something potentially worthwhile; to pursue something potentially worthwhile or of worth/value	**Stem 2:** (to be) a state of something being expedient or opportune; to be/do something expedient or opportune
CTE	(to be) a state of having value/being worthwhile/being valuable	**Stem 3:** (to be) a state of something being fitting, apropos or suitable; to be/do something fitting, apropos or suitable
CSV	(to be) a physical act of doing/pursuing something worthwhile/of value	
OBJ	(to be) something worthwhile; (be) something of value; (be) something valuable	

-RḶN- ‘ADVANTAGE / OPPORTUNITY / CONVENIENCE’		
BSC	**STEM 1:** (to be) a state of something being advantageous/favorable; to be advantageous, be to one’s advantage, to be in favorable or advantageous circumstances	**STEM 2:** (to be) a state of something being an opportunity / a chance to experience or do something normally unavailable; to pursue an opportunity, take advantage of an opportunity **STEM 3:** (to be) a state of something being a convenience / something convenient; to be convenient / do something convenient
CTE	(to be) a state of being advantageous/favorable	
CSV	(to be) a physical act of pursuing an advantage; to pursue an advantage	
OBJ	(to be) something advantageous/favorable	

-MÇ- ‘WELL-FORMEDNESS / AESTHETIC APPEAL / STYLISHNESS / PERFECTION / SPLENDOR’		
BSC	**Stem 1:** (to be) a state of being well-formed/shapely/well-proportioned; to be/make well-formed/shapely/well-proportioned	**Stem 2:** (to be) a state of something being aesthetically/sensually pleasing or satisfying (e.g., a fine meal, a glass of wine, a thrill ride, a sexual experience, etc.); to experience a sense of aesthetic/sensual satisfaction/pleasure from something **Stem 3:** (to be) a state of something being stylish/fine (as in a “finely”-made, “finely”-crafted); to be/make something stylish/fine, etc.
CTE	(to be) a state of shapeliness/well-formedness	
CSV	(to be) an act of making something well-formed/shapely/well-proportioned	
OBJ	(to be) that which is shapely/well-formed/well-proportioned	

-MÇK- ‘PERFECTION / OPTIMALITY / EXQUISITENESS / SUBLIMITY / SPLENDIDNESS’		
BSC	**STEM 1:** (to be) a state of something being perfect/optimal; be/make perfect/optimal (i.e., cannot possibly be better)	**STEM 2:** (to be) a state of something being exquisite/superb/sublime; to be/make something exquisite/superb/sublime **STEM 3:** (to be) a state of something being magnificent/splendid; be/make something magnificent/splendid
CTE	(to be) a state of perfection/optimality	
CSV	(to be) an act of making something perfect/optimal; to perfect, to optimize	
OBJ	(to be) that which is perfect/optimal	

-KÇT- ‘DECORATION / FANCINESS / ORNATENESS / ORNAMENTATION’		
BSC	**STEM 1:** (to be) a state of something being decorative; be/make decorative, to decorate	**STEM 2:** (to be) a state of something being fancy/showy; to be/make something fancy/showy **STEM 3:** (to be) a state of something being ornate/ornamental; be/make something ornate/ornamental
CTE	(to be) a state of decorativeness/decoration; to have a decorative look	
CSV	(to be) an act of making something look decorative; to do something that decorates	
OBJ	(to be) that which is decorative-looking	

-ŘŘ- ‘INTENSIFICATION / AMELIORATION / IMPROVEMENT / AGGRAVATION / WORSENING’		
BSC	**Stem 1:** (to be) a state/act of intensifying“heightening”; to intensify/“heighten” (i.e., make the effect of something stronger or more effective/impactful)	**Stem 2:** (to be) a state/act ameliorating/bettering/enhancing/improving; to ameliorate/better/improve (i.e., cause a situation or the condition/state of something to improve or become more desirable/optimal) **Stem 3:** (to be) a state/act of aggravation/worsening; to aggravate/worsen
CTE	(to be) a state of intensification	
CSV	(to be) an act of intensifying; to do something that causes intensification	
OBJ	(to be) that which is intensified	

-TḶT- ‘ADJUSTMENT / REPAIR / MENDING / RESTORATION / CORRECTION’		
BSC	**STEM 1:** (to be) an act/instance/process of adjustment; to adjust [i.e., make a minor change to the state/condition/shape/appearance/workings of something in order to change its function/efficiency/appearance/usability]	**STEM 2:** (to be) an act/instance/process of repairing/fixing/mending/ restoring; to repair/fix/mend/restore [i.e., bring something in a broken/damaged/worn state back to a functioning operative/usable/ presentable state **STEM 3:** (to be) an act/instance/process of correction, rendering something that is in an incorrect/improper/inaccurate state into a correct/proper/accurate state
CTE	(to be) a state of adjustment/being adjusted/having been adjusted	
CSV	(to be) a physical act of adjustment/adjusting; to perform a physical act of adjustment	
OBJ	(to be) the adjustment one makes; the specific change to an entity made during an adjustment	

-ŢJ- ‘EMPHASIS / HIGHLIGHT / UNDERSCORE / IMPRESS UPON / RAVE ABOUT’		
BSC	**Stem 1:** (to be) a state/act of highlighting/bringing attention to; to highlight, to bring attention to, to focus upon	**Stem 2:** (to be) a state/act of emphasizing/underscoring/impressing upon others about; to emphasize, underscore, to impress upon others about (i.e., to make another particularly aware or focused on something/someone) **Stem 3:** (to be) a state/act of going on about/raving about, not being able to say enough about; to go on about, rave about, can’t say enough about, sing the praises of
CTE	(to be) a state of being highlighted, having attention upon oneself	
CSV	(to be) an act of highlighting/bringing attention to; to do something that highlights or brings attention to an entity/party	
OBJ	(to be) the party/entity/object highlighted	

-VŠ- 'SHREWDNESS / ARTFULNESS / CRAFTINESS'		
BSC	**Stem 1:** (to be) a state/act of shrewdness; be/act shrewd(ly) [i.e., using unobtrusive cleverness and good judgement]	**Stem 2:** (to be) a state/act of artfulness; be/act artful(ly) (i.e., accomplish one's aims in a manner with the intention that others not realize/see/know exactly what one has done or how one has done it) **Stem 3:** (to be) state/act of craftiness/scheming/underhandedness/sneakiness; to act with craftiness, be crafty/scheming/underhanded, act in a crafty/scheming/underhanded manner (i.e., same as Stem 2 above but with the additional connotation that one's aims/goals are nefarious)
CTE	(to be) a state of shrewdness	
CSV	(to be) an act of shrewdness; to do something shrewd(ly)	
OBJ	(to be) that which is shrewd	

-KH- 'DEGREE OF SUBTLETY / NUANCE / CUNNING / GUILE / STEALTH / INSIDIOUSNESS' Affix: **SBT** (use SUF/EXN/EXD, etc.)		
BSC	**STEM 1:** (to be/manifest a) degree of subtlety/nuance (= slight but meaningful variation in appearance/manifestation/color/sound, etc.) — [both the quality and the entity manifesting such]	**STEM 2:** (to be/manifest a) degree of slyness / cunning / guile (= subtlety/nuance in terms of wordplay/behavior/actions, etc. with intention to manipulate a situation) **STEM 3:** (to be/manifest a) degree of stealth/insidiousness (= attempt to manipulate/act in an unseen/hidden/discreet manner for (quasi-)nefarious purposes)
CTE	(to be) something with a degree of subtlety/nuance	
CSV	(to be) a degree of subtlety/nuance	
OBJ	(to be) an entity/party impacted/affected by one's degree of subtlety/nuance	

-VSK- 'SURVEILLANCE / RECONNAISSANCE / UNDERCOVER OPERATION / ESPIONAGE'		
BSC	**Stem 1:** (to be) state/act/process of surveillance; to conduct surveillance [i.e., surreptitious monitoring/observation of something or someone]	**Stem 2:** (to be) an act/process of reconnaissance; to reconnoiter, conduct reconnaissance **Stem 3:** (to be) an act of undercover espionage; conduct undercover operation to learn secrets from a party, to spy, conduct espionage
CTE	(to be) a state of being secretly observed/monitored	
CSV	(to be) a physical act of secretly observing/monitoring; to physically observe or monitor as an act of surveillance	
OBJ	(to be) an observation made during surveillance; what is seen via surveillance	

-LČ- 'GUESS / SPECULATION / SUPPOSITION / ASSUMPTION'		
BSC	**Stem 1:** (to be) a state/act of guessing, state/act of speculation; to guess, to speculate, to venture to say (i.e., conclusion reached in the absence of sufficient evidence)	**Stem 2:** (to be) a state/act of supposition; to suppose (i.e., to consider as a possibility or possible interpretation) **Stem 3:** (to be) a state/act of assuming/presuming (i.e., believing one's interpretation or supposition to be correct in the absence of confirmatory evidence)
CTE	(to be) a state of speculation; ponder a situation as to how it might/should be interpreted or what conclusion one should come to about it	
CSV	(to be) an act of speculating, guessing; to make a guess	
OBJ	(to be) what on is guessing/speculating about	

-LČW- 'CONJECTURE / POSTULATING / THEORY / HYPOTHESIS'		
BSC	**STEM 1:** (to be) a state/act of postulating/making a conjecture; to postulate, make a conjecture	**STEM 2:** (to be) a state/act of theorizing; to theorize, create a theorem **STEM 3:** (to be) a state/act of hypothesizing; to hypothesize
CTE	(to be) a state of postulating or formulating a conjecture	
CSV	(to be) an act of postulating/conjecturing; to make a conjecture; to make a postulate	
OBJ	(to be) a conjecture, a postulate, what one postulates/conjectures	

-ŠH- 'IMPORTANCE / SIGNIFICANCE / PREVALENCE / CONSPICUOUSNESS / PROMINENCE'		
BSC	**Stem 1:** (to be) a state/act of importance/significance; to be/make important or significant	**Stem 2:** (to be) a state/act of prevailing/prevalence; to prevail upon, to make prevalent **Stem 3:** (to be) salience; to be/make salient/prominent/conspicuous/(pre-)eminent
CTE	(to be) a state of importance/significance	
CSV	(to be) an act of importance/significance; to do something important/significant	
OBJ	(to be) that which is important/significant	

-ŇČ- 'INFLUENCE / PERSUASION / PREDOMINANCE'		
BSC	**Stem 1:** (to be) a state/act of influencing/influence; to be influential, to influence	**Stem 2:** (to be) a state/act of persuasion; to persuade, sway, cajole, talk someone into, convince
CTE	(to be) a state of influence/being influential	
CSV	(to be) an act of influence; to do/say something that influences, do something influential	**Stem 3:** (to be) a state of being preponderant/predominant, being a preponderance; to have/carry weight/predominance, to hold sway, to be/constitute a preponderance of evidence in favor of
OBJ	(to be) an influence	

-RŇČ- 'INCITING / INSTIGATION / ENCOURAGEMENT / ADVOCACY'		
BSC	**STEM 1:** (to be) a state/act of stimulation/rousing/inciting/instigating; to stimulate/rouse/incite/instigate	**STEM 2:** (to be) a state/act of encouragement; to encourage
CTE	(to be) a state of being stimulated/roused/incited	**STEM 3:** (to be) a state/act of advocacy; to advocate for
CSV	(to be) an act of stimulating/instigating/inciting; to do something that incites/stimulates/instigates	
OBJ	(to be) that which a party/entity is stimulated/incited to be/do	

-ČPĻ- 'USE OF FORCE / COERCION / COMPULSION'		
BSC	**STEM 1:** (to be) a state/act of using the force of one's will/personality/demeanor/ power or social standing to make something happen; to force something to happen by means of one's will/personality/demeanor/power or social standing, to make something happen by force of will, to make somebody do something by force of will	**STEM 2:** (to be) a state/act coercing/compulsion; to coerce/compel [i.e., making someone do something against their will by means of an implied threat]
CTE	(to be) one's force of will/power/personality/demeanor/power or social standing as a (potential) means for making things happen	**STEM 3:** (to be) a state/act of using physical/brute force or physical manipulation/handling/attack against resistance to make something happen; to physically force/make something happen, to physically make/force somebody to do something against their will
CSV	(to be) an act of wielding/exercising one's force of will/power/personality/demeanor/ power or social standing to make something happen	
OBJ	(to be) what is made to happen by force of will/power/personality/demeanor/power or social standing	

-JR- ‘RIDE / DRIVE / TRANSPORT / PASSENGER’		
	[**NOTE:** This root focuses on operation/conveyance/carrying of an entity on a mode of transportation, NOT on the physical passage/trajectory through spacetime; for this latter notion, use an appropriate root/stem from Sec. 9.1.	
BSC	**Stem 1:** (to be) a state/act of riding/drive; to ride/drive (i.e., to operate/control a mechanical device, machine, or animal as a means of conveyance/transportation, e.g., a bike, a horse, an automobile) [NOTE: this stem does not mean ‘ride’ where one is simply being conveyed without being the controller/operator, e.g., ride a bus]	**Stem 2:** (to be) a a state/act of being a passenger conveyed/transported somewhere, riding; to ride (on) something (i.e., conveyance or transportation where one is NOT the one controlling/operating the device/machine/animal, e.g., ride a bus/train/escalator, etc.) **Stem 3:** (to be) a state/act of transportation via a means of transportation/conveyance; to transport (i.e., the conveying/carrying of an entity via a device or animal)
CTE	(to be) a state of riding/driving something (i.e., operating the means of conveyance/transportation)	
CSV	(to be) an act of driving/riding (i.e., operating/controlling) a means of personal conveyance/transportation; to perform the physical act(s) of riding/driving something (e.g., a horse, bike, automobile, skateboard, etc.)	
OBJ	(to be) the person being conveyed/transported, the transportee	

-PŠW- ‘STATION-KEEPING / PARKING / HOLDING AREA / WAITING AREA’		
BSC	**STEM 1:** (to be) a state/act of station-keeping, holding in place	**STEM 2:** (to be) a state/act of place-keeping something when not in use, storage-keeping something when not in use; to store something when not in use, to park (a vehicle) **STEM 3:** (to be) a state/act of temporarily holding or waiting in a holding area or waiting area; to temporarily hold/wait in a holding area/room/space or waiting area/room
CTE	(to be) a state of being held in place for station-keeping	
CSV	(to be) an act of holding something in place for station-keeping; to hold in place for station-keeping	
OBJ	(to be) where something is held in place for station-keeping	

-SC- ‘WASH / LAVE / BATHE / RINSE’		
BSC	**Stem 1:** (to be) a state/act of washing with water or liquid cleansing solution (i.e., to clean the surface of an entity by means of water or liquid cleansing solution)	**Stem 2:** (to be) a state/act of bathing; to bathe **Stem 3:** (to be) a state/act of rinsing/flushing/showering; to rinse/flush/shower (NOTE: by ‘shower’ is meant use of a controlled spray of water/cleansing fluid, not a rainshower or naturally occurring spray/pouring of water/liquid)
CTE	(to be) a state of being washed	
CSV	(to be) a physical act of washing (i.e., the physical motions/actions performed); to perform the physical motions/actions of washing	
OBJ	(to be) what is (being / to be) washed	

-MÇT- ‘EXAMPLE / EXEMPLAR / PRECEDENT / ARCHETYPE’		
BSC	**STEM 1:** (be) a state/instance of being an example of something; to exemplify something	**STEM 2:** (be) a precedent; i.e., an entity or situation which establishes a to-be-expected pattern for other/subsequent occurrences or instances of the entity or situation **STEM 3:** (be) an archetype; the most prototypical/definitive/desirable example/pattern/design of an entity or idea
CTE	(be) a state of being seen/used as, or performing the function of being, an example; to be an example	
CSV	(be) a state/act/process of exemplification; the perform an act/process of exemplification; to make an example out of	
OBJ	(be) that which is considered an example or made to be an example; an exemplar	

This root is also used as the **ARCHETYPAL (ACH) Bias Affix:** *what (a) ...!; how . . .! ; Boy! Did (does) X ever …*

	-VLL- ‘IMPLICATION / HINT / CONNOTATION / ALLUSION’	
BSC	**Stem 1:** (be) a (state/process of) implication [i.e., a statesituation which functions as evidence by which an observer may deduce/infer something; to imply something]	**Stem 2:** (process of) connotation; to connote **Stem 3:** (process of) allusion; to allude (to)
CTE	(be) an implication; that which is implied	
CSV	(be) a process of implying; offer/manifest evidence from which an inference can be made	
OBJ	(be) that which gives rise to, or is the basis for an implication; a hint, a “tip”	

This root is also used as the **IMPLICATIVE (IPL) Bias affix:** = *‘of course,’ ‘after all,’ or ‘needless to say’*

	-ÇÇK- ‘MAXIMIZATION / OPTIMALITY / SUPREMACY’	
BSC	**Stem 1:** (be at) the maximum point/stage/degree/state; to maximize	**Stem 2**: (be at) the optimal point/stage/degree/condition/state **Stem 3:** (be at) the supreme/utmost/”highest”/foremost point/stage/degree/state
CTE	(be) a state of maximality	
CSV	(be) a process/act of maximizing	
OBJ	(be) the entity which in a maximal state	

This root is also used as the **OPTIMAL (OPT) Bias affix:** = prolonged ‘so’ or ‘totally’ as in ‘*I so don’t care!’* or ‘*That is totally what I wanted.’*

-LXL- 'SCIENCE'		
BSC	**Stem 1:** (to be) an act/state of utilizing science; to do something scientifically (i.e., employ the analytical/rational/dispassionate methods/tools/resources of clinical science when analyzing, considering, planning, working on something)	**Stem 2:** (to be) an act/state that is based upon and/or consistent with the scientific method **Stem 3:** (to be) an act/state of relying upon science and/or scientific findings/conclusions when analyzing/judging a situation
CTE	(to be) a state of scientific rationality; for a state/act/situation to have science as its basis for existing/happening	
CSV	(to be) an act/state of doing something scientifically; to do something scientifically	
OBJ	(to be) what one does scientifically or the state/act that exists/takes place based upon science	

	-LDR- 'APPEAR / SHOW / DISPLAY / EXHIBIT'	
BSC	**Stem 1:** (to be) a showing, an appearance, i.e., be/make observable/discernible to others; to show, to manifest, to appear	**Stem 2:** (to be) an act/process of showing off, flaunting; to show off, to flaunt **Stem 3:** (to be) an act/process of displaying or exhibiting something; display, exhibition; to display, to exhibit
CTE	(to be) the state of appearing, being shown, being apparent	
CSV	(to be) the physical act/process of making something appear, or be shown, or be apparent	
OBJ	(to be) the object/entity/party being shown or appearing to others	

-DRR- 'CALLING ATTENTION TO / EXPOSURE / ADVERTISING'		
BSC	**STEM 1:** (to be) an act of bringing/calling attention to something, i.e., make others aware of the appearance/showing of something	**STEM 2:** (to be) an act/process of exposure, i.e., letting others observe/discern something they otherwise would not observe/discern **STEM 3:** (to be) an advertisement; to advertise
CTE	(to be) the state of being a focus of attention	
CSV	(to be) the physical act/process of calling attention to; to do/say something that calls attention to something	
OBJ	(to be) the object/entity/party/fact/situation that one calls attention to	

This root is also used as the **ANNUNCIATIVE (ANN) Bias Affix:** = *'Guess what!'* or *'Wait till you hear this!*

-ŢRR- ‘ANNOUNCEMENT / DECLARATION / PRONOUNCEMENT / PROCLAMATION’		
BSC	**STEM 1:** (to be) an act/instance of announcing something (verbally or in writing); to announce, to make an announcement	**STEM 2:** (to be) an act/instance of reporting something; to report [i.e., verbally letting others be aware of something that has heretofore not been so or has not been publicly known]
CTE	(to be) the state of being/having been announced	
CSV	(to be) the physical act/process of announcing; to do/say/write/publish something that announces something	**STEM 3:** (to be) an act/instance of declaring/pronouncing/proclaiming something to be so; to declare/pronounce/proclaim (i.e., where the act of doing so has the performative effect of making it so/true or causing it to be in effect)
OBJ	(to be) that which is announcement; an announcement	

-ŽW- ‘KIND / SORT / TYPE / GENRE / CLASS / CATEGORIZATION / MODE / MANNER / WAY’		
BSC	**Stem 1:** (to be) a state/act of categorization or being/placing in a category; to categorize, to place/label/identify something into or as being in a category/kind/sort /class/variety	**Stem 2:** (to be) a state/act of being/placing in a genre/style; to place/label/identify something into or as being in a particular genre/style
CTE	(to be) a kind/sort/class/variety/category	
CSV	(to be) an act of establishing/determining/identifying/creating a kind/sort/genre/category	**Stem 3:** (to be) a state/act of doing something in a certain way/manner/mode; to do something in a particular way/manner/mode
OBJ	(to be) a member of a kind/sort/class/variety/category	

-MTF- ‘LIST / SERIES / FILE’		
BSC	**Stem 1:** (to be) a state/act/instance of arranging or being arranged in a list; to list, arrange in a list, create/make a list	**Stem 2:** (to be) a an act/instance of making/placing in a series; to arrange/order in a sequential series
CTE	(to be) the order of items on a list	
CSV	(to be) an act/process of physically listing or making a list	**Stem 3:** (to be) an act/instance of arranging in a file; to file, create/make a file
OBJ	(to be) an entity named on a list, be a member of a list	

-ǍŢ- **'DEGREE OF CORRECTNESS/ACCURACY/ERROR/SUITABILITY/FITNESS'** Affix: **ERR** (use SUF/EXN/EXD, etc.)		
BSC	**STEM 1:** (to be/manifest a) degree of correctness/accuracy (= non-erroneous information) — [both the quality and the manifestation thereof]	**STEM 2:** (to be/manifest a) degree of suitability / fitness / propriety / effectiveness (= best choice for the context at hand) **STEM 3:** (to be/manifest a) degree of legitimacy/orthodoxy (= adherence to legal / ritualistic / societal conventions)
CTE	(to be) a state of being correct/accurate to a certain degree	
CSV	(to be) a degree of correctness/accuracy; to manifest a certain degree of correctness/accuracy	
OBJ	(to be) something that has/manifests a degree of correctness/accuracy	

This root is also used as the **CORRECTIVE (CRR) Bias Affix:** *'that is to say…,' 'What I mean(t) to say is…' 'I mean….'*

-ǍḌ- **'DEGREE OF EXACTITUDE'** Associated Affix: **EXT**		
BSC	**STEM 1:** (to be) something having a degree of exactitude [i.e., expected identity/equivalence to some expectation)	**STEM 2:** (to be) something of a particular degree of exactitude (i.e., accuracy or identical functionality/performance in comparison to some standard) **STEM 3:** (to be) something that serves/functions/substitutes [comparably] (in comparison to some standard)
CTE	(to be) the degree to which an entity meets an expectation of exactitude; to perform an appraisal/comparison of such	
CSV	(to be) the expectation of exactitude to which an entity is compared; to apply/establish such an expectation/standard	
OBJ	(to be) the entity so appraised/compared	

This root is usually used with affixes such as **EXN**, **SUF**, **EXD**, or similar to indicate the particular degree of the named quality.

-**RX**- **'DEGREE OF SIMILARITY / RESEMBLANCE'** Associated Affix: **SIM** (use with the EXN, SUF, EXD, etc. affixes if necessary)		
BSC	(to be/manifest) something with a particular degree of similarity/resemblance	**STEM 1:** similarity/resemblance to another/different concrete/tangible entity/instance **STEM 2:** similarity/resemblance to oneself on another/different occasion **STEM 3:** similarity/resemblance to another/different abstract situation or set of circumstances
CTE	(to be/manifest) the degree to which an entity meets an expectation of similarity/resemblance	
CSV	(to be/manifest) the expectation of similarity/resemblance to which an entity is compared	
OBJ	(to be) the entity being appraised/compared to the expectation of similarity/resemblance	

-RK- 'DEGREE/RANGE/EXTENT/INTENSITY'		Associated Affix: **EXN**
BSC	**STEM 1:** (to be) the degree/extent of something (= scalar amount of the effect/impact/capacity of something) [both the degree/extent and the entity manifesting that degree/extent]	**STEM 2:** (to be) the range of something (= measure of the "upper" or "outer" limit of the effect/impact of something) **STEM 3:** (to be) the intensity of something (= measure of the strength of the effect/impact of something)
CTE	(to be) something with a degree/extent/amount of effect/impact/capacity [focus on the entity itself]	
CSV	(to be) a degree/extent (= the amount of effect/impact/capacity)	
OBJ	(to be) the entity/party affected/impacted by the degree/extent of something	

The following root has the same Specification pattern as the root -**RK**- above:

-RT- SUFFICIENCY/ADEQUACY Associated Affix: **SUF**
Stem 1: (to be) a sufficient amount/extent/degree of something; to suffice, to be enough, to be sufficient
Stem 2: (to be) an adequate amount/extent/degree of something; to be adequate, to be good enough, to do (= to suffice adequately)
Stem 3: (to be) something to a satisfactory degree/extent; to be satisfactory, to meet the requirements

-MH- 'DEGREE OF CHANGE/STABILITY/VICISSITUDE/FLUCTUATION'		Associated Affix: **FLS** (use SUF/EXN/EXD, etc.)
BSC	**STEM 1:** (to be/manifest a) degree of change(-ableness)/mutability/constancy/alteration (= degree to which something becomes altered in form/behavior) [both the degree and the potential for, or nature of, the change	**STEM 2:** (to be/manifest a) degree of stability/instability/vicissitude (= degree of predictability of future pattern/behavior based on past pattern/behavior) **STEM 3:** (to be/manifest a) degree of fluctuation / wavering / variance / deviation (= degree to which normally stable/predictable pattern/behavior suddenly fluctuates/varies/wavers/deviates unpredictably)
CTE	(to be) something with a degree/extent/amount of change(-ableness/mutability [focus on the entity itself]	
CSV	(to be) the degree/extent of change(-ableness)/mutability (= the amount/extent of actual or potential change)	
OBJ	(to be) the entity/party affected/impacted by the degree/extent of (potential) change	

The following root has the same Specification pattern as the root -**MH**- above:

-RG- 'DEGREE OF STEADFASTNESS/MOBILITY/MOTILITY/ACTIVITY/MOVEMENT' Affix: **MVT** (use SUF/EXN/EXD, etc.)
STEM 1: (to be/manifest) a degree of steadfastness/durability (= [in]ability to be altered in form/behavior)
STEM 2: (to be/manifest) a degree of mobility/motility/activity (= [in]ability to move or to be made to move)
STEM 3: (to be/manifest) a degree of indelibility/fixedness (= [in]ability to be removed/erased/eliminated)

-ČH- **'DEGREE OF AFFECT (CUTENESS/GRANDEUR)'** Associated Affix: **AFT** (use SUF/EXN/EXD, etc. to specify degree)		
BSC	**STEM 1:** (to be/manifest a) degree of grandeur, grandiosity, greatness [both the quality and the entity manifesting such]	**STEM 2:** (to be/manifest a) degree of elegance, loveliness, grace **STEM 3:** (to be/manifest a) degree of endearment, cuteness
CTE	(to be) something with a degree of grandeur, grandiosity, greatness	
CSV	(to be) a degree of grandeur, grandiosity, greatness	
OBJ	(to be) an entity/party impacted/affected by one's degree of grandeur, grandiosity, greatness	

-TH- **'DEGREE OF RELIABILITY/INFALLIBILITY/QUALITY'** Affix: **QUA** (use SUF/EXN/EXD, etc. to specify degree)		
BSC	**STEM 1:** (to be/manifest a) degree of reliability/infallibility/quality (in terms of ruggedness/durability of workmanship or mechanical craftsmanship) [both the quality and the entity manifesting such]	**STEM 2:** (to be/manifest a) degree of quality, fineness, quality of effort (in terms of artisanality, artistry, artistic craftsmanship) **STEM 3:** (to be/manifest a) degree of durability / lastingness / permanence / persistence
CTE	(to be) something with a degree of reliability/infallibility/quality	
CSV	(to be) a degree of reliability/infallibility/quality	
OBJ	(to be) an entity/party impacted/affected by one's degree of reliability/infallibility/quality	

-ŘP- **'DEGREE OF SPECIALNESS/UNIQUENESS/MARVELOUSNESS'** Associated Affix: **UNQ** (use SUF/EXN/EXD, etc.)		
BSC	**STEM 1:** (to be/manifest a) degree of specialness, extraordinariness (in terms of variance with the usual/expected) [both the quality and the entity manifesting such]	**STEM 2:** (to be/manifest a) degree of uniqueness (= the degree to which something is unlike anything else, one-of-a-kind) **STEM 3:** (to be/manifest a) degree of marvelousness/wonder/awesomeness
CTE	(to be) something with a degree of specialness, extraordinariness	
CSV	(to be) a degree of specialness, extraordinariness	
OBJ	(to be) an entity/party impacted/affected by one's degree of specialness, extraordinariness	

-ŘV- **'CONSEQUENCE/OUTCOME/RESULT'** Affix: **CNQ** (use SUF/EXN/EXD, etc. to specify degree)		
BSC	**STEM 1:** (to be/manifest) a consequence of something (= a change in pre-existing circumstances resulting from a state/act/event/occurrence) [both the consequence itself and the state of consequentiality]	**STEM 2:** (to be/manifest a) result of something (= a concrete/tangible "product" or specific/nameable abstract entity resulting from a state/act/event/occurrence) **STEM 3:** (to be/manifest a) abstract result/outcome (= an abstract set of non-preexisting circumstances arising out of an occurrence/event/act/state)
CTE	(to be) a specific consequence	
CSV	(to be) a state of consequentiality; a state of there being a consequence; to manifest consequentiality	
OBJ	(to be) an entity/party from which arises a consequence	

-ŘG- **'DEGREE OF CONFORMITY/TYPICALNESS'** Affix: **TYP** (use SUF/EXN/EXD, etc. to specify degree)		
BSC	**STEM 1:** (to be/manifest a) degree of conformity to a norm, stereotype or standard [both the quality and the entity manifesting such]	**STEM 2:** (to be/manifest a) degree of being commonplace / typical / run-of-the-mill; to typify to a particular degree, to be typical of to a particular degree **STEM 3:** (to be/manifest a) degree of being original / innovative / "out-of-the-box"
CTE	(to be) something with a degree of conformity/typicalness	
CSV	(to be) a degree of conformity/typicalness	
OBJ	(to be) an entity/party impacted/affected by one's degree of conformity/typicalness	

-RD- **'DEGREE OF TRUTH/GENUINENESS/VERACITY/VALIDITY/FACTUALITY'** Affix: **REA** (use SUF/EXN/EXD, etc.)		
BSC	**STEM 1:** (to be/manifest a) degree of truth/veracity/reality (in terms of something actually being ontologically valid as true or real) [both the quality and the entity manifesting such]	**STEM 2:** (to be/manifest a) degree of authenticity/genuineness (in terms of an entity's identity/nature/behavior/essence actually being what it is purported/assumed/believed to be) **STEM 3:** (to be/manifest a) degree of validity/factuality/actuality (in terms of being an accurate assessment, description, representation, or identification)
CTE	(to be) something with a degree of truth/veracity/reality	
CSV	(to be) a degree of truth/veracity/reality	
OBJ	(to be) an entity/party impacted/affected by one's degree of truth/veracity/reality	

-ŢM- **'DEGREE OF INTENSITY'** Affix: **ITY** (use SUF/EXN/EXD, etc.)		
BSC	**STEM 1:** (to be/manifest a) degree of intensity/power (= ratio of physical manifestation/change/activity/energy to period of time) — [both the quality and the entity manifesting such]	**STEM 2:** (to be/manifest a) degree of activity/acuteness **STEM 3:** (to be/manifest a) degree of agitation/ebullience/bustle
CTE	(to be) something with a degree of intensity/power	
CSV	(to be) a degree of intensity/power	
OBJ	(to be) an entity/party impacted/affected by one's degree of intensity/power	

-FŢ- **'DEGREE OF EFFICIENCY/ADEQUACY'** Affix: **EFI** (use SUF/EXN/EXD, etc.)		
BSC	**STEM 1:** (to be/manifest a) degree of efficiency (= ratio of amount of effort/energy/resources to results)— [both the quality and the entity manifesting such]	**STEM 2:** (to be/manifest a) degree of adequacy (= extent to which something serves sufficiently) **STEM 3:** (to be/manifest a) degree of reward/value/ "pay-off"/ "bang-for-the-buck" (= extent/value of return on investment of resources/effort/energy)
CTE	(to be) something with a degree of efficiency	
CSV	(to be) a degree of efficiency	
OBJ	(to be) an entity/party impacted/affected by one's degree of efficiency	

-KŢ- **'DEVELOPMENT / GROWTH / MATURATION / DECLINE'** Affix: **MAT**		
BSC	**STEM 1:** (to be/manifest a) developmental stage, stage of developmental or maturational cycle; stage of a life-cycle— [both the stage and the entity manifesting such]	**STEM 2:** (to be/manifest) goal-oriented development/maturation, growth; develop, grow, mature (growth not necessarily in size, but rather in maturational development as part of a life-cycle); CPT Version = ripe(n), ripening **STEM 3:** (to be/manifest) the declining stage(s) of a life-cycle; decline, fade, age; CPT Version = die
CTE	(to be) something manifesting a developmental stage, stage of developmental or maturational cycle; stage of a life-cycle	
CSV	(to be/manifest) a developmental stage, stage of developmental or maturational cycle; stage of a life-cycle	
OBJ	(to be) an entity/party impacted/affected by one's developmental stage, stage of developmental or maturational cycle; stage of a life-cycle	

-PH- **'DEGREE OF PRECISION / DEFINITION / SCRUTINY'** Affix: **PCN** (use SUF/EXN/EXD, etc.)		
BSC	**STEM 1:** (to be/manifest a) degree of precision (= exactness of fine-tuning, fineness of or attention to detail, carefulness in workmanship/craftsmanship) — [both the quality and the entity manifesting such]	**STEM 2:** (to be/manifest a) degree of definition, being well-defined (= exactness of distinction, contrast with background, separability from background milieu) **STEM 3:** (to be/manifest a) degree of scrutiny / attention / scrupulousness (= thoroughness/exactitude in procedure / task / activity)
CTE	(to be) something with a degree of precision	
CSV	(to be) a degree of precision	
OBJ	(to be) an entity/party impacted/affected by one's degree of precision	

-ŢX- **'DEGREE OF SOPHISTICATION/CRUDENESS/REFINEMENT/PRIMITIVENESS'** Affix: **SPH** (use SUF/EXN/EXD, etc.)		
BSC	**STEM 1:** (to be/manifest a) degree of physical/technical refinement/sophistication (vs. crudeness, i.e., level to which an entity manifests a high level of technical workmanship/craftsmanship/artisanality) — [both the quality and the entity manifesting such]	**STEM 2:** (to be/manifest a) degree of evolvement (vs. primitiveness, i.e., temporal, evolutionary perspective over time) **STEM 3:** (to be/manifest a) degree of cultural refinement/sophistication/finesse (vs. cultural primitiveness/savagery/simplicity)
CTE	(to be) something with a degree of physical/technical refinement/sophistication	
CSV	(to be) a degree of physical/technical refinement/sophistication	
OBJ	(to be) an entity/party impacted/affected by one's degree of physical/technical refinement/sophistication	

-ŇŘ- **'IMPEDIMENT / BLOCKAGE / RESTRICTION / CONSTRAINT / LIMITING / PREVENTION'** Affix: **PRV**		
BSC	**STEM 1.** (to be) a state/process of something under restriction; something impeded or blocked from occurring or manifesting — [both the act/process/state of restriction and the method/means used to restrict]	**STEM 2.** (to be) a state/process of something being constrained or limited in its path/direction/area; to constrain, to limit, to narrow **STEM 3.** (to be) a state/process of preventing something from happening; the prevention of something; to prevent
CTE	(to be) a process/state of restricting/impeding/blocking from manifesting or occurring	
CSV	(to be) a the means/method of restricting/impeding/blocking from manifesting or occurring	
OBJ	(to be) an entity/party/state being restricted/blocked/impeded	

-ČM- **'DEGREE OF EASE vs. DIFFICULTY'** Affix: **DIF** (use SUF/EXN/EXD, etc.)		
BSC	**STEM 1:** (to be/manifest a) degree of ease vs. difficulty (= effort in terms of physical hardship/toil and/or perplexity/complexity/mental challenge involved) — [both the quality and the entity manifesting such]	**STEM 2:** (to be/manifest a) degree of ease vs. difficulty (= effort in terms of practicality, return/value-for-energy expended, likelihood of desired outcome) **STEM 3:** (to be/manifest a) degree of ease vs. difficulty (= effort in terms of being faced with a quandary / dilemma / predicament)
CTE	(to be) something with a degree of ease vs. difficulty	
CSV	(to be) a degree of ease vs. difficulty	
OBJ	(to be) an entity/party impacted/affected by one's degree of ease vs. difficulty	

-KP- **'DEGREE OF EFFORT'** Affix: **EFT** (use SUF/EXN/EXD, etc.)		
BSC	**STEM 1:** (to be/manifest a) degree of effort (= in terms of physical harship/toil and/or perplexity/complexity/mental challenge involved) — [both the quality and the entity manifesting such]	**STEM 2:** (to be/manifest a) degree of effort (= in terms of mental challenge involved) **STEM 3:** (to be/manifest a) degree of effort (= in terms of organizational complexity)
CTE	(to be) something with a degree of effort	
CSV	(to be) a degree of effort	
OBJ	(to be) an entity/party impacted/affected by one's degree of effort	

-VH- **'DEGREE OF DESIGN / ORDER / ORGANIZATION / SYSTEMATIZATION'** Affix: **DSG** (use SUF/EXN/EXD, etc.)		
BSC	**STEM 1:** (to be/manifest a) degree of design/planning — [both the quality and the entity manifesting such]	**STEM 2:** (to be/manifest a) degree of order(liness) / organization / arrangement / structure **STEM 3:** (to be/manifest a) degree of systematization (= extent to which components/parts are arranged for optimal interrelationships/functionality)
CTE	(to be) something with a degree of design/planning	
CSV	(to be) a degree of design/planning	
OBJ	(to be) an entity/party impacted/affected by one's degree of design/planning	

-JH- **'DEGREE OF PHYSICAL CONTROL'** Affix: **CRL** (use SUF/EXN/EXD, etc.) NOTE: This root does not mean control in terms of hierarchical authority, oversight or influence.		
BSC	**STEM 1:** (to be/manifest a) degree of physical control (= degree of brute strength / agility / prowess / forbearance, etc. needed to physically handle/manipulate something, e.g., a tool, a machine, etc.) — [both the quality and the entity manifesting such]	**STEM 2:** (to be/manifest a) degree of physical fine motor control / dexterity / finesse (= degree of ability to manipulate/handle delicate or fragile entity) **STEM 3:** (to be/manifest a) degree of control via potential force (= having the ability/means/resources to physically hold/contain/restrain/restrict something from breaking free, running wild, causing damage, attacking, etc.)
CTE	(to be) something with a degree of physical control	
CSV	(to be) a degree of physical control	
OBJ	(to be) an entity/party impacted/affected by one's degree of physical control	

-PT- **'DEGREE OF RISK / DANGER'** Affix: **DNG** (use SUF/EXN/EXD, etc.)		
BSC	**STEM 1:** (to be/manifest a) degree of physical risk or danger (= threat or potential for physical harm) — [both the quality and the entity manifesting such]	**STEM 2:** (to be/manifest a) degree of emotional risk or danger **STEM 3:** (to be/manifest a) degree of socially-oriented risk or danger (= threat to one's status, plans, life circumstances, assumptions, finances, property, etc.)
CTE	(to be) something with a degree of physical risk or danger	
CSV	(to be) a degree of physical risk or danger	
OBJ	(to be) an entity/party impacted/affected by one's degree of physical risk or danger	

-ŘČ- **'DEGREE OF OPERATIONAL FUNCTIONALITY / ACTIVITY / EFFICACY'** Affix: **OPF** (use SUF/EXN/EXD, etc.)		
BSC	**STEM 1:** (to be/manifest a) degree of operational functionality; degree of being in working order (= extent to which something is functioning/operating/working normally or as per expectation or design) — [both the quality and the entity manifesting such]	**STEM 2:** (to be/manifest a) degree of activity (= amount / extent to which entities are acting/functioning/moving) **STEM 3:** (to be/manifest a) degree of efficacy (= extent to which an activity / operation achieves/accomplishes/delivers what it is supposed/expected to)
CTE	(to be) something with a degree of operational functionality; degree of being in working order	
CSV	(to be) a degree of operational functionality; degree of being in working order	
OBJ	(to be) an entity/party impacted/affected by one's degree of operational functionality; degree of being in working order	

-BŇ- **'DEGREE OF NATURAL CAPACITY / TALENT / SKILL'** Affix: **TAL** (use SUF/EXN/EXD, etc.)		
BSC	**STEM 1:** (to be/manifest a) degree of natural capacity/ability to — [both the quality and the entity manifesting such]	**STEM 2:** (to be/manifest a) degree of natural talent / aptitude / "gift" for; be adept at **STEM 3:** (to be/manifest a) degree of natural virtue (= desired behavioral quality/characteristic/attribute)
CTE	(to be) something with a degree of natural capacity/ability to	
CSV	(to be) a degree of natural capacity/ability to	
OBJ	(to be) an entity/party impacted/affected by one's degree of natural capacity/ability to	

-SP- **'DEGREE OF NEWNESS'** Affix: **NEW** (use SUF/EXN/EXD, etc.)		
BSC	**STEM 1:** (to be/manifest a) degree of newness (= freshly made/born/developed, newly created, not previously existing, e.g., *a new book just published; a new baby; a brand new shirt*) — [both the quality and the entity manifesting such]	**STEM 2:** (to be/manifest a) degree of newness (= not previously seen/functioning/experienced in current context, e.g., *the new student; a new book, i.e., that I've not owned/read before*) **STEM 3:** (to be/manifest a) degree of newness (= another / different / replacement / substitute / revision, e.g., *Please turn to a new page in your sketchpad; Hand me a new fork, this one is dirty*.)
CTE	(to be) something with a degree of newness	
CSV	(to be) a degree of newness	
OBJ	(to be) an entity/party impacted/affected by one's degree of newness	

-Ž- **'CAPACITY / ABILITY / CAPABILITY'** Affix: **-CPC-**		
BSC	**STEM 1:** (to be/manifest) the physical capability/ability to do something particular; to be able (i.e., or have the physical capability/ability) to do something particular	**STEM 2:** (to have) the physical stamina, fortitude or strength (needed) to do something **STEM 3:** (to have) the physical prowess or finesse or specialized physical skill (needed) to do something
CTE	(to be) the physical capability/ability; to be able (= have the or physical capability/ability)	
CSV	(to be) the particular (nature of the) physical capacity/ability needed	
OBJ	(to be) the task/endeavor requiring a certain physical capacity/ability to perform/accomplish	

-ÇN- **'SOURCE / ORIGIN / START* / BEGINNING*'** Affix: **OAU** * This root refers to origination, first-time occurrence, first-time causation only. For 'beginning/initiation' meaning 'activate, start/initiate from a stopped/dormant state, or the commencement of a common/recurring/pre-arranged activity, see the root -**TḶ**-.		
BSC	**Stem 1:** (to be) a state/act of origination; to come into being, to get one's start, to originate	**Stem 2:** (to be) a state/act of begining, initiating, starting; to begin, to start, to initiate **Stem 3:** (to be) a state/act of causation; to cause (i.e., primary/direct, not secondary or enabling cause)
CTE	(to be) the origin / source of something — [both the point in spacetime and the fact of being the initial manifestation of something] (to be) an act/process of origin(ating) or functioning as a/the source	
CSV	(to be) a physical act of originating; to originate [i.e., come into existence or be formed at a paticular place and time]	
OBJ	(to be) that whose source/origin is being identified	

The following root has the same Specification pattern as the root -**ÇN**- above:

-**RÇN**- **'ELEMENT / FUNDAMENTAL BASIS / FOUNDATIONAL PRINCIPLE'**
STEM 1: (to be/manifest a) basic element / fundamental starting point or principle
STEM 2: (to be/manifest a) fundamental reason for something; fundamental reason why; a basis for something
STEM 3: (to be a) first principle / foundational principle / axiom

-**R**- **'NEGATION / DENIAL / ABSENCE'** Affix: **NEG**		
BSC	**Stem 1:** (to be/manifest) an instance/act of negation; make something negative; to negate, i.e., to make/render the status of an entity so that it is as if it does not exist / manifest / apply / matter / act / take effect or have relevance [both the act/process of negation and the status of being (rendered) negative]	**Stem 2:** (to be) an instance/act of denial; to deny -- to act as if something is not true or non-existent or inapplicable **Stem 3:** (to be) an instance of being absent; an absence; to be absent
CTE	(to be) an instance/act of denial; to deny -- to act as if something is not true or non-existent or inapplicable	
CSV	(to be) the physical/tangible process of negation, whether verbal, physical, by written proclamation, by memory loss, etc.	
OBJ	(to be) that which is negated	

-RXW- ‘NULLIFICATION / ABROGATION / CONTRADICTION / CONTRAVENTION’
Stem 1: (to be) an instance/act of nullification or abrogation; to nullify, render null, abrogate, render as having no effect
Stem 2: (to be) an instance/act of contradiction; to contradict
Stem 3: (to be) an instance/act of contravention; to contravene

-PǍ- ‘ACCESS / PASSAGE(WAY) / DOORWAY / GATEWAY / PORTAL / ENTRY(WAY) / EXIT’ Affix: **ACS**		
BSC	**STEM 1:** (to be/manifest) a 2-way access point or passage way/gateway/portal; to pass through to a different area or physical space — [both the passage way itself and the act of passage]	**STEM 2:** (to be/manifest a) an entry point, a point of ingress, an entry portal, entry gateway, a passageway by which to enter **STEM 3:** (to be/manifest a) an exit point, a point of egress, an exit portal, a passageway by which to exit
CTE	(to be) an act of passage through, an act of accessing another physical space; to pass through, to enter or exit	
CSV	(to be) a passage way or access point or doorway/gateway/portal*	
OBJ	(to be) the party passing through an access point or doorway/gateway/portal *	

* **NOTE:** The word for a “door” itself would be the BSC stem for ‘access/passageway’ plus the MDF_2/3 affix ‘that which impedes/prevents X’, so that a ‘door’ is literally a “passage-impeding implement’. Consequently, one does not say “open door” in the language — one says “unrestricted passageway” or “unrestricted doorway.”

-ŢD- ‘PHYSICAL CONTACT’ Affix: **CTC**		
BSC	**STEM 1:** (to be) a state/act of physical contact between entities; to make contact; touch — [both the state/act of contact and its effect/consequence]	**STEM 2:** (to be) an application of physical pressure/force + effect/consequence **STEM 3:** (to be) a move or setting in motion via physical contact + resulting motion; to set in motion + resulting motion
CTE	(to be) the effect/consequence of a state/act of physical contact between entities	
CSV	(to be) a state/act of physical contact between entities; make contact; touch	
OBJ	(to be) a party/entity in physical contact with another	

-SK- ‘CONTAIN / HOLD / CONTENTS / ENCLOSURE / SURROUNDING / ENVELOPMENT’ Associated affix: **CNM**		
BSC	**STEM 1:** (to be) an instance/act/state of containment via gravity (e.g., basin); to contain (via gravity) [both the container and the state of containment]	**STEM 2:** (to be) an instance/act/state of containment via an enclosing/surrounding/enveloping/encircling medium/conveyance; to enclose, to suround, to envelop, to encircle, to circumscribe **STEM 3:** (to be) an instance of containment via imbuement/mixing/blending in (i.e., as an ingredient/component); to mix something into, to blend something into [see also the root -**SJ**-]
CTE	(to be) the state of containment (via gravity)	
CSV	(to be) the container (via gravity)	
OBJ	(to be) that which is contained (via gravity); the content(s)	

The following roots for specific types of containers have the following Specifications:

BSC	(to be) an instance/act/state of containing/holding/keeping something in a container
CTE	(to be) the state of containment
CSV	(to be) the container itself
OBJ	(to be) that which is contained (via gravity); the content(s)

-VSKW- 1. bag/sack 2. sealable plastic "baggie" 3. satchel

-VSKY- 1. purse 2. wallet 3. coin purse

-VSKL- 1. box 2. carton 3. custom-shaped carton/storage inlay

-VSKR- 1. can 2. drum 3. tank/cylinder

-VSKF- 1. bottle 2. jar 3. lidded pot

-VSKŢ- 1. bucket 2. tub 3. catch basin

-VSKÇ- 1. bin 2. crate 3. shipping container

-VSKV- 1. vial 2. flask 3. beaker

-VSKH- 1. tube 2. sealed wrapper 3. blister pack

-VŠT- 1. test tube 2. pipette 3. eye-dropper

-THW- 'COMPONENT / PART / INGREDIENT'		
BSC	**STEM 1:** (to be) the state/process of being/constituting a component/part/piece of an entity [i.e., a quasi- or semi-autonomous object/entity that can be (at least theoretically) separated from the whole or was such prior to being joined/connected/fused/built in to the larger whole]	**STEM 2:** (to be) the state/process of being/constituting an ingredient within a larger whole [i.e., an entity which (theoretically) has/had an autonomous existence prior to being merged with the whole but is now imbued within /inseparable from the whole **STEM 3:** (to be) a state/act/process of being an instruction
CTE	(to be) the state of constituting/being composed of/being made from a collection of parts	
CSV	(to be) an act of separating out a component/part from a whole; to isolate/have/hold/remove/separate a component/part	
OBJ	(to be) a particular component/part/piece of a whole	

NOTE: For positionally-defined component parts of an entity or system (e.g., the "front", "back", "side", "bottom" of an object), see Sec. 3.5.

2.1 TOOL USE

NOTE: Previous versions of this document indicated that the naming of most tools would be accomplished using various affixes attached to verbal stems indicating the desired action or result. Such an approach was subsequently determined to be unrealistic and not viable. Roots/stems for various tools are now given toward the end of this section.

2.1.1 Acts/Processes Accomplished by Means of Tools/Implements

This section lists roots associated with actions/states which readily lend themselves to the various instrumentative/tool-using affixes:

-CǍ- 'GRIND / PULVERIZE / GRAVEL / GRANULE / POWDER'			
	Stem 1	**Stem 2**	**Stem 3**
BSC	(be) an act of grinding to a gravel-like consistency; grind to a gravel-like consistency	(be) an act of grinding to a granular, sand-like consistency; grind to a granular consistency	(be) an act of grinding to a powder; pulverize
CTE	(be) the entity/object/substance that is (to be) ground to a gravel-like consistency	(be) the entity/object/substance that is (to be) ground to granular, sand-like consistency	(be) the entity/object/substance that is (to be) pulverized
CSV	(be) the mechanical process of an act of grinding to a gravel-like consistency	(be) the mechanical process of an act of grinding to a granular, sand-like consistency	(be) the mechanical process of an act of pulverizing
OBJ	(be) a piece of a gravel-like substance; (be) a piece of the product of an act of grinding to a gravel-like consistency	(be) a piece of a granular, sand-like substance; (be) a piece of the product of an act of grinding to a granular, sand-like consistency	(be) a grain of a powder(y substance)

-KÇ- 'CUT / STAB / CRACK / FISSURE' (i.e., quasi-linear puncturing or quasi-planar breaking of surface integument)		
BSC	**Stem 1:** (be) an act of cutting something with a (quasi-) bladed instrument or force ("cut" = to make a quasi-linear, parallel-to-the-surface break in the structural/cohesive integrity of the surface integument of an entity/object by means of a bladed (or blade-like) instrument)	**Stem 2:** (be) an act of stabbing (i.e., puncturing in a quasi-linear manner perpendicular to the surface integument of an object/entity using a sharp-pointed quasi-linear object). **Stem 3:** (be) an act of making a quasi-linear rent or crack or fissure in the surface integument of an object/entity; to (make a) rent, to (make a) crack
CTE	(be) the physical cut itself; a quasi-linear break in the structural/cohesive integrity of the surface integument of an entity/object	
CSV	(be) the physical act/process of cutting; to cut	
OBJ	(be) a blade (the portion of a knife/sword/axe/scissors, etc. that effectuates a cut)	

-ÇR- 'SEPARATION / SEVERANCE / DETACHMENT'		
BSC	**Stem 1:** (be) an act of separation; to separate something from something else; make something be physically situated apart from something else	**Stem 2:** (be) an act of detachment or physical disjoining/disconnection; to detach/disjoin/disconnect **Stem 3:** (be) an act of severance; to sever
CTE	(be) a state of being separated; be apart from something due to being/having been separated	
CSV	(be) an act of separation; the physical act of separating entities	
OBJ	(be) an implement, procedure, process, command, situation, etc. which effectuates an act of separation	

-ZKR- 'SLICE / PLANE / SHAVE / SHRED'		
BSC	**Stem 1:** (be) an act of slicing/sectioning something with a bladed (or blade-like) instrument (i.e., cut quasi-perpendicularly relative to the surface plane of an entity into quasi-planar piece/section)	**Stem 2:** (be) an act of making a planed/shaved slice; to plane; to shave off a piece (i.e., cut parallel to the surface plane of an entity into a quasi-planar piece/strip) **Stem 3:** (be) an act of shredding (i.e., using a specialized instrument to render a three-dimensional volume/portion of an object entity into quasi-linear ribbon-like or thread-like pieces)
CTE	(be) the physical gash/divide caused by an act of slicing; a quasi-planar break in the structural/cohesive integrity of the surface integument of an entity/object and the three-dimensional volume beneath	
CSV	(be) the physical act/process of slicing; to slice	
OBJ	(be) a blade (the portion of a knife that effectuates slice/section)	

-PPŠ- 'SCRATCH * / SCRAPE / PEEL'		
BSC	**Stem 1:** (be) an act of scratching, i.e., using a pointed or (quasi-)chisel-headed instrument or body part (e.g. fingernails) to imbue a surface with shallow quasi-linear rents; to scratch something [= to make a scratch/scratches in a surface, NOT to relieve an itch]	**Stem 2:** (be) an act of scraping, i.e., using an implement to remove a shallow layer of material from a surface; to scrape **Stem 3:** (be) an act of peeling; to peel
CTE	(be) a scratch	
CSV	(be) the physical act/process of scratching; to scratch	
OBJ	(be) the means (i.e., implement or body part) utilized to effectuate an act of scratching	

* for the meaning of scratching for grooming purposes or to relieve an itch, see **-VZD-;** for the sensation of being scratched, see -**VZW**-

-ZC- 'CHOP / DICE / MINCE'		
BSC	**Stem 1:** (be) an act of chopping, i.e., using a (quasi-)bladed instrument to make a quasi-planar gash/divide in the three-dimensional volume of an object/entity, perpendicularly to its surface integument)	**Stem 2:** (be) an act of dicing (i.e., slicing into quasi-cubic pieces) **Stem 3:** (be) an act of mincing (i.e., slicing in cross-hatched fashion to the point of making an entity's texture a slurry or paste-like)
CTE	(be) the physical gash/divide/separation caused by an act of chopping; a quasi-planar gash/divide in the structural/cohesive integrity of the 3-dimensional volume of an entity/object and the three-dimensional volume beneath	
CSV	(be) the physical act/process of chopping; to chop	
OBJ	(be) a blade (the portion of a knife/cleaver/axe, etc. that effectuates an act of chopping)	

-ẒČ- 'CARVE / GUT / EVISCERATE'		
BSC	**Stem 1:** (be) an act of (surface-)carving, i.e., using a (quasi-)bladed instrument to remove (specific) surface material from an object entity)	**Stem 2:** (be) an act of interior carving, carving out, i.e., careful/precise removal of specific interior tissue/volume of an entity using a (quasi-)bladed instrument; to carve out **Stem 3:** (be) an act of gutting or evisceration; to gut/eviscerate (i.e., remove most or all of the interior volume of an entity
CTE	(be) the surface area/volume removed due to an act of (surface-)carving	
CSV	(be) the physical act/process of surface-carving; to carve the surface of	
OBJ	(be) a blade (the portion of carving implement that effectuates an act of carving)	

-ẒF- 'PIERCE / PUNCTURE / BREACH / HOLE / TUNNEL'		
BSC	Stem 1: (be) an act of piercing through, puncturing through, punching through, breaching, breaking through, tunneling (i.e., create an access point/passageway in a 2- or 3-dimensional medium to the other side or to another separate space)	**Stem 2:** (be) an act of removing material by burrowing or tunneling* [Specification pattern is like Stem 2 of the root 'DIG'] **Stem 3:** (be) an act of leaving behind a hole, puncture, tunnel, i.e., an access point or passageway through a medium to another side or separate space [Specification pattern is like Stem 3 of the root 'DIG']
CTE	(be) the material removed via an act of piercing, puncturing	
CSV	(be) an act of piercing, puncturing	
OBJ	(be) the appendage/implement/tool used for piercing, puncturing	

-JJ- 'DIG / HOLE' *			
	Stem 1	**Stem 2**	**Stem 3**
BSC	(be) an act of digging; to dig, i.e., to use an implement or one's appendage to break the surface of the 3-dimensional volume of a substance or entity, for the purpose of remove an amount of material from the volume, leaving behind a concave depression/pit/hole*)	(be) the act of removing material via digging	(be) an act of creating/leaving behind a depression or concavity by means of an act of digging; to make/create/leave a depression/concavity/pit/hole* by means of digging
CTE	(be) the state/process undergone by an entity when material is dug from it and a depression or concavity left behind	(be) the state/process undergone by an entity when material is removed from it by digging	(be/have) a depression or concavity created by an act of digging
CSV	(be) a physical act of digging; remove material by digging leaving a depression or concavity behind	(be) a physical act of digging; remove material by digging	(be) an act of making a depression/concavity/hole* by digging; make a depression or concavity by digging
OBJ	(be) the appendage/implement used for digging	(be) the material removed by an act of digging	(be) the medium/firmament in which a depression or concavity is made

* Use of the word "hole" in translating the stems of this root is within the narrow context of being a synonym for "scooped-out depression/concavity within a 3-dimensional medium; it does not mean "hole" as an access point between two spaces or through some two- or 3-dimensional medium as in "a hole in my jeans" or "a hole through the wall" (use the root **-ẒF-** below instead).

NOTE: Use the above root with the **SVS** affix to derive words for scoop, excavate and to derive words for 'drill', 'bore', 'cavern'. Combine it with other appropriate morphology to render the word for 'delve' and 'worm one's way into'.

-RẔČ- 'EXCAVATION / DREDGING / MINING'		
BSC	**STEM 1:** (to be) an act/process of excavation; to excavate/exhumation [i.e., remove a relatively large amount material from a solid substrate for the purpose of reshaping that substrate or to find something within the removed material	**STEM 2:** (to be) an act/process of dredging; to dredge (i.e., remove material from an underwater surface to improve the use of that substrate as a conduit/passage or to find something within the material removed] **STEM 3:** (to be) an act/process of mining; to mine [i.e., to use an implement or machinery to remove desired ore or other desired material from a surrounding subterranean or underwater subtrate]
CTE	(to be) the material removed/carved out from an act/process of excavation/exhumation	
CSV	(to be) a physical act of; to physically perform an act of excavation/exhumation	
OBJ	(to be) an implement/tool/process/device used in the act/process of excavation/exhumation	

-ẒDW- ‘BURROW / HOLLOW OUT’		
BSC	Stem 1: (be) an act of hollowing out, burrowing/tunneling*; to burrow/tunnel (i.e., create a tube-like space within a 3-dimensional medium	**Stem 2:** (be) an act of removing material by hollowing out, burrowing or tunneling* [Specification pattern is like Stem 2 of the root ‘DIG’] **Stem 3:** (be) an act of leaving behind a hollow, burrow or tunnel* [Specification pattern is like Stem 3 of the root ‘DIG’]
CTE	(be) the material removed via an act of hollowing out or burrowing	
CSV	(be) an act of hollowing out or burrowing	
OBJ	(be) the appendage/implement/device/machine used for hollowing/burrowing/tunneling	

* This root refers to the creation of a tube-like space within a 3-dimensional medium; it does not necessarily imply that the tube-like space functions as a conduit or passageway to another separate space or medium -- thus, use of the translation “tunnel” here is in a limited context. If one wishes to signify a tunnel-like conduit connecting two separate spaces, use the root **-ẒF-** above instead.

-GṬ- ‘TIGHTNESS / FITTEDNESS / COMPATIBILITY’		
BSC	**Stem 1:** (be) a state of being tight; tightness; be/make tight, i.e., difficult to disjoin or open or separate	**Stem 2:** (be) a state of one entity physically fitting into/with another **Stem 3:** (be) a state of one entity being physically compatible with another so that they operate/function/exist better as, or ideally as, a unit
CTE	(be) the state undergone by an entity or entities having a physical bond/connection/juncture/fitting which is tight	
CSV	(be) the physical act of tightening	
OBJ	(be) the juncture itself which is (to be) tight(ened)	

-FŠ- ‘FITTING / ACCOMMODATION / NICHE’		
BSC	**Stem 1:** (be) an act/instance of one entity physically fitting into/with another so that one is “carried” along as the other moves/operates	**Stem 2:** (be) an act/instance of accommodating something (i.e., finding a place for something within a surrounding context so that it can become a functioning/contributing/associated part of that context) **Stem 3:** (be) a act/instance of finding/establishing a niche for something (i.e., establishing an existence/lifestyle/role which becomes part of and contributes to the encompassing contextual environment)
CTE	(be) the state of fitting into another	
CSV	(be) the physical act of fitting	
OBJ	(be) the juncture itself which one establishes or seeks to establish by an act of fitting something into something else	

-JP- 'STRAIGHTENING / ALIGNMENT'		
BSC	**Stem 1:** (be) an act/instance of making something straight/rectilinear [by design]; make something straight by design (i.e., make rectilinear as opposed to making it bent, curved, crooked)	**Stem 2:** (be) an act/instance of straightening something out; to straighten out (i.e., fix something that was crooked/bent/curved/warped by making it rectilinear/straight as it should be) **Stem 3:** (be) a act/instance of aligning/alignment; to align (i.e., make an entity or collection of entities line up in a rectilinear shape/pattern for aesthetic reasons or for proper/optimal functioning)
CTE	(be) the state of being straight/rectilinear	
CSV	(be) the physical act of straightening; make something straight/rectilinear	
OBJ	(be) that which is straight(ened)	

-CGV- 'JUNCTURE OF KINEMATIC PAIR'		
BSC	**Stem 1:** (be) an act/instance of one entity physically being connected to another via a structural juncture or pivot point	**Stem 2:** (be) an act/instance of being able to pivot due to the placement of a structural juncture **Stem 3:** (be) a act/instance of structural/physical tension caused by pivoting within the constraints allowed by a structural juncture
CTE	(be) the state of being connected to another via a structural juncture or pivot point	
CSV	(be) the physical act of connecting via a structural juncture/pivot point	
OBJ	(be) the implement itself which establishes such a juncture or pivot point (e.g., a hinge, a universal joint, a ball joint, prismatic joint, cylindrical joint, snake-like flexible joint, etc.)	

-TKF- 'MECHANICAL JOINT/PIVOT POINT FOR KINEMATIC PAIR' (allowing limited movement in one or two dimensions)		
BSC	**Stem 1:** (be) a connection utilizing a prismatic joint [allowing linear back-and-forth (or up-and-down) movement only]	**Stem 2:** (be) a hinge-like joint allowing relative rotation about a single axis, a.k.a. pin joint, revolute joint **Stem 3:** (be) a cylindrical joint allowing circular rotation upon an axis
CTE	(be) the state of being connected via a prismatic joint	
CSV	(be) the physical act of connecting via a prismatic joint	
OBJ	(be) the prismatic joint implement itself	

-LTKF- 'MECHANICAL JOINT/PIVOT POINT FOR KINEMATIC PAIR' (allowing 3-dimensional movement)		
BSC	**Stem 1:** (be) a ball-and-socket type of bending/swivel point or joint of an entity	**Stem 2:** (be) a universal joint **Stem 3:** (be) a snake-like flexible entity
CTE	(be) the state of being connected via a ball-and-socket type of joint	
CSV	(be) the physical act of connecting via a ball-and-socket type of joint	
OBJ	(be) the ball-and-socket type of joint implement itself	

-MZK- 'ENERGY OUTPUT DIFFERENTIAL OR REDIRECTION VIA GEAR MECHANISM' (allowing 3-dimensional movement)		
BSC	**Stem 1:** (be) a change in ratio of speed or energy output caused by a geared differential mechanism	**Stem 2:** [same as Stem 1 except that the gear mechanism is designed to redirect energy into a different direction as opposed to a change in energy output ratio or speed] **Stem 3:** [combination of Stems 1 and 2; both change in energy output ratio or speed plus change in direction of energy output/utilization
CTE	(be) the state of a having one's energy output or speed changed via a gear mechanism	
CSV	(be) the physical act of changing the kinematic ratio or speed of energy output via a gear mechanism	
OBJ	(be) the gear mechanism which creates a change in kinematic ratio of energy output or speed	

-ŘN- 'SUSPEND / HANG (UP) / DANGLE / PENDENCY'		
BSC	Stem 1: (be) an act/instance of hanging something (up); to hang, suspend (i.e., supporting something continuously against gravity	**Stem 2:** (be) an act/instance of dangling something; to dangle **Stem 3:** (be) an act/instance of overhanging, jutting (out)
CTE	(be) a state of hanging/suspension; be hanging (up/on) something	
CSV	(be) the physical act of hanging something up or onto	
OBJ	(be) the implement/connecting medium used to keep something hanging (e.g., hook, knotted string/cord/rope, strap, harness, etc.)	

Morphological derivatives: swing

-ḶX- 'FLOTATION / HOVERING / DRIFTING / ALOFT'		
BSC	Stem 1: (to be) an act/state/instance of flotation/drifting upon the surface of a liquid or semi-liquid medium; to float upon something, to drift upon or along something [i.e., subject to uncontrolled, unpowered, random drifting]	**Stem 2:** (to be) an act/instance of being/floating/drifting aloft; be aloft, soar, swim within or through, i.e., to have free movement or move freely within a 3-dimensional medium (e.g., air, water, space, etc.); **Stem 3:** (to be) an act/instance of hovering in place within a medium (e.g., air, water, space, etc.); to hover [i.e., controlled station-keeping in one place within a 3-dimensional medium against the force of gravity]
CTE	(to be) that which floats or drifts	
CSV	(to be) an act/state of flotation/drifting upon a (semi-)liquid surface; to (make) float/drift, to be afloat, to be adrift	
OBJ	(to be) the liquid or semi-liquid surface upon which one floats/drifts	

-LŘ- 'FLATTEN / SMOOTH / LEVEL'		
BSC	Stem 1: (be) an act/instance of flattening;, make flat(ter), render flat(ter) (i.e., make have a more planar surface without significant perturbations)	**Stem 2:** (be) an act/instance of making smooth (i.e., not having surface roughness, projections, or perturbations) **Stem 3:** (be) an act/instance of make (more) level or even or parallel (i.e., make a quasi-linear or quasi-planar surface be as parallel to another surface or measuring line/plane as possible)
CTE	(be) a state of being flat	
CSV	(be) the physical act of flattening	
OBJ	(be) the implement/tool used to flatten something	

Morphological derivatives: polish

-GF- 'HOLD TOGETHER / HOLD STILL / BIND / RESTRAIN'		
BSC	**Stem 1:** (be) an act/instance of holding something together on a long-term or (quasi-) permanent basis (i.e., so that it doesn't separate, come apart, or disassemble)	**Stem 2:** (be) an act/instance of binding something (i.e., keeping it from coming apart, coming loose, or escaping by internal motion in-situ) **Stem 3:** (be) an act/instance of holding something still; restrain (i.e., keeping it from moving from one place to another; keeping it in one place)
CTE	(be) a state of being held together	
CSV	(be) the physical act of holding together	
OBJ	(be) the implement/tool/medium used to keep something held together (e.g., clip, clasp, clamp, vice, clothespin, string/cord, rubber band, bookend(s), etc.)	

This root refers to long-term, (quasi-)permanent states of holding things together. For incidental, temporary contexts, see the next root below.

-TV- 'HOLD UP / SUPPORT / BEAR UP / KEEP ALOFT'		
BSC	**Stem 1:** (be) an act/instance of supporting something up or holding something up so that it does not collapse; to support/hold up/bolster something against collapse	**Stem 2:** (be) an act/instance of bearing something so that it remains raised high, to hold high, to bear up (i.e., keeping it from falling or returning to a lowered height) **Stem 3:** (be) an act/instance of holding something aloft; to hold aloft, keep aloft (i.e., keeping it from returning/falling to the ground)
CTE	(be) a state of being held up or supported against collapse	
CSV	(be) the physical act of holding up or supporting against collapse	
OBJ	(be) the implement/tool/medium used to keep something held up (e.g., crutch, strut, hanging wire/rope/string, negative pressure, etc.)	

-ḌJ- 'ERECT / BRING TO A VERTICAL POSITION / RAISE UP' (i.e., perpendicular to the ground)		
BSC	**Stem 1:** (be) an act/instance of erecting something to a vertical position perpendicular to the ground (or parallel to the direction of gravity)	**Stem 2:** (be) an act/instance of letting rise up to an erect position **Stem 3:** (be) an act/instance of raising something up to a high(er) level, involving a use of energy and manipulation (as opposed to letting it rise freely on its own); to physically raise something to a high(er) level
CTE	(be) a state of erecting	
CSV	(be) the physical act of erecting	
OBJ	(be) the means/implement/procedure/command used to erect something (e.g., pulley)	

-MJ- 'FELL / BRING DOWN / LOWER'		
BSC	**Stem 1:** (be) an act/instance of making something fall via gravity from its (quasi-) perpendicular stance; to fell	**Stem 2:** (be) an act/instance of letting something fall; to drop **Stem 3:** (be) an act/instance of lowering something down to (quasi-) ground level from on high, involving a use of energy and manipulation (as opposed to letting it fall freely via gravity); to physically lower something to one's level
CTE	(be) a state of falling	
CSV	(be) the physical act of felling	
OBJ	(be) the means/implement/procedure/command used to make something fall	

-PD- 'GRAB / GRASP / PICK UP / PICK OUT / HANDLE / MANIPULATE'		
BSC	**Stem 1:** (be) an act/instance of grabbing/grasping something with one's hand or manipulative appendage(s)	**Stem 2:** (be) an act/instance of picking something up or picking something out of something else (i.e., physically isolating and removing something from its surface/medium/substrate) **Stem 3:** (be) an act/instance of handling/manipulating something with one's hands or manipulative appendage(s)
CTE	(be) a state of being in another's grasp	
CSV	(be) the physical act of grabbing/grasping	
OBJ	(be) the hand or appendage with which one grabs/grasps	

-SČ- 'BREAK APART / BREAK INTO PIECES / CRUMBLE'		
BSC	**Stem 1:** (be) an act/instance of breaking something apart or breaking something open	**Stem 2:** (be) an act/instance of breaking something into multiple pieces **Stem 3:** (be) an act/instance of crumbling something into non-discrete pieces (i.e., powder, dust, gravel, crumbs, etc.)
CTE	(be) a state of being broken apart or broken open	
CSV	(be) the physical act of breaking apart or broken open	
OBJ	(be) the implement/tool/medium used to break something apart (e.g., pickaxe, crowbar, sledgehammer, mallet, chisel, etc.	

-RKS- 'SPLIT IN HALF / DIVIDE INTO PARTS / SPLIT OFF A PIECE'		
BSC	**Stem 1:** (be) an act/instance of splitting/dividing something in two/half; to split into two, to split in half	**Stem 2:** (be) an act/instance of splitting/dividing something into three or more parts/portions **Stem 3:** (be) an act/instance of splitting/dividing off a smaller piece of something from the larger whole
CTE	(be) a state of being split/dividing in half	
CSV	(be) the physical act of splitting/dividing in half	
OBJ	(be) the implement/tool/medium used to split/dividing something in half	

-ZN- 'MANUAL MANIPULATION'		
BSC	**Stem 1:** (be) an act/instance of incidentally or temporarily holding something together (i.e., so that it doesn't separate, come apart, or disassemble)	**Stem 2:** (be) an act/instance of incidentally or temporarily touching, holding or handling something for the purpose of carrying, conveying, protecting, inspecting, displaying, admiring it, etc. **Stem 3:** (be) an act/instance of manipulating something for the purpose of changing/activating/reshaping/stimulating/operating it (e.g., by squeezing, twisting, pressing, pushing, pulling, rubbing, etc.)
CTE	(be) a state of being incidentally or temporarily held together	
CSV	(be) the physical act of incidentally or temporarily holding together	
OBJ	(be) the implement/tool/medium used to keep something incidentally or temporarily held together (e.g., pliers, tweezers, forceps, etc.)	

-ČN- 'MOVE APART / DRAW APART / OPEN'		
BSC	**Stem 1:** (be) an act/instance of moving two or more entities apart (to create a gap or opening or passageway)	**Stem 2:** (be) an act/instance of pushing (i.e., applying force/pressure toward) an entity in order to move it apart so as to create a gap, opening, or passageway **Stem 3:** (be) an act/instance of pulling (i.e., drawing force/pressure away from) an entity in order to move it apart so as to create a gap, opening, or passageway
CTE	(be) a state of being/having been moved apart	
CSV	(be) the physical act of moving two or more entities apart (e.g., via pulling, pushing, or other force or means)	
OBJ	(be) the means/implement/procedure used to effectuate an act of moving apart	

-MC- ‘JUNCTION / CONNECTION / FASTENING / BONDING’		
BSC	**Stem 1:** (be) an act/instance of joining two or more entities together so that they become a set (i.e., bringing two or more entities spatially close or in contact so that they move/function or can be considered as a set)	**Stem 2:** (be) an act/instance of connecting or attaching; a connection; to connect or attach (i.e., utilize a tangible part of an object, or a dedicated implement, e.g., wire, rope, glue, staple, etc., to form a physical link between two or more entities **Stem 3:** (be) an act/instance of fastening/bonding; a bond; to fasten or bond (i.e., utilize a dedicated implement or means, e.g., wire, rope, glue, staple, etc., to physically attach two or more entities together in a (quasi-)permanent or long-term bond so that one entity is now considered part of another or that the entities are considered a unit)
CTE	(be) a state of being a member of a set	
CSV	(be) the physical act of joining together into a set	
OBJ	(be) the juncture itself which one establishes or seeks to establish by an act of joining	

The Specification pattern of the above root applies to the following root as well:

-RMC- ‘LINKAGE / UNIFICATION / FUSION / MERGING / COALESCING / MELDING’
Stem 1: (be) an act/instance of linking; a link, linkage; to link (i.e., a joining together of entities with a focus on the individual members of the resulting set maintaining their individual autonomy/identity)
Stem 2: (be) an act/instance of unifying, uniting (i.e., joining entities together so that they are objectively or subjectively a unit, a single gestalt entity
Stem 3: (be) an act/instance of fusion/merging/coalescing/melding (i.e., joining entities together so that each loses its individual boundaries or compositional integrity and becomes part of or mixed with others into a (quasi-)indivisible physical whole.

-JṬ- ‘PENETRATION / DRIVING INTO / INJECTION’		
BSC	**Stem 1:** (be) an act/instance of penetration through a (quasi-)two-dimensional surface (i.e., the passing of an external entity into or through a quasi-planar surface in which the structural integrity of the surface is broken/interrupted)	**Stem 2:** (be) an act/instance of driving into (i.e., forceful insertion or penetration through a resistant/hard surface by breaking/interrupting its surface integrity and passing into the resistant/hard three-dimensional volume beyond via pure force and quasi-violent breaking/distortion/displacement of the structural integrity of that volume **Stem 3:** (be) an act/instance of injection
CTE	(be) a state of being/having been penetrated	
CSV	(be) the physical act of penetration	
OBJ	(be) the entity/object employed to penetrate (e.g., a nail, screw, blade, pointed object, etc.)	

NOTE: This root refers only to the act/process of penetration of an external entity through a two-dimensional medium or into a three-dimensional medium; it is not focused on the resulting state of interiority itself. Thus, for translations of English words focused on the resulting state of interiority rather than the interruption of the surface integrity, e.g., *‘insert, infuse, immerse, instill, imbue, implant’*, use an appropriate SPATIO-TEMPORAL Root instead, e.g., -**XW**-, -**XL**-, -**CW**-, -**ṬP**-, -**ḌB**- and/or appropriate Spatio-Temporal affixes associated with these roots.

-ŇÇ- 'EXTRACTION'		
BSC	**Stem 1:** (be) an act/instance of extraction; to extract, draw out, draw forth (i.e., to remove or separate out something from something else by using physical force, causing it to exit from some surrounding or encasing medium or substrate, e.g., pulling, sucking, reversal of pressure, etc.)	**Stem 2:** (be) an act/instance of removing something from some surrounding medium/substrate by physically invading/cutting into that medium and forcibly extracting (e.g., surgery, digging for something, immersing one's hands or appendage(s) into and grabbing, etc.) **Stem 3:** (be) an act/instance of obtaining or drawing out something from a surrounding medium/substrate by chemical or other non-manual means (e.g., osmosis, titration, burning, chemical reaction, etc).
CTE	(be) a state of being/having been extracted, drawn out, withdrawn out of something	
CSV	(be) the physical act of extraction; to do something that physically extracts, draws something forth/out	
OBJ	(be) the entity/object employed to extract	

-FFR- BENDING / DISTORTION / TWISTING / WARPING		
BSC	**Stem 1:** (to be) an act/instance/process of bending; to undergo bending	**Stem 2:** (to be) an act/instance/process of distortion; to undergo distortion [i.e., physical bending/twisting of an object/entity into a different, less functional shape] **Stem 3:** (to be) an act/instance/process of twisting or warping; to undergo twisting or warpage
CTE	(to be) a state of being bent	
CSV	(to be) the physical act of bending; to bend	
OBJ	(to be) the part/area of an entity/object that is bent	

-RTV- TEMPLATE-BASED FORMATION / DRAWING (= mechanical process) / EXTRUSION		
BSC	**Stem 1:** (to be) an act/instance/process of forming/shaping/molding an object/entity by means of a template or die	**Stem 2:** (to be) an act/instance/process of drawing [i.e., the mechanical process of pulling a material through a die for the purpose of making it into a particular shape or form, e.g., the drawing of wire] **Stem 3:** (to be) an act/instance/process of extrusion [i.e., the mechanical process of pushing a material through a die for the purpose of making it into a particular shape or form]
CTE	(to be) a state of being/having been formed/shaped/molded via a die or template	
CSV	(to be) the physical act of formation via a die or template; to form something via extrusion	
OBJ	(to be) a die or template used for forming/shaping/molding something	

-DKH- BULGING / PROLAPSE / ERUPTION / EXPLOSION		
BSC	**Stem 1:** (to be) an act/instance/process of bulging outward	**Stem 2:** (to be) an act/instance/process of prolapsing **Stem 3:** (to be) an act/instance/process of eruption or explosion
CTE	(to be) a state of bulging outward	
CSV	(to be) the physical act of bulging; to make something bulge outward	
OBJ	(to be) the part/area of an entity/object that bulges outward	

-DBW- COLLAPSING / SINKING / IMPLOSION		
BSC	**Stem 1:** (to be) an act/instance/process of collapsing inward	**Stem 2:** (to be) an act/instance/process of sinking inward or downward **Stem 3:** (to be) an act/instance/process of implosion/imploding
CTE	(to be) a state of collapsing inward	
CSV	(to be) the physical act of bulging; to make something collapse inward	
OBJ	(to be) the part/area of an entity/object that collapse inward	

-ḌǍ- 'STRETCH / ELONGATE / SPREAD' NOTE: This root refers to expansion in size **without** an entity's individual component members/contents/nodes themselves expanding, but rather only the space [or underlying substrate or connective medium] between them		
BSC	**Stem 1:** (be) an act/instance of linear (i.e. unidimensional) stretching or elongation; to stretch, to elongate [i.e., to expand in size along a single dimensional axis without the individual component molecules/members/contents themselves expanding, but rather only the space between them)	**Stem 2:** (be) an act/instance of planar (two-dimensional) stretching, expansion or spreading; to spread (two-dimensionally), to expand in terms of surface area **Stem 3:** (be) an act/instance of quasi-spheroidal (3-dimensional) stretching or expansion; to expand (3-dimensionally)
CTE	(be) a state of being/having been [unidimensionally] stretched or elongated	
CSV	(be) the physical act of [unidimensionally] stretching or elongation	
OBJ	(be) the means/implement/procedure used to [unidimensionally] stretch	

The Specification pattern of the above root applies to the following root as well:

-ŘḌǍ- 'STRETCH / ELONGATE / SPREAD [under tension]'
The three stems of this root are identical to the stems of the above root -**ḌǍ**- except that they connote a concurrent physical strain/tension with the act of stretching/spreading/expansion, so that the elongated/expanded entity is subject to potential recoil or collapse back to its original state if the stretching/expansion force being applied is terminated.

-CY- ‘CLUSTERING / CLUMPING / FLOCK / SWARM / MOB’		
BSC	**Stem 1:** (to be) an act/process of entities (e.g., molecules, bacteria, rodents, leaves, etc.) clustering/clumping together into a cluster/clump; to cluster/glump together	**Stem 2:** (to be) an act/process of entities gathering/flocking together into a large group/flock **Stem 3:** (to be) an act/process of entities swarming together into a crowd/mob or swarm
CTE	(to be) a cluster/clump	
CSV	(to be) an act/process of clumping/clustering together, to participate in a process of clumping/clustering, to cluster, to clump	
OBJ	(to be) the entity/means/procedure utilized to achieve clustering/clumping	

-D- ‘GROUP / GATHERING / ASSEMBLY / MEETING / CONGREGATION / CONFERENCE / CONGRESS’		
BSC	**Stem 1:** (to be) an act/process of entities (e.g., people, animals, molecules, etc.) assembling or gathering together into a group; to assemble/gather together/group	**Stem 2:** (to be) an act/process of entities congregating/meeting together for a purpose or shared activity; to meet (up), to meet together, to congregate **Stem 3:** (to be) an act/process of entities convening for an official/formal conference or congress
CTE	(to be) a state of entities assembled/gathered together in a group	
CSV	(to be) an act/process of gathering/bringing entities together, to perform an act which gathers/helps to gather entities together, to muster, to round up	
OBJ	(to be) the entity/means/procedure utilized to achieve gathering/grouping	

-CḌ- ‘ADHESION / STICKINESS / AGGREGATION / AGGLOMERATION’		
BSC	**Stem 1:** (be) an act/instance of adhesion; to adhere (i.e., something sticking to something else via some adhesive medium	**Stem 2:** (be) an act/instance of stickiness; be sticky; to stick (i.e., to physically adhere to something via a sticky medium where the focus is on the textural stickiness of the adhesion, not necessarily the fact of adhering) **Stem 3:** (be) an act/instance of aggregation/agglomeration/conglomeration (i.e., bringing separate entities physically together for a unified purpose, where the focus is on their functioning together as opposed to the nature of the physical connection itself)
CTE	(be) a state of adhesion	
CSV	(be) the physical act of adhering	
OBJ	(be) an entity/substance used for adhering; an adhesive	

-TḶ- 'COMMENCEMENT / ACTIVATION / INITIATION / INSTIGATION' (NOTE: This root refers to 'beginning/initiation' meaning 'activate, start/initiate from a stopped/dormant state, or the commencement of a common/recurring/pre-arranged activity; for the 'beginning/initiation' meaning "origination, first-time occurrence, first-time causation", see the root -**ÇN**-.		
BSC	**Stem 1:** (be) an act/instance of commencing/starting an act/event/state (i.e., the transition moment/period between a non-active vs. active state	**Stem 2:** (be) an act/instance of initiating/activating something; to activate, to initiate, to start up (i.e., the act/process/procedure necessary to bring energy to a non-active state/entity so that it becomes active, e.g., starting a motor, activating a device, initiating a complex procedure, etc.) **Stem 3:** (be) an act/instance of instigation; to instigate (i.e., set in motion a series of events or arranging a precursor state/act/event that will lead to a specific outcome)
CTE	(be) a state of commencement/starting	
CSV	(be) the physical act of commencing/starting	
OBJ	(be) an entity/object used to commence/start something	

-PḶ- 'TURN / VEER / PITCH'		
BSC	**Stem 1:** (be) an act/instance of turning something; to turn something (i.e., cause an object to move in an arc-like motion so that its direction/orientation relative to the surrounding space//surface/medium/substrate changes	**Stem 2:** (be) an act of veering; to veer (i.e., a sudden, unexpected turn) **Stem 3:** (be) an act of changing one's pitch; to pitch upward or downward (i.e., to turn one's orientation on a vertical axis upward or downward relative to one's (quasi-)static horizontal frame of reference)
CTE	(be) a state of being turned	
CSV	(be) the physical act of turning	
OBJ	(be) an entity/object used to make something turn	

NOTE: Use this root with various spatial affixes to add specific detail, e.g., SPR, CRC, SPM, SWM, SPN, IPE, IPF, PFL, etc.

The Specification pattern of the above root applies to the following root as well:

-RPḶ- 'SPINNING / REVOLVING / ORBIT / ROTATION'
Stem 1: (be) an act/instance of spinning/revolving (i.e., revolving in place on one's own rotational axis)
Stem 2: (be) an act/instance of revolving/orbiting; to revolve/orbit (i.e., to follow a (quasi-)circular motion around another entity/object)
Stem 3: (be) an act of rotation; to rotate (e.g., twisting a handle or turning a steering wheel)

-LXR- 'ROLL / WHEELED TRANSLATIVE MOTION / WIND UP OR AROUND'		
BSC	**Stem 1:** (be) an act/instance of rolling an entity [i.e., vertical spinning motion relative to a quasi-stationary substrate resulting in vectored movement]	**Stem 2:** (be) an act/instance of moving an object/entity translatively by means of wheel(s) **Stem 3:** (be) an act/instance of winding something into a coil (e.g., string, wire, rope, etc.) by continuous circular turning; to wind (up) into a coil (e.g., rolling up a carpet)
CTE	(be) a state of rolling	
CSV	(be) the physical act of rolling; to roll something or to make something revolve	
OBJ	(be) an entity/object used for rolling (e.g., log, wheel, bobbin, reel, spool, etc.)	

-ČPH- 'STRIKE / HIT / BLOW / KNOCK / TOPPLE'		
BSC	Stem 1: (be) an act/instance of striking/hitting something using one's body/appendages or an implement/device used as an extension of one's body/appendages, with the intention of changing its location, placement, movement, state, arrangement, etc. (e.g., as in hitting a ball in sport, or striking a nail with a hammer)	**Stem 2:** (be) an act/instance of striking/hitting something, using one's body/appendages or an implement/device used as an extension of one's body/appendages, either as an offensive attack or in self-defense, with the intention of causing physical harm or physically thwarting a perceived attack **Stem 3:** (be) an act/instance of knocking something down or over by means of striking, hitting or delivering a blow, whether by using one's body/appendages or an implement/ device used as an extension of one's body/appendages; to knock over, to knock down, to push/shove to the ground, to physically topple something
CTE	(be) a state of being struck/hit	
CSV	(be) the physical act of striking/hitting	
OBJ	(be) the implement/tool used to strike/hit something	

Derivations: tap, punch, beat, kick, pat, tamp, cudgel, nudge

-ŽN- 'IMPULSE / PUSHING / PROPULSION'		
BSC	**Stem 1:** (be) an act/instance of pushing something; to push (i.e., make something move by exerting an impulsive force upon it via one's hands or appendages or via a manually-controlled object/device)	**Stem 2:** (be) an act/instance of shoving, bumping, butting, prodding (i.e., using one's hands or body, or a manually-controlled extension thereof, to deliver a sudden push to another entity for the ostensible purpose of making it move out of one's path or to gain the entity's attention) **Stem 3:** (be) an act/instance of thrusting or propulsion; to make something (or oneself) move quickly via a sudden, violent use of impulsive force
CTE	(be) a state of being pushed	
CSV	(be) the physical act of pushing	
OBJ	(be) an entity/object used to push	

-ŽR- 'THROWING / LAUNCHING / SHOOTING'		
BSC	**Stem 1:** (be) an act/instance of throwing something (i.e., using one's arms or other appendage to suddenly and forcefully impel/project an object into the surrounding air/space where its path of motion then becomes subject to gravitational/environmental forces)	**Stem 2:** (be) an act/instance of discharging, firing off, shooting a projectile from a firearm or other device using violently explosive/propulsive force; to shoot, fire off, discharge **Stem 3:** (be) an act/instance of launching an airborne or spaceborne projectile, rocket or craft using a large amount of force (e.g., chemical rockets, nuclear reaction, anti-gravity, magic, etc.); to launch a weapon/missile/aircraft/spacecraft
CTE	(be) a state of being thrown	
CSV	(be) the physical act of throwing	
OBJ	(be) an entity/object used to throw	

-ÇV- 'TRACTION / DRAWING / PULLING / DRAGGING'		
BSC	**Stem 1:** (be) an act/instance of pulling or drawing (i.e., make something move by exerting a tractive force upon it via one's hands or appendages or via a manually-controlled object/device)	**Stem 2:** (be) an act/instance of dragging, tugging, towing, lugging (i.e., to pull/draw with resistance from the entity being towed either due to gravity, friction, or the pulled entity's own resistive efforts) **Stem 3:** (be) an act/instance of hauling, trawling, raking (i.e., using pulling/drawing force to carry/convey/transport a load/object/entity)
CTE	(be) a state of being pulled or drawn	
CSV	(be) the physical act of pulling or drawing	
OBJ	(be) an entity/object used to pull or draw, e.g., rope, reins, chain, etc.	

Morphological derivative: hoist

The Specifications of the above root are applied to the following root:

-RÇV- 'MANUAL MEANS OF CARRYING & MOVING A LOAD'
Stem 1: (be) a hand-held stretcher
Stem 2: (be) a dragged platform (e.g., with one end tied to a pack animal and the other end dragged along the ground to carry supplies or as a stretcher)
Stem 3: (be) a sled [for moving, not recreation]

-FJ- 'IMPEDE / HINDER / BLOCK / STOP / INTERFERE / INTERRUPTION'		
BSC	**Stem 1:** (be) an act/instance of impeding/hindering the path or flow or course of an entity or impede/hinder some activity or active state [CPT version = to block, to stop, make cease]	**Stem 2:** (be) an instance of a channel or conduit through which something flows becoming clogged/obstructed/blocked; to clog, to block **Stem 3:** (be) an act/instance of interfering, interrupting or obstructing some entity or activity or state, with the effect of (potentially) disrupting it or rendering it ineffectual; to hinder, to interfere, to interrupt, to obstruct [CPT version = to disrupt]
CTE	(be) a state of being impeded/hindered	
CSV	(be) the physical act of impeding/hindering	
OBJ	(be) an entity/object used to impede/hinder	

The Specification pattern of the above root applies to the following root as well:

-RFJ- 'CONSTRICTION / SEALING / PLUGGING / PREVENTION'
Stem 1: (be) an act of constricting or choking something; a constriction; to constrict, to choke [CPT = to cut off the flow of something]
Stem 2: (be) an act/instance of sealing or plugging ; to seal, to plug
Stem 3: (be) an act/instance of prevention; to prevent something from happening [CPT version = to thwart]

-ṬS- 'COVERING / SUPERIMPOSITION / ENCASEMENT / SHEATH'		
BSC	**Stem 1:** (be) an act/instance of covering something; to cover (i.e., superimposing an entity over/onto another for the purpose of protection, concealment, etc.)	**Stem 2:** (be) an act/instance of placing an encasing medium/material around something; to encase, to wrap **Stem 3:** (be) an act/instance of sheathing; to sheathe (i.e., to place something into its associated/customized encasing cover/holder/container
CTE	(be) a state of being covered	
CSV	(be) the physical act of covering	
OBJ	(be) an entity/object used to cover (e.g., lid, blanket, tarp, screen, etc.)	

The Specification pattern of the above root applies to the following root as well:

-RṬS- 'TOP / ROOF / VENEER / COATING / MASK'
Stem 1: (be) an act/instance of placing a top or roof(-like covering); to roof, to tent, to put a top upon something
Stem 2: (be) an act/instance of placing a veneer, facing, coating, crust, or protective layer (of a substance/material) onto something
Stem 3: (be) an act/instance of masking something; put a mask on something; to mask (i.e., a false face or false exterior covering designed to hide the appearance of something)

-ŇM- 'SURFACE APPLICATION / OVERLAY / SPRAY'		
BSC	**Stem 1:** (be) an act/instance of applying/spreading/overlaying a (quasi-)liquid or viscous substance or material to the surface of something; to apply, spread on(to), to overlay, to coat, to smear a (quasi-)liquid substance/material to something (e.g., frosting, plaster, stucco, peanut butter, etc.)	**Stem 2:** (be) an act/instance of applying/spreading/overlaying a gel-like substance or material to the surface of something; to apply, spread on (to), coat, smear a gel-like substance or material [CPT = foam up, froth] **Stem 3:** (be) an act/instance of spraying a substance or medium onto another so that it is flush with, in complete contact with, or adheres to the underlying entity; to spray on, to apply via a spray(er) OBJ Specification = 'aerosol'
CTE	(be) a state of having (had) a substance/material applied	
CSV	(be) the physical act of applying a (quasi-)liquid or viscous substance or material	
OBJ	(be) the material or substance (to be) applied	

The following stems have the same Specification pattern as the root -**ŇM**- above:

-**ŇVW**- 1. housepaint/industrial paint 2. artist's paint 3. watercolor paint
-**ŇVY**- 1. varnish 2. polymer-based sealant 3. shellac
-**ŇVR**- 1. facial make-up / rouge 2. body paint/body make-up 3. theatrical make-up
-**ŇVL**- 1. eyeshadow 2. eye-liner 3. false eyelash
-**ŇVŘ**- 1. mascara 2. lipstick 3. lip gloss

-**ŇFW**- 1. cream 2. lotion 3. ointment
-**ŇFY**- 1. salve/balm 2. lacquer 3. butter (indicating a type of topical application, e.g., tanning butter, not the edible milk-based subtance)
-**ŇFL**- 1. disinfectant 2. insecticide/pesticide 3. insect repellent
-**ŇFR**- 1. moisturizer/emollient 2. sunscreen 3. bronzer/tanning topical application

-ÇDR- 'LEVERAGE'		
BSC	**Stem 1:** (be) an act/instance of leveraging; to leverage (i.e., to move/hoist something by means of the principle of the lever)*	**Stem 2:** (be) an act/instance of being the balancing point for an act of leverage; function as/like a fulcrum **Stem 3:** (be) an instance of being the force/pressure/power applied to move/hoist something by means of leverage
CTE	(be) a state of being leveraged	
CSV	(be) the physical act of leveraging	
OBJ	(be) a lever	

*If using this stem figuratively (as is common in contemporary business English), it must be morphologically marked as being metaphorical.

-ŢŇ- ‘FILL / INUNDATE / FILL-IN / STUFF’		
BSC	**Stem 1:** (be) an act/instance of filling something (up) (i.e., use an empty volume of space as a container/holder for some something)	**Stem 2:** (be) an act/instance of filling something in, stuffing (i.e., to fit/move/insert something into a contained space to the point where the contained space is completely filled) **Stem 3:** (be) an instance of inundating a space with some entity/substance; to inundate, to flood, cover over/up (e.g., with a liquid, gas, or malleable substance which completely supplants, covers, or renders inaccessible a surface, an object, or a contained volume of space
CTE	(be) a state of being filled	
CSV	(be) the physical act of filling	
OBJ	(be) the substance/entity with which a contained space is filled	

-ŠŇ- ‘INFLATION / PRESSURIZATION / VENTILATION’		
BSC	**Stem 1:** (be) an act/instance of inflating or puffing up with air or other gaseous substance; to inflate, to puff up with air/gas [both the act/process of inflation and the state of being inflated with air/gas]	**Stem 2:** (be) an act/instance of pressurization **Stem 3:** (be) an instance of ventilation [i.e., maintain a state of pressurization/inflation via continuous flow /replacement of air or gas]
CTE	(be) a state of being inflated/puffed up	
CSV	(be) the physical act of inflation/puffing up; to inflate, puff up	
OBJ	(be) the substance/gas with which a contained space is inflated or puffed up	

-ŠḌ- ‘REMOVAL / ELIMINATION / EMPTYING / CLEARING / VACUUM’		
BSC	**Stem 1:** (be) an act/instance of removal/elimination of something from somewhere; to remove, to eliminate something from somewhere	**Stem 2:** (be) an act/instance of emptying or clearing an area, container, or volume of space **Stem 3:** (be) an instance/state of vacuum; to render a volume of space into a state of vacuum
CTE	(be) a state of having been removed/eliminated	
CSV	(be) the physical act of removal/elimination; to perform an act which removes or eliminates something	
OBJ	(be) the means/implement/procedure by which something is removed or eliminated	

-XV- 'COMPRESSION / COMPACTING / CONCENTRATION / POTENTIAL ENERGY STORAGE' Associated Affix: **DCP**		
BSC	**Stem 1:** (to be) a state/act of compression/compaction; to compress, to compact, to squeeze [i.e., decrease the dimensional volume of something via externally applied pressure]	**Stem 2:** (to be) a state/act of compression/compaction/concentration/ condensing; to compress, to compact, to concentrate, to condense [i.e., to fit a greater amount of a material, substance, content, into a fixed space via increasing its density per volume] **Stem 3:** (to be) a state/act of energy storage; to store potential energy [OBJ = 'battery']
CTE	(to be) a state of being compressed/compacted/squeezed	
CSV	(to be) the physical act of compressing/compacting/squeezing	
OBJ	(to be) the means/implement/procedure by which something is compressed/compacted/squeezed	

-TÇV- 'DISSIPATION / DISPERSION / SCATTERING / THINNING / CULLING'		
BSC	**Stem 1:** (to be) a state/act of dissipating/dispersing; to dissipate/disperse something (i.e., render a conglomerate entity less viable or concentrated by increasing the distance between its members or component entities)	**Stem 2:** (to be) a state/act of scattering; to scatter [i.e., dissemble or destroy the symmetry/order/arrangement/viability of a multi-member or conglomerate entity by physically separating its members/components in (quasi-)random fashion **Stem 3:** (to be) a state/act of thinning/culling [i.e., rendering a conglomerate entity less viable or less concentrated by removing/eliminating some of its members or component entities
CTE	(to be) a state of being / having been dissipated/dispersed	
CSV	(to be) the physical act of dispersion/dissipation; to dissipate/disperse	
OBJ	(to be) the means/implement/procedure by which something is dissipated/dispersed	

-SMY- 'SORTING / STRAINING / SIFTING / AGGREGATION'		
BSC	**Stem 1:** (to be) a state/act of sorting through something in order to find/separate out a particular kind of entity from it; to sort through something in order to find/separate out a particular kind of entity from it [i.e., separate one or more of a particular kind of entity from a group of multiple kinds of entities]	**Stem 2:** (to be) a state/act of straining/sifting something in order to separate out certain ingredients/elements/components/contaminants, etc.; to strain/sift something out of something else **Stem 3:** (to be) a state/act of locating and setting aside, bringing together, or aggregating members of a particular kind of entity from various/separate locations; to aggregate members of a particular kind of entity from various/separate locations
CTE	(to be) a state of being / having been sorted through	
CSV	(to be) the physical act of sorting; to sort	
OBJ	(to be) the entity/object or collection of entities through which one sorts in order to separate out a particular kind of entity	

-SXL- ‘FILTRATION / DISCRIMINATION / SEGREGATION’		
BSC	**STEM 1:** (to be) an act/state of filtering or filtration; to filter something out of something else	**STEM 2:** (to be) an act of discrimination; to discriminate **STEM 3:** (to be) an act/state of segregation; to segregate
CTE	(to be) a state of being / having been filtered	
CSV	(to be) the physical act of filtering; to filter	
OBJ	(to be) the entity/object or collection of entities being filtered	

-SXR- ‘COMBING / BRUSHING / RAKING / CARDING’		
BSC	**STEM 1:** (to be) an act/state of combing (through) hair/fur; to comb (through) hair/fur	**STEM 2:** (to be) an act/state of brushing hair/fur; to brush hair/fur **STEM 3:** (to be) an act/state of raking or carding (to remove unwanted elements from a substrate (e.g., fallen leaves from grass, knots from wool, etc.); to rake or card
CTE	(to be) a state of being combed	
CSV	(to be) the physical act of combing hair/fur; to run a comb through hair/fur	
OBJ	(to be) the hair/fur being combed	

-VZD- ‘RUBBING / MASSAGING / SCRATCHING’ *		
BSC	**STEM 1:** (to be) an act/state of rubbing something with one’s hand(s) or other body part	**STEM 2:** (to be) an act/state of massaging with one’s hand(s) or other body part **STEM 3:** (to be) an act/state of scratching with one’s fingernails/claws or other sharp body part **
CTE	(to be) a state of being rubbed with an entity’s hand(s) or other body part	
CSV	(to be) the physical act of rubbing with an entity’s hand(s) or other body part	
OBJ	(to be) what is being rubbed	

* for the sensation of being rubbed, massaged, or scratched, see -**VZW**-

** this stem refers only to scratching for grooming purposes or to relieve an itch; for the meaning of “to make/leave a scratch or scratches in something” see **-PPŠ-**

-LṬR- ‘DRAINAGE’		
BSC	**Stem 1:** (to be) a state/act of draining a liquid via gravity; to drain a liquid via gravity [i.e., remove/relocate a volume of liquid by means of pouring it or channeling it via gravity through a hole/conduit/pipe/channel, etc.	**Stem 2:** (to be) an act of drainage via osmosis/soaking/absorption into a surrounding media (e.g., the ground, a bandage, absorptive cotton, etc.) **Stem 3:** (to be) an act/state of drainage via pressure/suction/pumping; to drain via pressure/suction/pumping
CTE	(to be) a state of being / having been drained	
CSV	(to be) the physical act of draining; to drain	
OBJ	(to be) the liquid which is removed/relocated by being drained	

NOTE: the following root uses the same Specification pattern as the above root:

-FṬR- ‘RINSING / DRYING / EVAPORATION’
Stem 1: (to be) a state/act of rinsing; to rinse [i.e., pour/spray water or other liquid upon/over an object/entity for the purpose of removing superficial material]
Stem 2: (to be) a state/act of drying; to dry [i.e., remove all presence of wetness or moisture]
Stem 3: (to be) an act/state of evaporation; to evaporate

-ZṬ- ‘ABSORPTION* / IMBUEMENT / INFILTRATION’		
BSC	**Stem 1:** (be) an act/instance of absorption*; to absorb (i.e., for a [quasi-]liquid substance to be soaked up into a porous material)	**Stem 2:** (be) an act/instance of imbuement/infusion; to imbue, to infuse (i.e., to introduce a quality or characteristic into an entity so that it spreads throughout and becomes part of that entity’s character/essence/identity, etc.) **Stem 3:** (be) an instance of infiltration
CTE	(be) a state of being (or having been) absorbed	
CSV	(be) the physical act of absorbing	
OBJ	(be) a porous material/object/entity which absorbs	

* This stem refers to the drawing up of a (quasi-)liquid substance into an absorbable material; for a focus on removing/cleaning up a spilled substance, see -**ŠNY**- below.

-RZṬ- ‘SOAK / DIFFUSION / PERMEATION / OSMOSIS’
Stem 1: (be) an act/instance of soaking/sousing [CPT version = saturate]
Stem 2: (be) an act/instance of diffusion/suffusion; to diffuse, to suffuse [CPT version = permeate]
Stem 3: (be) an act/instance of osmosis

-ZW- ‘CLEANLINESS / PURITY / DECONTAMINATION / DISINFECTION / SANITATION / HYGIENE’		
BSC	**Stem 1:** (be) a state of cleanliness; be clean, to (make) clean (i.e., a state of being relatively free from dirt/grime/germs/mold, etc.) *	**Stem 2:** (be) a state of purity, being pure; be/make pure, purify, decontaminate, disinfect (i.e., to be/make free from foreign/invasive/polluting substances) **Stem 3:** (be) a state/act of proper sanitation/hygiene; be sanitary/hygienic, exercise proper sanitation/hygiene (i.e., practices which help to ensure an entity/party/environment remains clean or pure)
CTE	(be) a state of cleanliness	
CSV	(be) a physical act of cleaning	
OBJ	(be) the entity/party/object (to be/being) cleaned	

* This stem does not mean ‘organize’ or ‘de-clutter’ as in “to clean a room” — see the next root below.

-ṬB- ‘TIDINESS / ORDERLINESS / GROOMING’		
BSC	**Stem 1:** (be) a state of tidiness/orderliness/neatness, being tidy/orderly;neat; to tidy (up), put in order, to neaten (i.e., to organize/arramge the location/placement of entities in an environment so that they are logically and neatly arranged for ease of use and so the environment has a positive aesthetic appearance)	**Stem 2:** (be) a state of pristineness, pristine condition; to be/keep/maintain (in) pristine (condition) (i.e. in original condition) **Stem 3:** (be) a state of being well-groomed; to groom (i.e. be/make free from slovenliness)
CTE	(be) a state of orderliness/tidiness/neatness	
CSV	(be) an act of tidying/neatening; to make tidy or neat, to tidy up	
OBJ	(be) that which is (to be) tidy/neat/orderly	

-ŠNY- ‘WIPING / MOPPING / CLEANING VIA ABSORPTION’		
BSC	**Stem 1:** (be) a state/act of wiping using an absorbable cloth-like material (for the desired purpose of removal/cleaning a stain, spilled substance, etc.) [CPT = to wipe up, clean up via wiping, remove via wiping up]	**Stem 2:** (be) a state/act of mopping; to use a mop, to mop (up) **Stem 3:** (be) a state/act/process of removing/cleaing an undesired (quasi-)liquid substance via an absorptive material (e.g., a towel, a tissue, gauze, etc.)
CTE	(be) a state of (desired) cleanliness (to be) achieved by wiping (up)	
CSV	(be) an act of wiping (up)	
OBJ	(be) that which is removed via wiping (up); a stain, a spill, etc.	

The Specifications of the above root are applied to the following two roots to specify the means/tool used for cleaning/wiping/absorbing:

-ŠMW- ‘MEANS USED FOR WIPING/CLEANING’ I
Stem 1: (be) a rag **Stem 2:** (be) a towel **Stem 3:** (be) a mop

-ŠMY- ‘MEANS USED FOR WIPING/CLEANING’ II
Stem 1: (be) a sponge **Stem 2:** (be) an absorbent pad or bandage **Stem 3:** (be) an absorbent substance or material (e.g., sand, powder, cellulose granules, etc.)

-MBW- ‘SUCTION / PUMPING’		
BSC	**Stem 1:** (be) a state/act of orally sucking; to orally suck (i.e., create a velaric ingressive airstream in one’s oral cavity in order to suck in liquid, food, air, etc.)	**Stem 2:** (be) a state/act of suction; to suction, create suction **Stem 3:** (be) a state/act/process of storing/transferring/draining liquid or gas by means of pumping; to pump a liquid or gas for purpose of storage/transfer/drainage [i.e., via artificially created pressure differential]
CTE	(be) a state of being sucked	
CSV	(be) an act of sucking; to physically suck something	
OBJ	(be) that into which something is suck	

-SJ- 'COMBINE / (INTER-)MINGLE / MIX / BLEND / SYNTHESIS'		
BSC	**Stem 1:** (be) an act/instance of combining two or more entities (i.e., bring separate entities together to function as one, for to achieve a more effective/desirable result/outcome than would/could be achieved by having the entities remain separate); to combine	**Stem 2:** (be) an act/instance of mixing or (inter-)mingling two or more parties/entities (i.e., for the purpose of having the entities interact/associate with each other in some way); to mix **Stem 3:** (be) an instance of synthesizing; to synthesize (i.e., to combine elements to form a new gestalt)
CTE	(be) a state of being (or having been) combined	
CSV	(be) the physical act of combining	
OBJ	((to be) the rule(s)/pattern(s)/procedure(s) one follows in order to achieve a meaningful or coherent combination of elements	

NOTE: for the meaning of the old FORMAL stems previously associated with this root, use the following new root:

-RSJ- 'BLENDING / FUSION / AMALGAMATION / SYNERGY'
Stem 1: (be) an act/instance of mixing/blending two or more entities together (i.e., same as Stem 1 of the above root except that the two entities coalesce/merge/dissolve into a new gestalt entity); to blend, to fuse, to amalgamate
Stem 2: (be) an act/instance of synergy
Stem 3: (be) an act/instance of synergy with emergent properties

-RČ- 'DAMAGE / CORRUPTION / SUBVERSION / RUIN / DESTRUCTION / DEMOLISH / ELIMINATION / ERADICATION'		
BSC	**Stem 1:** (be) an act/instance of physically damaging [CPT version = wreck, destroy physically, i.e., damage to the point of physical irreparability]	**Stem 2:** (be) an act/instance of rendering less-than-fully operable/functional; corrupt, subvert, undermine [CPT Version = ruin, devastate, break up, destroy operationally, i.e., damage, corrupt, or subvert to the point of being unable to function/operate] **Stem 3:** (be) an act/instance of reducing the presence/existence of; to reduce, lessen [CPT version = eradicate, eliminate, annihilate, "disappear", i.e., to render something non-existent]
CTE	(be) a state of being (or having been) physically damaged	
CSV	(be) the physical act of damaging	
OBJ	(be) that which is used to physically damage	

-BN- 'CHANNEL / STEER / AIM'		
BSC	**Stem 1:** (be) an act/instance of channeling; to channel (i.e., direct one's translative course or that of another through a channel or conduit)	**Stem 2:** (be) an act/instance of steering; to steer (i.e., direct one's course via manually-manipulated means) **Stem 3:** (be) an act/instance of aiming; aim for, take aim at
CTE	(be) a state of being moved through/within/along a channel/conduit	
CSV	(be) the physical act of moving through spacetime via a channel/conduit	
OBJ	(be) that which is used to channel the course of something (e.g., a funnel, chute, conduit, pipe, tube, trough, shaft, tunnel, etc.)	

-ḌC- 'SAFETY / SECURITY / LOCK'		
BSC	Stem 1: (be) an act/instance of making something safe (i.e., removing the threat of other entities from harming something or someone)	**Stem 2:** (be) an act/instance of securing something, safekeeping, i.e., establishing preventative practices/protocols/procedures with the goal of keeping something safe from external harm/damage; to secure, keep safe **Stem 3:** (be) a act/instance of locking something, i.e., establishing a physical means by which other entities cannot reach or have contact with something, or by which an entity is prevented from reaching or having contact with external parties
CTE	(be) the state of being safe	
CSV	(be) the physical act of making something/someone safe	
OBJ	(be) the object/entity/procedure/rule/law, etc. by which one effectuates a state of safety	

-ÇMY- 'BUFFER / FAIL-SAFE / LIAISON'		
BSC	Stem 1: (be) an act/instance of buffering; to buffer (i.e., place an intervening medium between two entities as a protective measure)	**Stem 2:** (be) an act/instance of functioning as a fail-safe; to function/act as a fail-safe measure **Stem 3:** (be) a act/instance of being/functioning as a go-between or liaison; to function as a go-between or liaison (i.e., a neutral entity or party for purposes of diplomacy)
CTE	(be) the state of being buffered	
CSV	(be) the physical act of buffering; to do something that acts as or creates a buffer	
OBJ	(be) the object/entity/procedure/rule/law, etc. used to buffer	

-VTḶ- ‘HANDLE / MANIPULATIVE ASSISTIVE COMPONENT’		
BSC	**Stem 1:** (to be) a state/act/process of supporting oneself by holding onto a handle (i.e., a protruding extension, ring, bar, etc. of an entity designed for one to hold onto)	**stem 2:** (to be) a state/act of physically controlling another entity by holding onto its handle (e.g., a pot handle, the handle of a hammer or shovel, the handle of a gun, etc.) **stem 3:** (to be) a remote control device (e.g., a joystick controller, a detonation plunger, etc.)
CTE	(to be) a handle (e.g., of a cane)	
CSV	(to be) an act/process of making use of a handle; to grab/hold (onto) a handle	
OBJ	(to be) the entity being supported by holding onto a handle	

-TVY- ‘POLARITY / NEUTRALITY’		
BSC	**STEM 1:** (to be) a state of positive polarity, be positive in terms of polarity	**STEM 2:** (to be) a state of negative polarity, be negative in terms of polarity **STEM 3:** (to be) a state of neutral polarity, be neutral in terms of polarity
CTE	(to be) a state of being positive in terms of polarity	
CSV	(to be) an act of making something positive in terms of polarity	
OBJ	(to be) that which is of positive polarity	

-VPW- ‘REFLECTION / REFRACTION / DIFFRACTION’		
BSC	**STEM 1:** (to be) a state of reflecting (of electromagnetic radiation); to reflect [i.e., change in the direction of EM waves/particles via contact with a barrier]	**STEM 2:** (to be) a state of refraction; to refract [i.e., the bending of, and change in speed and/or wavelength of electromagnetic energy due to passage from one medium to another] **STEM 3:** (to be) a state of diffraction; to diffract [i.e., the bending of, and change in direction of electronic energy to due to passage through an opening or around a barrier]
CTE	(to be) a state of reflection; for there to be reflecting going on	
CSV	(to be) a physical act/process of reflecting; to have/use a reflective surface that causes electromagnetic energy to change direction	
OBJ	(to be) a reflection, i.e., what gets reflected (i.e., a beam of light, a visual image)	

The following root has the same Stem and Specification structure as the root -**VPW**- above:

-VPY- ‘MACRO=LEVEL REFLECTION / REFRACTION / DIFFRACTION’
[The three stems of this root are identical to the stems of the root -**VPW**- above except that they deal with the reflection/deflection/bending of macro-level phenomena, e.g., water waves, a tennis ball against a raquet, a billiard ball against a snooker cushion, a person diving into a swimming pool, a metal object deflected from its trajectory by a magnet, etc.]

2.1.2 Tools and Implements/Materials/Substances Used To Maintain Completed States Created via Tools/Devices/Machines/Manual Processes

-CŇ- PIN / TACK / STAPLE		
BSC	**Stem 1:** (be) a state/act of maintaining a state between entities by means of a pin, (be) a pin in use, a pin being used; to utilize/make use of a pin	**Stem 2:** (be) a state/act of maintaining a state between entities by means of a tack, (be) a tack in use, a tack being used; to utilize/make use of a tack **Stem 3:** (be) a state/act of maintaining a state between entities by means of a staple, (be) a staple in use, a staple being used; to utilize/make use of a staple
CTE	(be) a pin	
CSV	(be) an act of utilizing/making use of a pin; use/make use of a pin	
OBJ	(be) what a pin is holding together	

The following roots/stems follow the same Specification pattern as the root PIN/TACK/STAPLE above:

-**CPY**- 1. nail 2. spike 3. pile (i.e., a foundational support structure)

-**CPW**- 1. screw 2. bolt 3. rivet

-**DBL**- 1. stopper 2. plug/cap 3. seal/suture

-**GDY**- 1. cement 2. mortar 3. plaster

-**GDW**- 1. putty 2. caulk 3. grout

-**GDL**- 1. resin 2. polymer 3. solder

-**GDR**- 1. adhesive 2. paste 3. glue

-**KŢT**- 1. clip 2. clamp 3. magnet

-**KŢW**- 1. string/twine 2. wire 3. strap

-**KŢY**- 1. button 2. zipper 3. velcro fastener

-**FSM**- 1. brace 2. girder 3.crossbeam

-**FSN**- 1. hook 2. frame 3. rigging/scaffolding

-**JDY**- 1. stent 2. tubing 3. valve

-**MZT**- 1. strand of cord 2. strand of rope 3. strand of wicker/thatching

-**ÇMW**- 1. nut, lug nut 2. washer 3. drywall anchor

-**DDŘ**- 1. needle [as perforator] 2. needle [as thread-drawer/puller] 3. needle [as probe or delivery mechanism]; syringe

-**RBB**- 'LIQUID MEDIA USED AS TOOL' 1. soaking agent/medium 2. hydration agent/medium 3. lubricant/lubricating agent/medium

-**ŽDV**- 'GUIDING ALONG/THROUGH A CHANNEL' 1. funnel 2. trough 3. chute

-**XBL**- 'LIQUID AS RESOURCE CONSUMED IN A PROCESS 1' 1. fuel 2. machine oil 3. cooking oil

-**XBR**- 'LIQUID AS RESOURCE CONSUMED IN A PROCESS 2' 1. cleansing agent 2. hydraulic fluid 3. solvent

-ČV- **'TOOL / INSTRUMENT / PIECE OF EQUIPMENT'** Affix: **TDM**		
BSC	**Stem 1:** (to be) a tool, instrument, implement, piece of equipment as extension of bodily appendage(s) or sensory organs; use a tool (e.g., as per force, strength, precision, reach, pressure, instrument as labor-saving or efficiency-increasing means etc.) — [both implement and the manner/process of its function]	**Stem 2:** (to be) a tool, instrument, implement, piece of equipment as a means of "defying" the forces of Nature (e.g., as with a wheel, fulcrum, wedge, clamp, etc.); use such a tool — [both implement and the manner/process of its function] **Stem 3:** (to be a) supply-like resource/implement/piece of equipment used by itself or in association with specific tool to maintain/sustain a constructed state (e.g., a nail, tack, hook, rivet, staple, screw, clothespin, etc.)
CTE	(to be) a tool, instrument or implement; use a tool	
CSV	(to be) the manner/process (i.e., how it works) of using a tool or implement; to engage in the manner/process of using a tool	
OBJ	(to be) the party/entity using a tool or implement	

The following stems for tools have the same Specification pattern as the root -**ČV**- above:

-**GPW**-	1. hammer 2. mallet 3. sledgehammer
-**GPŘ**-	1. ball-peen hammer 2. half-hatchet hammer 3. specialized hammer (e.g., body mechanic's hammer, drywall hammer, tack hammer, etc.))
-**GPY**-	1. hatchet 2. axe 3. specialized axe (e.g., fireman's axe, executioner's axe, double bit axe, etc.)
-**TGW**-	1. (round-point) awl 2. bradawl 3. birdcage awl
-**TGY**-	1. sewing awl 2. stitching awl 3. other specialty awl (e.g.,saddler's, sailmaker's, scratching, closing, etc.)
-**ZXW**-	1. chisel 2. dovetail chisel 3. other specialized hand-driven chisel (e.g., skew chisel, framing chisel, bolster chisel, socket chisel, cranked-handled chisel)
-**ZXL**-	1. concrete chisel 2. cold chisel 3. other specialized hammer- or power-driven chisel (i.e., not hand-driven)
-**ŢST**-	1. adjustable/slip-joint pliers 2. non-adjustable/needle-nose pliers 3. locking pliers
-**KŠVW**-	1. fixed-size open-ended wrench 2. box-end wrench 3. combination wrench
-**KŠVY**-	1. Allen wrench / hex wrench 2. lug wrench 3. other non-adjustable wrench
-**KŠVL**-	1. adjustable wrench 2. monkey wrench / pipe wrench 3. ratcheting wrench
-**KŠVR**-	1. socket wrench 2. other adjustable wrench 3. specialized wrench (e.g., basin/sink wrench, strap wrench, star-head key)
-**ŽTŘ**-	1. (flathead) screwdriver 2. Phillips screwdriver 3. specialty-head screwdriver (e.g., torx, hex, pozidrive, tri-wing, etc.)
-**ŠŠX**-	1. rasp / file 2. plane (jack, block, surform) 3. sanding block
-**KSFW**-	1. crosscut saw 2. backsaw 3. hacksaw, coping saw, bow saw
-**KSFY**-	1. jigsaw 2. circular saw 3. chainsaw
-**GŽK**-	1. crowbar / ripping bar 2. cat's paw 3. flat bar

-**ŽBV**-	1. hand drill 2. carpenter's brace 3. power drill / motorized drill
-**ŽBL**-	1. twist drill bit 2. brad-point drill bit 3. auger drill bit
-**ŽBR**-	1. self-feed drill bit 2. spade drill bit / paddle drill bit 3. Forstner drill bit
-**ŽBW**-	1. hole saw 2. countersink drill bit 3. plug cutter
-**ŽBY**-	1. step drill bit 2. tile drill bit 3. other specialty drill bit (e.g., masonry drill bit, installer drill bit)
-**FSV**-	1. C-clamp / G-clamp 2. hand-screw clamp 3. bench vise
-**FSVW**-	1. spring clamp 2. bench clamp 3. other specialty clamp (e.g., mitre clamp, trigger clamp, drill press clamp, bar clamp, vise-grip, etc.)
-**DFW**-	1. trowel 2. shovel / spade 3. pick
-**DFY**-	1. pitchfork / garden fork / potato fork / compost fork 2. hoe 3. rake
-**DFL**-	1. auger 2. planting dibble / hand seeder 3. post-hole pincer
-**DFR**-	1. aerator 2. tiller 3. plow
-**FÇT**-	1. hand-held brush (e.g., paint brush) 2. cleaning brush, whisk broom 3. broom
-**ÇNW**-	1. scissors 2. pruning shears / snips / topiary shears 3. gardening shears / hedge shears
-**ÇNY**-	1. bolt cutters 2. lopper 3. pole pruner / tree pruner
-**ŇHW**-	1. fireplace poker / stoker 2. bellows 3. anvil
-**LZDR**-	1. measuring ruler 2. measuring tape 3. tailor's measuring tape
-**NZDR**-	1. measuring square 2. measuring compass 3. protractor
-**ZZDR**-	1. calipers 2. micrometer 3. angle gauge
-**ŘZDR**-	1. level 2. bubble inclinometer 3. angle locator
-**DGR**-	1. scale 2.thermometer 3. pressure gauge
-**FŠŘ**-	1. clock / watch / timepiece 2. speedometer 3. odometer
-**LÇF**-	1. wheel 2. roller, rolling shaft 3. ball bearing
-**NZH**-	1. ramp 2. gangway, portable bridge 3. bracing, crossbeam
-**KŠB**-	1. key 2. lockpick 3. other lock-opening device/instrument/means (e.g., fingerprint scanner)
-**VPÇ**-	1. mirror 2. lens 3. prism

-**ĻTH**-	1. net 2. mesh 3. web
-**KŠŇ**-	1. (spirally-coiled) compression spring 2. (spirally-coiled) torsion spring 3. constant force spring
-**ŠPH**-	1. worm gear (of a worm drive) 2. worm screw (of a worm drive) 3. Archimedes screw
-**LČF**-	1. section of a roller chain 2. section of a leaf chain 3. sprocket/chainwheel
-**BKH**-	1. winch 2. crankshaft (to convert back-and-forth motion into rotary motion 3. slider-crank linkage (to convert rotary motion into linear motion) [Stem Zero = crank(ing mechanism)]

-RČV- 'UTENSIL / IMPLEMENT / FUEL'
Stem 1: (to be a) utensil, specialized implement for assisting in refined task
Stem 2: (to be a) mechanical / motorized / electric / electronic implement/utensil
Stem 3: (to be a) supply-like resource/implement used as fuel or input for a mechanical / electrical / electronic implement/utensil

-ŢŢ- 'TRY OUT'		
BSC	**Stem 1:** (to be) an act/process of trying something out (e.g., a tool, a device, a vehicle, a machine, etc.) to see if it works/functions	**Stem 2:** (to be) an act/process of trying something out (e.g., a tool, a device, a vehicle, a machine, etc.) to observe/learn/understand what it does or how it works/functions or whether it improves efficiency/productivity
CTE	(to be) a state of wearing something as a trial to see if it works/functions	
CSV	(to be) an act of trying something out to see if it works/functions	
OBJ	(to be) that which is being tried out to see if it works/functions	**Stem 3:** (to be) an act/process of trying something out (e.g., a tool, a device, a vehicle, a machine, etc.) to see if one can successfully operate it, work with it, or learn to use it skillfully

NOTE: For trying on an article of clothing or other wearable implement (e.g., a spacesuit), see the root -**FF**- instead.

-SST- 'DEVICE / APPARATUS / MACHINE / ENGINE / MOTOR'		
BSC	**Stem 1:** (to be) a state/act/process of a mechanical device/apparatus/contraption functioning/operating; for a mechanical device, apparatus, or contraption to operate/function	**stem 2:** (to be) a state/act/process of an engine/motor (whether electrical, chemically-fueled, steam-driven, wind-driven, solar-powered, nuclear-powered, etc.) to be operating/functioning **stem 3:** (to be) a state/act/process of a powered machine (whether electrical, chemically-fueled, steam-driven, wind-driven, solar-powered, nuclear-powered, etc.) to be operating/functioning
CTE	(to be) a state/act/process of a mechanical device/apparatus/contraption designed or used to do/perform/accomplish something to be in operation/functioning; for there to be such a device/apparatus/contraption functioning /operation	
CSV	(to be) a state/act of operating/utilizing/using a device/apparatus/contraption; to operate/use/make use of a device/apparatus/contraption	
OBJ	(to be) a mechanical device/apparatus/contraption itself (seen as a gestalt entity)	

-SSV- 'DEVISE / CONTRIVANCE / INVENTION'		
BSC	**Stem 1:** (to be) an act/process/instance of devising something; to devise something [i.e., to bring an idea for a device or mechanical process into reality]	**stem 2:** (to be) an act/process/instance of contriving/jury-rigging; to contrive/jury-rig [i.e., to bring a simple or quickly-formed idea for a device or mechanical process into reality as a temporary or stop-gap solution] **stem 3:** (to be) a state/act/process of inventing; to invent something
CTE	(to be) a state/act/process of having/coming up with an idea for devising something	
CSV	(to be) a physical state/act of devising something based upon the idea one has for it	
OBJ	(to be) that which one devises (or has the idea for devising)	

-LPL- 'TECHNOLOGY'		
BSC	**Stem 1:** (to be) a state/act/process of using a technological invention or inventions to do/perform something (i.e., accomplishing something by means of, or with the assistance of an invented machine/device or a discovered process/procedure)	**stem 2:** (to be) a process/procedure based upon a biological/medical/scientific discovery or scientific research/experiments. **stem 3:** (to be) a state/act of doing/performing something using digital/electronic/cyber- technology (e.g., running of algorithmic programs, digitized data, interconnected/networked computers, etc.)
CTE	(to be) a state of there being a technological invention (potentially) available in order to do/perform something	
CSV	(to be) an act/process of using a technological invention; to use a technological invention	
OBJ	(to be) the specific/particular technological invention (to be) used	

The stems of the following three roots have the same Specification pattern as the root - **LPḶ**- above:

-KṬG- ‘ELECTRONIC COMPUTER / COMPUTER ACCESSORY’
1. personal electronic computing device (e.g., PC, laptop, tablet, touchpad, etc.)
2. keyboard, keypad or other finger-manipulated alphanumeric interface to a computing device
3. mouse, pen, or other non-alphanumeric interface to a computing device

For computer monitor/screen, concatenate Stem 1 of this root with Stem 3 of -**ṬLW-.**

-KṬGW- ‘COMPUTERIZED DEVICE’
1. computerized, algorithmically controlled device
2. central processor (e.g., microchip, central gearing mechanism of a Babbage difference engine, etc.) of an algorithmic computing device
3. input-output interface for an algorithmic computing device

-KṬGY- ‘COMPUTER PROGRAM / APPLICATION’
1. computer program, computer application, software
2. firmware
3. basic input-output instruction program for a computer, computer operating system

-LṬV- ‘ELECTRONIC COMPONENT’
1. transistor
2. diode
3. LED

-KṬGL- ‘AUTOMATON (ABSTRACT SELF-PROPELLED COMPUTING DEVICE)’		
BSC	**Stem 1:** (to be) an automatic computing device (i.e., an automaton per automata theory – not necessarily electronic or numerically/digitally-based); to run an automatic computational device	**Stem 2:** (to be) a Turing machine or the lambda calculus representation thereof **Stem 3:** (to be) a finite-state machine / finite-state automaton
CTE	(to be) the internal configurational state of an automaton/self-propelled computing device	
CSV	(to be) the process of transitioning between states/forms in an automaton; to transition between states/forms in an automaton	
OBJ	(to be) the input or saved memory used by an automaton to transition between states/forms	

-KṬGR- ‘ALGORITHMIC COMPUTATION / PROGRAMMING / RUN A COMPUTER PROGRAM’		
BSC	**Stem 1:** (to be) an act/process of producing a computer-implementable instruction, a “line of computer code”; to write a line of programming code	**Stem 2:** (to be) the act/process of transformation/compilation of instructions/text into a different format for a computer to execute **Stem 3:** (to be) a state/process of computation; to implement/run a computer program
CTE	(to be) a computer-implementable instruction, to be a line of programming code	
CSV	(to be) an act of creating/composing a line of programming code	
OBJ	(to be) that which a line of programming code accomplishes; what a line of computer code does/performs when implemented or run	

-MFK- ‘RADIO BROADCASTING AND RECEPTION’			
	Stem 1	**Stem 2**	**Stem 3**
BSC	(to be) an act/instance of using (i.e., listening to) radio broadcast	(to be) an act/instance of broadcasting via radio	(to be) a radio (set or receiver)
CTE	(to be) a state of receiving/processing information learned (i.e., heard) from a radio broadcast	(to be) a state of there being information (having been) disseminated/broadcast via radio	(to be) a state of receiving/processing information learned (i.e., heard) on a particular radio set or receiver
CSV	(to be) an act/process of tuning in / finding a radio broadcast to listen to	(to be) an act/process of broadcasting information on a particular radio frequency	(to be) an act/process of tuning in / finding a radio broadcast to listen to on a particular radio set or receiver
OBJ	(to be) the information heard on the radio	(to be) the information broadcast via radio	(to be) the information heard on a particular radio set or receiver

The Stem and Specification patterns of the above root apply to the following roots as well:

-MFŇ- commercial television

-LMḌ- commercial internet-based streaming

-LMḌL- private/independent internet-based streaming/podcasting

-LMḌR- internet-based social media platform

-**KSŇ**- a newspaper

2.1.3 Common Household Appliances and Electronic Devices

The list of Roots/Stems below have the following Specification pattern:

BSC	(to be) an act/process of using from an electronic device
CTE	(to be) the desired state/situation/improvement to be achieved by use of an electronic device CPT = to be the resulting state thereof
CSV	(to be) an act of physically interacting with/using an electronic device
OBJ	(to be) the actual electronic device being used

-**KSMW**- stove

-**KSMY**- oven

-**KSC**- refrigerator

-**KSČ**- freezer

-**LVB**- washing machine

-**LVD**- clothes dryer

-**MŽK**- air conditioner

-**MŽX**- air purifier

-**JJW**- blender

-**ŢCW**- calculator

-**PÇM**- coffee maker

-**ŢMY**- doorbell camera

-**ẔKW**- telephone

-**ẔKL**- smartphone

-**ẔKR**- digital tablet/pad

-**ČNW**- clock

-**ŠDR**- DVD player

-**ŠDL**- CD player

-**LÇN**- charging unit for electronic device

-**ŢNW**- curling iron

-**ŢNY**- hair dryer

-**GPḶ**- earphones/headphones

-**TFW-** video game controller

-**TFR**- garage door opener

-**ČČW-** lawn mower

-**RČT**- inkjet/laser printer

-**RČTW**- photocopier

-**RČTY**- scanner

-**RČTR**- FAX machine

-**RČTF**- 3-D printer

-**LCŘ**- mp3 player

-**KŠČ**- remote control

-**ŢČW**- generator

-**KÇP**-	rice cooker	-**CCW**-	sewing machine	-**ŠDW**-	stereo receiver
-**KÇPL**-	pressure cooker	-**ŠČW**-	bathroom scale	-**ŠDY**-	stereo speaker
-**KÇPR**-	crockpot	-**PÇN**-	juicer	-**FGW**-	water purifier
- **KÇPW**-	microwave cooker	-**LPH**-	vacuum cleaner	-**DBY**-	flashlight
-**LČR**-	toaster	-**FGL**-	water heater	-**CBW**-	memory stick/flash memory unit
-**FGY**-	water pump	-**FGR**-	heating unit / radiator		

2.1.4 Objects/Implements Used to Hold/Contain Foodstuffs During Preparation, Cooking, and Serving

-**ŠXR**- **'MANIPULATION & HANDLING OF FOODSTUFF WHILE PREPARING OR COOKING IT'**		
BSC	**STEM 1:** (to be) an act/state of handling/manipulating food while preparing it for cooking	**STEM 2:** (to be) an act/state of handling/manipulating food while cooking it or processing it for ingestion **STEM 3:** (to be) an act/state of handling manipulating food while serving it or ingesting it
CTE	(to be) in a state of being handled/manipulated while under preparation for cooking	
CSV	(to be) the physical act of handling/manipulating food while cooking it	
OBJ	(to be) the tool/utensil used to handle/manipulate food while cooking/preparing it	

The following roots utilize the same Specification pattern as the root -**ŠXR**- above:

-**ČKR-** 1. knife 2. serrated knife 3. chef's knife (use Type-2 SIX affix to distinguish knives by funciton, e.g., paring knife, bread knife
-**ČXR**- 1. vegetable peeler 2. potato masher 3. whisk
-**JXR**- 1. colander / strainer 2. grater 3. salad spinner
-**ZXR**- 1. fork 2. "spork" 3. ladle
-**MXR**- 1. spoon 2. slotted spoon 3. measuring spoon
-**NSXR**- 1. spatula 2. chopstick 3. tongs
-**ŘXR**- 1. can opener 2. bottle opener 3. citrus juicer
-**LSXR**- 1. garlic press 2. mushroom slicing press / strawberry slicing press 3. knife sharpening rod / stone
-**MZP**- 1. mortar 2. pestle 3. grinding stone

-ŠGV- 'HOLDING & CONTAINING OF FOODSTUFF FOR PREPARING OR COOKING OR SERVING IT'		
BSC	**STEM 1:** (to be) an act/state of holding/containing food while preparing it for cooking	**STEM 2:** (to be) an act/state of holding/containing food while cooking it or preparing it for ingestion **STEM 3:** (to be) an act/state of holding/containing food for the purpose of serving it or as a result of serving it
CTE	(to be) in a state of being held/contained while under preparation for cooking	
CSV	(to be) the physical act of holding/containing food while cooking it	
OBJ	(to be) the object/implement used to hold/contain food while cooking/preparing it	

The following roots utilize the same Specification pattern as the root -**ŠGV**- above:

-**SBW**- 1. skillet / frying pan 2. sauté pan 3. grilling pan

-**SBY**- 1. saucepan 2. pot / stockpot 3. casserole dish

-**SBR**- 1. baking sheet 2. muffin pan 3. broiler pan

-**SBŘ**- 1. pie tin 2. baking dish or mold 3. other metallic or ceramic baking dish/container/platform

-**SBV**- 1. barbecue grille 2. griddle 3. rotisserie

-**ŠGW**- 1. cup 2. bowl 3. measuring cup (use Type-2 SIZ affix to distinguish bowls by function, e.g., serving-, soup-, salad bowl, etc.)

-**ŠGY**- 1. plate / dish 2. ceramic drinking vessel (e.g., cup, mug) 3. glass or crystal drinking vessel

-**ŠGL**- 1. serving pitcher 2. decanter 3. flagon

-**ŠGR**- 1. cutting board 2. serving board 3. trivet

2.1.5 Fixtures Used To Facilitate/Maintain States Created via Tools/Devices/Machines/Manual Processes/Powered Processes

-LCW- 'BUILDING FIXTURE / ARCHITECTURAL FIXTURE'			
	STEM 1	**STEM 2**	**STEM 3**
BSC	(to be) a state/act/of serving/functioning as a built-in/permanent fixture in a building/domicile for the purposes of maintaining/providing a continuous mechanical, electrical, plumbing, or living state or process; to serve/function as such a fixture (e.g., wall-mounted or built-in feature, fireplace, chimney, window shutter, ceiling fan, tapware, sink/basin, tub, toilet, water or gas pipe, electrical conduit, wiring, etc.)	(to be) a state/act of utilizing/making use of a built-in/permanent fixture in a building/domicile in order to obtain some needed/desired function/service	(to be) a state/act of doing without, being deprived of, or being unable to utilize a needed/desired fixture in a building/domicile
CTE	(to be) the state/service maintained/delivered by a fixture	(to be) the state of a fixture being in use	(to be) the state of a needed/desired fixture being absent/unavailable
CSV	(to be) an act/process of a building fixture functioning or in operation; for a fixture to function/operate	(to be) an act of a person utilizing/making use of a fixture	(to be) a physical act/state of a person having to do/go without a needed/desired fixture
OBJ	(to be) the fixture itself	(to be) the fixture used	(to be) the consequence(s) of a needed/desired fixture being absent/unavailable

The following roots for both built-in/permanent fixtures and temporary/portable fixtures have the same Stem & Specification pattern as the root above:

-**MBL**- water tap / water faucet / tap valve
-**VBL**- pipe (for water, gas, sewage, etc.)
-**BBL**- sink
-**BBR**- washbasin
-**BBW**- bathtub
-**BBY**- shower stall
-**BBŘ**- bidet
-**LČŘ**- toilet / loo
-**ŘČŘ**- urinal
-**GŢR**- cistern
-**JŢR**- artificial fountain
-**VBR**- electrical conduit
-**VBŘ**- electrical wire/wiring
-**VBŽ**- electrical outlet / wall outlet
-**RVB**- power system protection unit (e.g., fusebox / circuit breaker box)
-**ŇSL**- electrical socket
-**LČL**- lamp (i.e., powered local light source)
-**NKV**- air-conditioning unit
-**ŢGV**- ceiling fan
-NŽX- built-in heater, heating unit
-**RŽX**- HVAC unit
-**ŽXL**- fireplace

-**BŢR**- outdoor swimming pool
-**GZŇ**- sewer
-**GZV**- cesspool
-**GZX**- septic tank
-**PÇT**- bucket / pail / portable basin
-**VPŘ**- wall-mount for a wall-hanging or for a wall-mounted device
-**ŽXR**- chimney
-**ŠFL**- built-in storage shelf
-**ŠFR**- built-in storage drawer
-**MFW**- railing / handrail / banister / balustrade
-**LMTR**- hat rack / hat stand / hat tree
-**LSTL**- coat rack

The following affix is useful in conjunction with the above roots for fixtures:

-řst	LCM Life-Cycle of a Mechanism/Device/Fixture/Machine
1	designing X, drafting a blueprint of X; to design X, to draft a blueprint/design of X
2	constructing/forming/crafting/making X; to make/build/construct/craft X
3	determining placement/location of X within/among/in association with its surrounding infrastructure
4	installation/connection/set-up of X; to install/connect/set up X
5	operational test of X; to test whether X is functioning/operating correctly
6	servicing/maintenance/adjustment of X; to perform maintenance service or adjustment on X
7	repair of X; to repair X
8	removal/retirement/discarding of X; remove/retire/discard X
9	replacement of X; replace X

2.1.6 Furniture

-PŢT- 'FURNITURE / FURNISHING / APPLIANCE'		
BSC	**Stem 1:** (to be) a piece of functional furniture; to furnish an environment with a piece of functional furniture (i.e., piece of furniture whose primary purpose is functional rather than decorative, e.g., chair, table, lamp, armoir, cabinet, chest of drawers, desk, bed, bookcase, etc.); utilize or make use of a piece of functional furniture	**Stem 2:** (to be) a piece of decorative furniture; to furnish an environment with a piece of decorative furniture (e.g., antique curio cabinet, fancy rug, designer furnishing, decorative mirror, chandelier, wallpaper, etc.) **Stem 3:** (to be) a major household or office appliance; to furnish an environment with a major functional appliance (e.g., stove, oven, refrigerator, generator, sewing machine, washing machine, freezer, heating unit, air conditioner, television set, etc.)
CTE	(to be) the function(al purpose) of a piece of furniture; to utilize/make use of a piece of furniture	
CSV	(to be) the physical structure/design of a piece of furniture; to design a piece of functional furniture (CPT = construct a piece of functional furniture)	
OBJ	(to be) the space in a room or environment taken up by a piece of functional furniture	

The following seven roots have a Specification pattern modeled after the root -**PŢT**- 'FURNITURE/FURNISHING/APPLIANCE' above:

-SKF- 'TABLE / SHELF / PLATFORM'	
STEM 1	(to be) a table; utilize a table (i.e., a personal-sized (quasi-) rectangular or symmetrically shaped flat surface perpendicular to the direction of gravity positioned to allow one to either stand or sit and use it as a surface for working, storage, or performing some function)
STEM 2	(to be) a shelf
STEM 3	(to be) a platform or large flat artificially-surfaced area to provide a large planar surface for goods or people engaged in some specialized function (e.g., a stage)

-KŠT- 'CHAIR / STOOL / THRONE'	
STEM 1	(to be) a chair; sit in a chair
STEM 2	(to be) a stool; sit on a stool
STEM 3	(to be) an ottoman, a footstool; rest oneself or one's feet on an ottoman

-KŠTL- 'BENCH / COUCH / FUTON'	
STEM 1	(to be) a bench for sitting multiple persons; sit on a bench
STEM 2	(to be) a couch/sofa/divan; sit on a couch/sofa/divan
STEM 3	(to be) a futon; sit or lie on a futon

-NKL- 'BED / SOFA BED / CHAISE LONGUE'	
STEM 1	(to be) a bed; to sit/lie in a bed
STEM 2	(to be) a sofa bed; to sit/lie on a sofa bed
STEM 3	(to be) a chaise longue; sit/lie on a chaise longue

-JKŢ- 'WORKBENCH / COUNTERTOP / DRAWING BOARD'	
STEM 1	(to be) a workbench
STEM 2	(to be) a countertop
STEM 3	(to be) a drawing board

-MTY- 'CABINET'	
STEM 1	(to be) a cabinet with doors for storage; to store in a cabinet
STEM 2	(to be) a display case; to display in a display case
STEM 3	(to be) a hutch or specialty cabinet for storage of specific items

-VTŘ- 'DRAWER / FILING SLOT / RACK'	
STEM 1	(to be) a drawer [container within a supporting/enclosing structure that slides or rolls out to allow access
STEM 2	(to be) a filing slot (i.e., as within the drawer of a filing cabinet)
STEM 3	(to be) a rack (i.e., a framework, lattice, or partition for holding or displaying objects, keeping them spatially separated from each other)

-ŽNW- ‘CARPET / RUG / MAT’	
STEM 1	(to be) a (wall-to-wall) carpet; to cover a floor with a carpet, to have carpeting on a floor
STEM 2	(to be) a rug; to cover an area of the floor/ground with a rug, to have a rug on the floor/ground
STEM 3	(to be) a mat; to cover an area of the floor/ground with a mat, to have a mat on the floor/ground

-ŽNY- ‘PILLOW / CUSHION / BOLSTER’	
STEM 1	(to be) a pillow; to use a pillow
STEM 2	(to be) a cushion; to use a cushion
STEM 3	(to be) a bolster; use a bolster

2.1.7 Building

-RDW- ‘BUILDING CONSTRUCTION’		
BSC	**Stem 1:** (to be) a state/act of constructing/erecting a building (i.e., a constructed space with enclosed rooms for residential and/or commercial occupancy or storage); to construct/erect a building	**Stem 2:** (to be) a state/act of constructing a commercial building **Stem 3:** (to be) a state/act of constructing a non-residential, non-commercial building used for storage or other activities (e.g., a workshop, a studio, etc.)
CTE	(to be) a state of a building under construction	
CSV	(to be) a physical act of building/constructing/erecting a building; to perform the work of building/constructing/erecting a building	
OBJ	(to be) the building under construction [CPT version = finished building]	

-SŘ- ‘ROOM / CHAMBER / COMPARTMENT’		
BSC	**Stem 1:** (to be) a room/chamber	**Stem 2:** (to be) a compartment **Stem 3:** (to be) a cubicle, partially partitioned-off section of a larger room
CTE	(to be) the state of a room/chamber being a partitioned off quasi-autonomous (sub-)section of a larger building interior space	
CSV	(to be) the boundary planar partitions/surfaces/elements that constitute the form of a room/chamber (i.e., the walls, ceiling, floor, doorway, etc.); to make/construct a room (by putting up dividing walls, an entryway, etc.)	
OBJ	(to be) what a particular room/chamber is (used) for, what function it serves	

-DBR- ‘TOWER / SPIRE’		
BSC	**Stem 1:** (to be) a tower room of a building where the enclosed room is intended as a normal dwelling space	**Stem 2:** (to be) a tower room of a building where the enclosed room is a specialized ancillary space (e.g., bell tower, clock tower, a room containing air conditioning or heating or electrical infrastructure, etc. **Stem 3:** (to be) a spire on a building (decorative/aesthetic only with no enclosed or functional interior room
CTE	(to be) the state of a building having a tower room	
CSV	(to be) the boundary planar partitions/surfaces/elements that constitute the form of a tower room	
OBJ	(to be) what a tower room is used for, what function it serves	

-KÇM- ‘BALCONY / LANDING / MEZZANINE / LOFT’		
BSC	**Stem 1:** (to be) a(n outdoor) balcony	**Stem 2:** (to be) a landing (at the end of a stretch of stairs) **Stem 3:** (to be) a mezzanine or loft (a partial floor or partial storey of a building between or above a full floor or storey
CTE	(to be) the state of there being an elevated outdoor space as part of a building	
CSV	(to be) the platform-like area with guard wall/railing that constitutes a balcony; to make/construct a balcony	
OBJ	(to be) what a balcony is used for, what function it serves	

-TTḶ- ‘WALL / PARTITION / BARRIER’		
BSC	**Stem 1:** (to be) a planar wall/partition/barrier as the boundary of an enclosure/container, the surface partition constituting the boundary of an inclosed/contained space	**Stem 2:** (to be) a planar wall/partition/barrier as a defense against visibility by or intrusion of elements/entities from/on the other side **Stem 3:** (to be) a planar wall/partition as a component of a building or architectural structure
CTE	(to be) the state of enclosure/containment due to the presence of a wall/partition	
CSV	(to be) the tangible material component of a planar wall/partition/barrier; to build/erect a wall/partition/barrier	
OBJ	(to be) what is contained/inside the boundaries established by a wall/partition/barrier	

-ṬLW- ‘SCREEN’		
BSC	**Stem 1:** (to be) a state/act of placing or there being a screen (as partition) between two areas of space; to partition using a screen, to place a screen between, to put up a screen	**Stem 2:** (to be) a state/act of screening as a means of filtering (e.g., light, air, water, etc.) ; to filter something (e.g., light, air, water) using a screen **Stem 3:** (to be) a state/act of using a screen to display or project an image; to display/project an image on(to) a screen
CTE	(to be) the state of partitioning due to the presence of a screen	
CSV	(to be) the screen itself; to put up/erect a screen as a partition	
OBJ	(to be) what is partitioned by there being a screen in place	

-KTH- ‘FIRMAMENT / GROUND / FLOOR’		
BSC	**Stem 1:** (to be) the (quasi-) planar surface on which an entity naturally rests due to gravity, i.e., firmament, “ground”, “surface of the earth”, “floor”	**Stem 2:** (to be) the Earth’s natural land surface, i.e., the ground [in a literal sense] (whether dirt, sand, grass, rock, forest floor, mud, swamp, etc.) **Stem 3:** (to be) a (manually constructed) floor (of a building, room)
CTE	(to be) the tangible material substance of the firmament/surface/ground/floor upon which an entity is situated due to gravity	
CSV	(to be) the state/act of being bound to, being (up)on, or falling toward a (quasi-)planar surface/firmament due to gravity; to be bound to or headed for the ground due to gravity, to fall	
OBJ	(to be) that which is bound to, situated (up)on, or falling toward the ground due to gravity	

-ṬKY- ‘WINDOW / VIEWING APERTURE’		
BSC	**Stem 1:** (to be) an aperture / “window” in a surface for the purpose of being able to visually view what is on the other side	**Stem 2:** (to be) an aperture/window in a surface for the purpose of letting in sunlight (or artificial light) from outside or another room **Stem 3:** (to be) an aperture/window in a surface for the purpose of limited physical access between one side of a partition and another (e.g., as in a ticket window, a pass-through slot, etc.)
CTE	(to be) the state of there being (potential) visibility to the other side of a partition/wall or planar surface due to there being a window	
CSV	(to be) the physical aperture within a (quasi-) planar surface which constitutes a window	
OBJ	(to be) the (quasi-) transparent/transluscent planar entity situated within the bounds of a window/aperture to function as a protective partition between one side of the window and the other, e.g., plane of glass, screen, plane of plastic/acrylic, etc.	

-ŽTḶ- ‘STAIRCASE / LADDER / PEDESTAL’		
BSC	**Stem 1:** (to be) a staircase	**Stem 2:** (to be) a ladder **Stem 3:** (to be) a pedestal, plinth, or dais (i.e., a shelf-like planar surface for the purpose of positioning an entity at a higher level than the surrounding firmament)
CTE	(to be) the state of ascent/descent permitted by use of stairs	
CSV	(to be) the act/process of ambulating upon stairs; to ambulate upon stairs, go up or down stairs	
OBJ	(to be) a single individual stair of a staircase	

NOTE: Stems 1 and 2 of this root represent the atypical lexification of a collective entity (i.e., a collection of individual stairs/steps) due to the fact that staircases and ladders as wholes are more semantically salient than individual stairsteps or ladder-steps. Therefore, use the **SBS** affix with these stems to signify an individual stair of a staircase or step of a ladder.

-NŽH- ‘BRIDGE / WINCH / BOOM		
BSC	**Stem 1:** (to be) a bridge [both the physical object itself and the accessibility between two point it allows]	**Stem 2:** (to be) a winch or windlass and the ability to move a superheavy object **Stem 3:** (to be) an extended/extendable boom and the ability to lift/move a superheavy object
CTE	(to be) in a state of two points being accessible to each other via a bridge; to cross a bridge	
CSV	(to be) an act/process of bridging/spanning; to bridge, to span	
OBJ	(to be) the physical bridge itself	

-RM- ‘ABODE / DWELLING / DOMICILE / HOUSE’		
BSC	**Stem 1:** (to be) an abode/dwelling, i.e., where an entity dwells/lives, e.g., burrow, hole, nest, cave, den, lair, warren, house, apartment, trailer, etc.; to dwell, to live in [a place]	**Stem 2:** (to be) a domicile (i.e., an artificially constructed quasi-permanent residential dwelling which provides protection from outside environmental elements) **Stem 3:** (to be) a house (i.e., a quasi-permanent constructed residence with conveniences for quasi-permanent residence)
CTE	(to be) a in a state of dwelling somewhere; to have a place to live	
CSV	(to be) an act of dwelling; to go about the day-to-day activities that constitute living in a particular place	
OBJ	(to be) the party/entity that dwells within a dwelling	

-RHW- ‘OCCUPANCY / TENANCY / RESIDENCY’		
BSC	**Stem 1:** (to be) a state/act of occupancy within a delineated/demarcated place and time; to occupy	**Stem 2:** (to be) an act/state of tenancy (i.e., long-term occupancy with no assumptions regarding property rights or ownership claims **Stem 3:** (to be) an act/state of residency (i.e., long-term occupancy with assumptions of certain property rights/ownership claims)
CTE	(to be) the location one occupies, the place/locaton one occupies	
CSV	(to be) an act of occupancy; to take up space for a certain amount of time in a particular place	
OBJ	(to be) the party that occupies a place/domicile/residence/burrow/nest, etc.	

Morphological derivatives: vacancy, be vacant, to vacate

-PHW- ‘URBAN RESIDENCY / SUBURBAN RESIDENCY / RURAL RESIDENCY’		
BSC	**Stem 1:** (to be) a state/act of residency or being situated or taking place in an urban(ized)/city environment or locale	**Stem 2:** (to be) an act/state of residency or being situated or taking place in a suburban environment or the suburbs of a city **Stem 3:** (to be) an act/state of residency or being situated or taking place in a non-urban/rural environment/the countryside
CTE	(to be) a place of urban residency (use appropriate Configuration and SIZ affix to specify village, town, city, etc.)	
CSV	(to be) an act of making happen/take place/be situated in an urban environment/city	
OBJ	(to be) the party/entity/situation that takes place/occurs/resides in an urban environment/city	

-ČHW- ‘RELOCATION / RESETTLEMENT / MIGRATION / REFUGE’		
BSC	**Stem 1:** (to be) a state/act of relocation, resettlement of an entity in a new quasi-permanent residence/domicile/home away from that entity’s previous residence/domicile/home	**Stem 2:** (to be) an act/state of migration/immigration **Stem 3:** (to be) an act/state of taking refuge
CTE	(to be) where one is or has been relocated/resettled	
CSV	(to be) an act of relocating/resettling; to physically move from one location to another as part of an act/process of relocation/resettlement	
OBJ	(to be) the party/entity/situation that undergoes relocation/resettlement	